

Burrowing into the forest: Phylogeny of the Asian forest scorpions (Scorpionidae: Heterometrinae) and the evolution of ecomorphotypes

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Abstract

Asian forest scorpions (Scorpionidae Latreille, 1802: Heterometrinae Simon, 1879) are distributed across South and Southeast Asia. All are fossorial, constructing burrows under stones or in open ground, in habitats differing in precipitation and vegetation cover, from rainforests and tropical deciduous forests to savanna and scrubland. The systematics of these scorpions has long been confused due to bad taxonomy and the absence of a phylogenetic framework. Although the monophyly of the group was previously confirmed as part of broader phylogenetic analyses based on exemplar species, the only quantitative analysis of species-level variation to date was based on overall similarity. This contribution presents the first species-level phylogenetic analysis of Asian Scorpionidae, based on 186 morphological characters and 4188 aligned base-pairs of DNA sequence data from two nuclear and three mitochondrial loci for 132 terminals including all 41 ingroup species and four outgroup species. Simultaneous analyses of the morphological and molecular datasets with parsimony, Maximum Likelihood and Bayesian Inference provided the framework for a revised classification presented elsewhere. In order to understand how adaptation following dispersal into new habitats has driven the morphological diversification of Asian forest scorpions, species were scored for 10 characters concerning morphology and burrow architecture, which contributed to an ensemble index of adaptation to habitat aridity. Species were classified into three ecomorphotypes based on the index, and ancestral state reconstruction of ecomorphotypes performed on the phylogeny. A pattern was recovered in which lineages and species occurring in different habitats on a continuum from wet (evergreen forest) to dry (savanna, scrubland) exhibited characters presumed to be adaptive and hence responsible for driving scorpion diversification.

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Introduction

Asian forest scorpions (Scorpionidae Latreille, 1802: Heterometrinae Simon, 1879) are distributed across South and Southeast Asia from Pakistan, India and Sri Lanka to Wallace's Line in the Indonesian archipelago (Fig. 1). These impressive scorpions are among the largest in the world, some reaching almost 180 mm in length (Pocock, 1900; Vachon, 1953; Couzijn, 1981; Tikader and Bastawade, 1983; Sissom, 1990).

Although their venom is relatively mild, these scorpions deter attackers by snapping with their powerful pedipalp chelae, lunging with the metasoma and telson, and by stridulation, an audible rasping hiss created by rubbing together the opposing surfaces of the cheliceral coxae and the coxae of the first pair of walking legs (Prendini et al., 2003).

Like other Scorpionidae, Asian forest scorpions are fossorial, constructing burrows in loamy riverbanks and other sloping terrain, in open ground, at the base of stones or among the roots of trees (Khataavkar and More, 1990; Tare et al., 1993; Prendini et al., 2003; Bastawade, 2004, 2005, 2006a, 2006b, 2008, 2009;

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Sureshan et al., 2007; Bastawade and Borkar, 2008; Javed et al., 2010; Mirza et al., 2012; Pande et al., 2012; Tahir and Prendini, 2014). Composite, multi-entranced burrows containing up to 15 related individuals of overlapping generations have been recorded in some species (Shivashankar and Veeresh, 1991; Shivashankar, 1992, 1994) whereas others live subsocially within large termitaria (Hull-Williams, 1989). The thickened metasoma and robust legs with stout, spiniform macrosetae distributed laterally and distally on the basi- and telotarsi, and curved telotarsal ungues, of most species are used to excavate burrows in the hard, clayey or loamy substrata of their habitats. Although all Asian forest scorpions burrow, some opportunistically inhabit holes in rock walls, tree trunks and logs, enabling them to survive in and around human habitation.

Across their broad distribution, Asian scorpionids occur in habitats differing in precipitation and vegetation cover, including tropical and subtropical rainforests, moist and dry tropical deciduous forests, tropical thorn forests, savanna and scrubland (Prendini et al., 2003; Tahir and Prendini, 2014, Prendini and Loria, 2020; Fig. 2). The morphology of these scorpions closely reflects their habitats. Species from humid, densely vegetated habitats, like *Heterometrus longimanus* (Herbst, 1800) from the rainforests of Southeast Asia, are mostly or entirely black in colour, with large median ocelli, gracile pedipalps, metasoma and telson, and elongated, sparsely setose legs, whereas species from arid, sparsely vegetated habitats, like *Decanometrus xanthopus* (Pocock, 1897) from scrub savanna on the Deccan Plateau of India, are reddish-brown in colour, with small median ocelli, robust pedipalps, metasoma and telson, and short, pale legs armed with rake-like rows of spiniform macrosetae (Figs. 2–5). Although ecomorphological adaptations for burrowing in diverse habitats have evidently driven the evolution and diversification of Asian forest scorpions, the mechanisms by which this occurred have remained as poorly understood as their phylogeny and classification.

The systematics of Asian scorpionids has been confused throughout most of the group's history. Since the first species, *Heterometrus spinifer* (Ehrenberg, 1828), was described almost two centuries ago, 85 infrageneric names have been applied to these scorpions and many changed ranks, were synonymized or revalidated multiple times (Prendini and Loria, 2020). Their supraspecific classification has been similarly eclectic with various genera and subgenera recognized over the same timeframe.

Heterometrus Ehrenberg, 1828, the genus with which most Asian forest scorpions have long been associated, was first revised by Kraepelin (1899), and accommodated 13 species. Pocock (1900) diagnosed 16 species

and six subspecies from what was then British India including three new species and five new subspecies. After these early studies, the taxonomy of Asian forest scorpions was ignored for almost a century until interest renewed in the 1970s.

In the first modern revision, Couzijn (1981) recognized 21 species and 31 subspecies of *Heterometrus*. Couzijn (1978, 1981) also presented the first quantitative analysis of morphological variation, a phenetic cluster analysis, on which basis five subgenera were defined (Table 1): *Chersonesometrus* Couzijn, 1978, *Gigantometrus* Couzijn, 1978, *Javanimetrus* Couzijn, 1981, *Srilankametrus* Couzijn, 1981, and the nominotypical *Heterometrus*. Couzijn (1981) proposed that the African scorpionid genus, *Pandinus* Thorell, 1876, was the closest relative of the Asian scorpionids, of which *Gigantometrus* was suggested to be the most basal, sister to a monophyletic group comprising two clades, one comprising *Chersonesometrus* and *Heterometrus*, the other comprising *Javanimetrus* and *Srilankametrus* (Fig. 6).

Couzijn's (1981) attempt to partition the morphological variation into widespread polymorphic species with many subspecies was less successful. Subsequent attempts, describing new species and elevating subspecies based on small samples comprising few or no adult male specimens from a limited number of populations (e.g., Tikader and Bastawade, 1983; Kovářík, 2004; Lourenço et al., 2005; Zhu and Yang, 2007; Javed et al., 2010; Mirza et al., 2012; Plíšková et al., 2016) fared little better.

Tikader and Bastawade (1983) adopted Couzijn's (1981) subgeneric classification, published keys and redescrptions for the species of *Heterometrus* occurring in South Asia, and described two new species. Despite examining many of the same specimens, Tikader and Bastawade (1983) differed from Couzijn (1981) in their opinions on the rank or validity of various taxa such that 30 species and 27 subspecies (including eight nominotypical forms) were subsequently recognized (Fet, 2000).

Prendini (2000a) and Prendini et al. (2003) tested the monophyly and phylogenetic position of *Heterometrus* in Scorpionoidea Latreille, 1802 and Scorpionidae, respectively, with phylogenetic analyses of morphology and multilocus DNA sequence data for exemplar species. *Heterometrus* was consistently recovered as monophyletic and sister to *Pandinus*, to the exclusion of two other predominantly African scorpionid genera, *Opisthophthalmus* C.L. Koch, 1837 and *Scorpio* L., 1758, based on the shared presence of a stridulatory organ, comprising a “rasp” (granular tubercles) and “scraper” (stridulatory setae or scaphotrix) on opposing surfaces of the coxae of the pedipalps and the first pair of legs; and similar counts of pro- and retroventral spiniform macrosetae

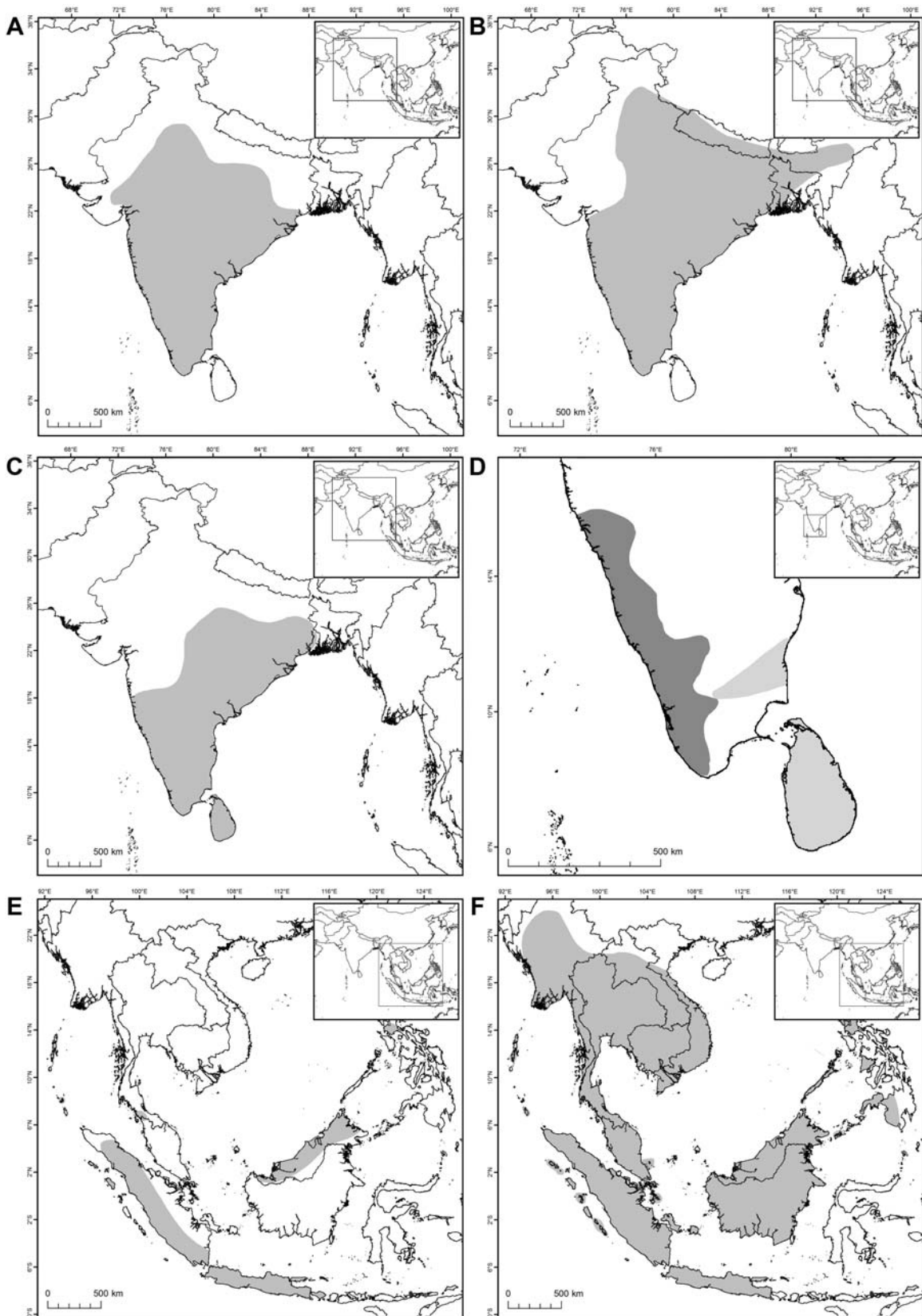


Fig. 1. Approximate distributions of the genera of Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879) in South (A–D) and Southeast Asia (E, F). A. *Chersonesometrus* Couzijn, 1978. B. *Deccanometrus* Prendini and Loria, 2020. C. *Gigantometrus* Couzijn, 1978. D. *Sahyadrimetrus* Prendini and Loria, 2020 (dark grey); *Srilankametrus* Couzijn, 1981 (light grey). E. *Javanimetrus* Couzijn, 1981. F. *Heterometrus* Ehrenberg, 1828.



Fig. 2. Ecomorphotypes and habitats of Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879). A. Wet silvicolous ecomorphotype: *Heterometrus longimanus* (Herbst, 1800), ♂, Kuching, Sarawak, Malaysia. B. Khao Kai Jae, Chumphon Prov., Thailand, evergreen forest habitat of *Heterometrus laevigatus* (Thorell, 1876). C. Dry silvicolous ecomorphotype: *Chersonesometrus fulvipes* (C.L. Koch, 1837), ♂, Gautala Wildlife Sanctuary, Maharashtra, India. D. Polonnaruwa, North Central Prov., Sri Lanka, deciduous forest habitat of *Srilankametrus gravimanus* (Pocock, 1894). E. Savannicolous ecomorphotype: *Deccanometrus xanthopus* (Pocock, 1897), ♂, Andori, Maharashtra, India. F. Saswad, Maharashtra, India, savanna habitat of *D. xanthopus*.

on the telotarsi of the first and second pairs of legs, and the third and fourth pairs of legs. Prendini et al. (2003) provided the first comprehensive diagnosis of *Heterometrus* and synthesized what was known about the biology of its species.

Kovářík (2004) presented the third revision of *Heterometrus*, in which many of the specimens examined by Pocock (1900), Couzijn (1981) and Tikader and Bastawade (1983) were again re-examined. Kovářík (2004) ignored the work of Prendini et al. (2003) and disagreed

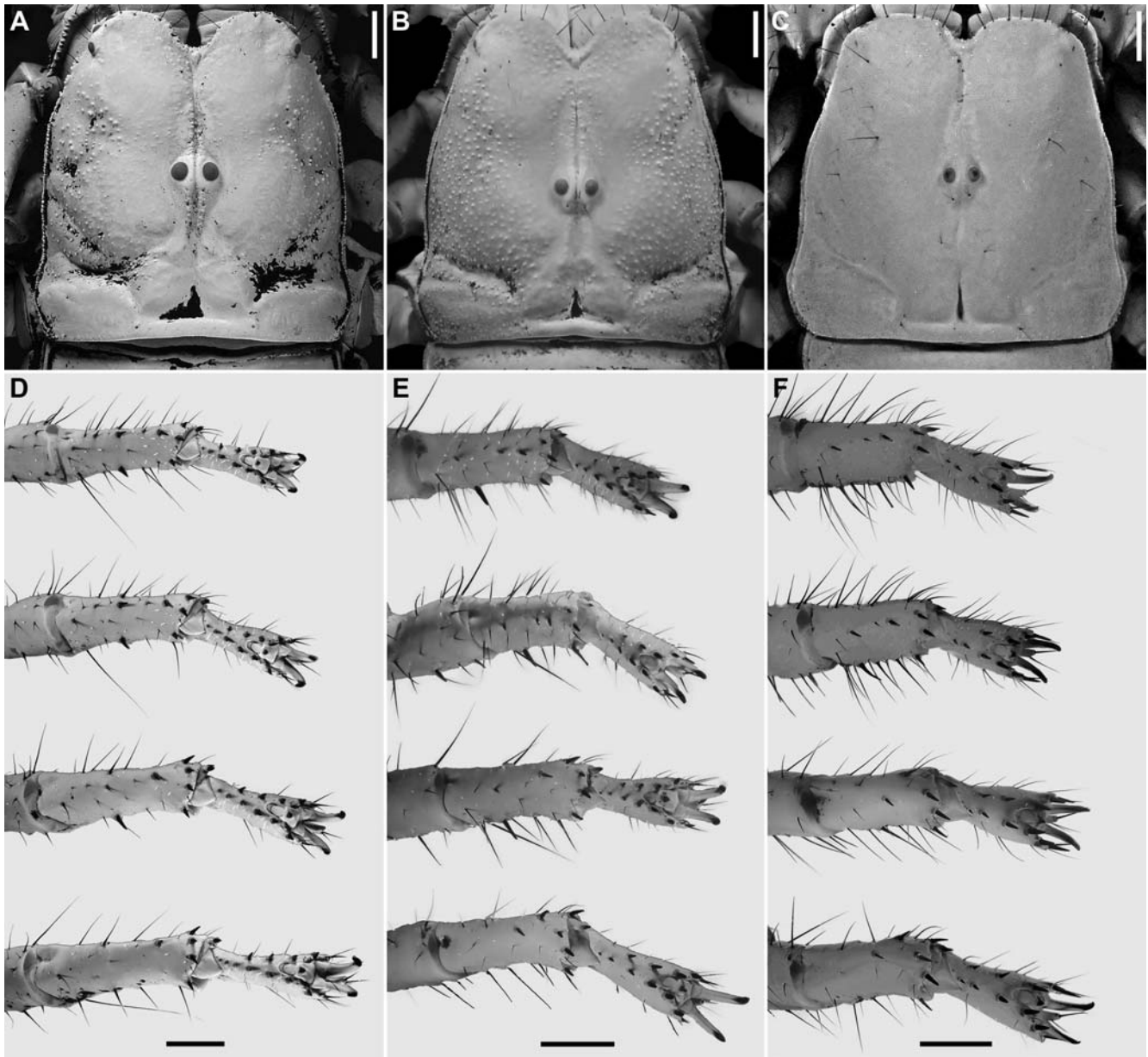


Fig. 3. Carapace, dorsal aspect (A–C), and legs I–IV, ventral aspect (D–F), of three ecomorphotypes of Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879). A, D. Wet silvicolous ecomorphotype: *Heterometrus longimanus* (Herbst, 1800), ♂ (AMNH), Kuching, Malaysia. B, E. Dry silvicolous ecomorphotype: *Chersonesometrus madraspatensis* (Pocock, 1900), ♂ (CAS 9071142). C, F. Savannicolous ecomorphotype: *Deccanometrus latimanus* (Pocock, 1894), ♂ (AMNH), Charbagh, Pakistan (C), ♀ (AMNH), Daggar, Pakistan (F). Scale bars = 5 mm (A–C), 1 mm (D–F)

with many of the taxonomic decisions of Couzijn (1981). Couzijn's (1981) subgenera were synonymized, along with five species and six subspecies, seven new species described, and one species revalidated, leaving the genus with 31 species and without subspecies.

Further changes after Kovařík (2004) included the description of six new species (Lourenço et al., 2005; Zhu and Yang, 2007; Kovařík, 2009; Javed et al., 2010; Mirza et al., 2012; Plíšková et al., 2016; Kovařík et al., 2019), the revalidation of one species (Kovařík

et al., 2016), and the revalidation of subgenus *Gigantometrus* (Rossi, 2016).

When the present study commenced, the classification of Asian forest scorpions did not accurately reflect either the taxonomic or cladistic diversity in the group. The basis for the confusion lay in two areas (Prendini et al., 2003; Tahir and Prendini, 2014): (1) bad taxonomy caused by small sample sizes of material examined (typically fewer than five, but often singletons, female or immature), failure to compare new material with

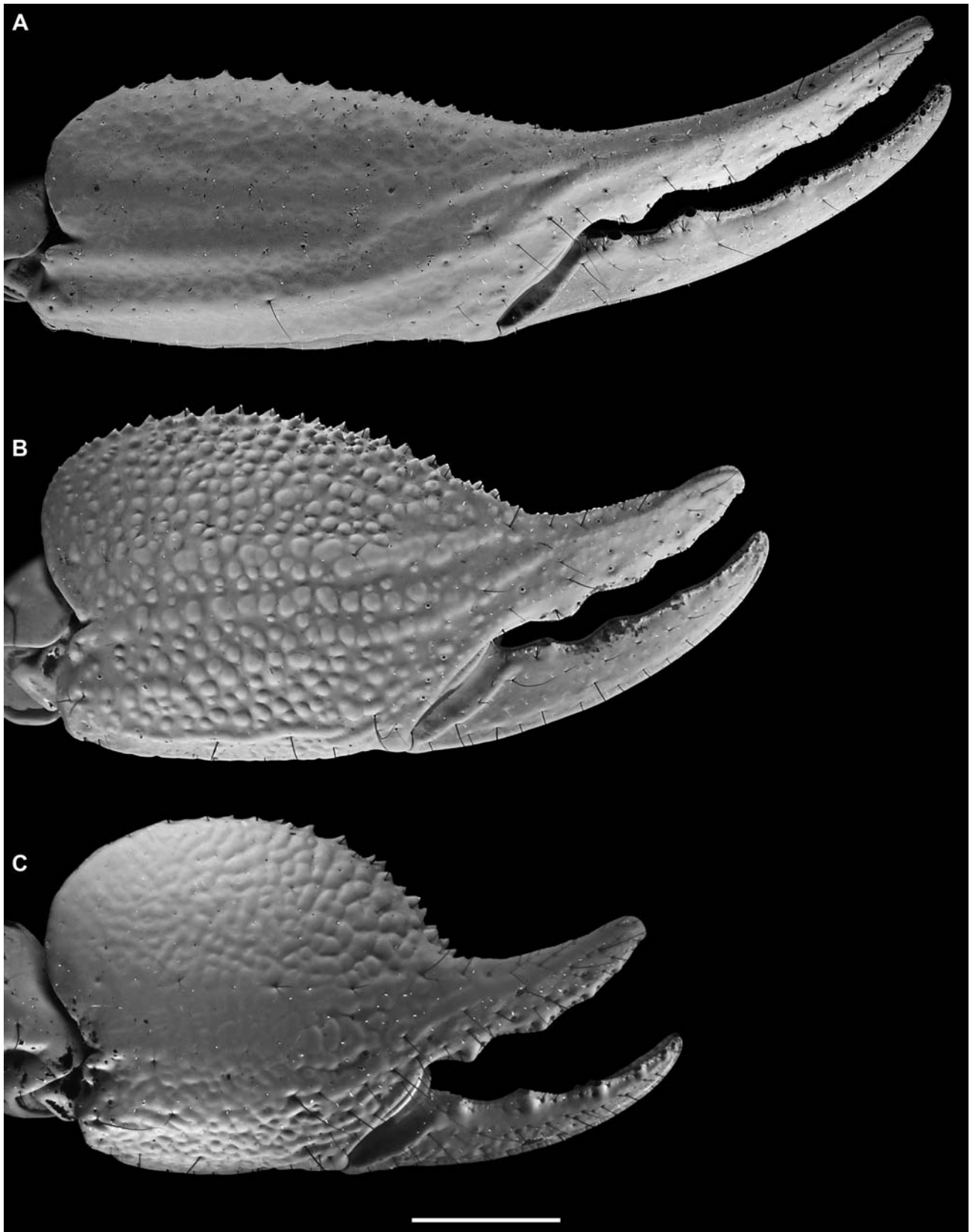


Fig. 4. Pedipalp chela, retrodorsal aspect, of three ecomorphotypes of Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879). A. Wet silvicolous ecomorphotype: *Heterometrus longimanus* (Herbst, 1800), ♂ (AMNH), Kuching, Malaysia. B. Dry silvicolous ecomorphotype: *Chersonesometrus madraspatensis* (Pocock, 1900), ♂ (CAS 9071142). C. Savannicolous ecomorphotype: *Deccanometrus latimanus* (Pocock, 1894), ♂ (AMNH), Daggar, Pakistan. Scale bar = 5 mm

previously described taxa, a lack of understanding of ontogenetic variation and sexual dimorphism, and weak diagnoses with inadequate illustrations; (2) the absence of a rigorous phylogenetic framework. Whereas the monophyly of the Asian scorpionids was previously tested and confirmed as part of broader phylogenetic analyses based on exemplar species (Prendini, 2000a; Prendini et al., 2003), the only quantitative analysis of species-level variation to date (Coutzjn, 1978, 1981) was based on overall similarity not synapomorphy.

Prendini and Loria (2020) addressed the first problem with a comprehensive revision of the Asian Scorpionidae, in which Coutzjn's (1981) subgenera were elevated to the rank of genus, two new genera described, 27 new synonyms presented, and 41 species diagnosed (Table 2). The present contribution focuses on the second problem in providing a phylogenetic framework for the revised classification of Prendini and Loria (2020), as well as testing the monophyly of most Asian scorpionid species. The phylogeny presented is based on a simultaneous analysis of 186 morphological characters and 4188 base-pairs of concatenated DNA sequence from two nuclear loci (18S rDNA and 28S rDNA) and three mitochondrial loci (Cytochrome *c* Oxidase Subunit I, 16S rDNA and 12S rDNA) for 132 terminals, including all 41 ingroup species and four outgroup species, representing three scorpionid genera and one genus of Diplocentridae Karsch, 1880, a close relative of Scorpionidae (Prendini, 2000a; Prendini et al., 2003).

In order to understand how adaptation following dispersal into new habitats may have driven the morphological diversification of Asian scorpionids, species were scored for 10 characters concerning morphology and burrow architecture, which contributed to an ensemble index of adaptation to habitat aridity. Species were classified into three ecomorphotypes based on their habitat aridity adaptation index (i.e., the summed scores of the characters), and ancestral state reconstruction of ecomorphotypes performed on the phylogeny.

Materials and Methods

Taxon sample

The taxon sample included four outgroup terminals and 128 ingroup terminals representing all genera and species of Asian Scorpionidae (Appendix 1). Ten ingroup species for which tissue samples could not be obtained were represented only by morphological data: *Chersonesometrus nathanorum* Prendini and Loria, 2020; *Chersonesometrus shivashankari* Prendini and Loria, 2020; *Chersonesometrus wroughtoni* (Pocock, 1899); *Deccanometrus liurus* (Pocock, 1897); *Deccanometrus phipsoni* (Pocock, 1893); *Deccanometrus ubicki* (Kovářík, 2004); *Sahyadrimetrus barberi* (Pocock, 1900); *Srilankametrus caesar* (C.L. Koch, 1841); *Srilankametrus couzjini*

Prendini and Loria, 2020; *Srilankametrus yaleensis* (Kovářík et al., 2019). Additional terminals were scored from the type material of *Chersonesometrus pelekomanus* (Coutzjn, 1981), *Deccanometrus bengalensis* (C.L. Koch, 1841), *Deccanometrus phipsoni*, *Deccanometrus xanthopus* (Pocock, 1897) and *Srilankametrus caesar* to test the phylogenetic position of the types, relative to other putatively conspecific terminals, resulting in a total of fifteen terminals represented only by morphological data.

The outgroup comprised exemplar species of three scorpionid genera, representing each of the other four subfamilies from Africa and the Middle East: *Opisthophthalmus capensis* (Herbst, 1800), representing Opisthophthalminae Rossi, 2016; *Pandinus imperator* (C.L. Koch, 1841), representing Pandininae Thorell, 1876; and *Scorpio fuscus* (Ehrenberg, 1829), representing Scorpioninae Latreille, 1802. Phylogenetic trees were rooted on *Nebo hierichonticus* (Simon, 1872), representing Diplocentridae, a family closely related to Scorpionidae (Prendini, 2000a; Prendini et al., 2003).

The ingroup comprised multiple terminals sampled from across the range of each genus and species. *Heterometrus*, the most widespread Asian scorpionid genus, was represented by 43 terminals, *Chersonesometrus* and *Sahyadrimetrus* Prendini and Loria, 2020 were each represented by 24 terminals, *Deccanometrus* Prendini and Loria, 2020 by 17, *Srilankametrus* by 10, and *Gigantometrus* and the monotypic genus, *Javanimetrus*, by five each. As several Asian scorpionid species are widely distributed and formerly comprised geographical subspecies, 31 were represented by two or more terminals from geographically separated locations. Only 10 ingroup species were represented by a single terminal, i.e., four species of *Srilankametrus*, three species of *Chersonesometrus*, two species of *Deccanometrus*, and one species of *Sahyadrimetrus*. One species each of *Gigantometrus* and *Heterometrus*, two species of *Deccanometrus*, and three species each of *Chersonesometrus* and *Srilankametrus*, were represented by two terminals; one species each of *Deccanometrus*, *Gigantometrus*, *Heterometrus* and *Sahyadrimetrus*, and three species of *Chersonesometrus*, by three; two species each of *Deccanometrus* and *Sahyadrimetrus*, by four; one species each of *Heterometrus*, *Javanimetrus* and *Sahyadrimetrus* by five; one species of *Chersonesometrus* and three species of *Heterometrus*, by six; one species each of *Heterometrus* and *Sahyadrimetrus*, by seven; and one species of *Heterometrus* by eight.

Material

Materials were obtained by field collections, donations, and loans from natural history collections (Appendix 1). Field collections involved turning logs and stones, or excavating burrows during the day, or searching at night using portable ultraviolet (UV) lamps, comprising a pair of mercury-vapor tubes attached to a chromium parabolic reflector and powered by a rechargeable 7 Amp/hr, 12 V battery, or UV light-emitting diode (LED) flashlights. Material was preserved and injected with 95% ethanol.

Tissue samples are stored at -150°C in the Ambrose Monell Cyrocollection for Molecular and Microbial Research (AMCC) at the American Museum of Natural History (AMNH).

A portable Garmin™ GPS etrex 30 device was used for recording the geographical coordinates (WGS84 datum) of some collection localities in the field; others were retroactively georeferenced using Google Earth.

Morphology

Morphological characters were scored from adult male and female specimens of each species, preferably vouchers from the same locations as tissue samples, when available, or the nearest geographical locations. Specimens were examined with Nikon SMZ1500 and Leica



Fig. 5. Metasoma and telson, lateral aspect, of three ecomorphotypes of Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879). A. Wet silvicolous ecomorphotype: *Heterometrus longimanus* (Herbst, 1800), ♂ (AMNH), Kuching, Malaysia. B. Dry silvicolous ecomorphotype: *Chersonesometrus madraspatensis* (Pocock, 1900), ♂ (CAS 9071142). C. Savannicolous ecomorphotype: *Deccanometrus latimanus* (Pocock, 1894), ♂ (AMNH), Charbagh, Pakistan. Scale bar = 10 mm

MZ16 stereo dissecting microscopes. Photographs were taken in visible light as well as under long wave UV light using a Microptics™ ML-1000 digital photomicrography system.

A morphological data matrix, comprising 186 morphological characters scored for all 132 terminal taxa (Appendices 2, 3), and incorporating characters from previously published matrices by Lamoral (1979), Couzijn (1981), Prendini (2000a), and Prendini et al. (2003), was prepared using WinClada ver. 1.00.08 (Nixon, 2002), and deposited in Morphobank (<http://morphobank.org/permalink/P3719>). Characters scored in the matrix included a variety of traits capturing the morphological diversity within Scorpionidae, most pertaining to

Table 1

Couzijn's (1981) classification of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879

fulvipes group
<i>Heterometrus</i> (<i>Chersonesometrus</i>) <i>fastigosus</i> Couzijn, 1981 *
<i>H. (C.) fulvipes</i> (C.L. Koch, 1838)
<i>H. (C.) granulomanus</i> Couzijn, 1981
<i>H. (C.) pelekomanus</i> Couzijn, 1981
<i>H. (C.) scaber obscurus</i> Couzijn, 1981
<i>H. (C.) s. rugosus</i> Couzijn, 1981
<i>H. (C.) s. scaber</i> (Thorell, 1877)
<i>H. (C.) xanthopus</i> (Pocock, 1897)
phipsoni group
<i>H. (C.) collinus</i> (Pocock, 1900)
<i>H. (C.) liurus</i> (Pocock, 1897)
<i>H. (C.) phipsoni kanarensis</i> (Pocock, 1900)
<i>H. (C.) p. phipsoni</i> (Pocock, 1892)
<i>H. (C.) tristis</i> (Henderson, 1919)
<i>H. (C.) wroughtoni</i> (Pocock, 1899)
<i>Heterometrus</i> (<i>Gigantometrus</i>) <i>swammerdami flavimanus</i> Pocock, 1900
<i>H. (G.) s. swammerdami</i> Simon, 1872
<i>H. (G.) s. titanicus</i> Couzijn, 1981
<i>Heterometrus</i> (<i>H.</i>) <i>bengalensis</i> (C.L. Koch, 1841) *
<i>H. (H.) laoticus</i> Couzijn, 1981
<i>H. (H.) liophysa laevifrons</i> Roewer, 1943
<i>H. (H.) l. liophysa</i> (Thorell, 1888)
<i>H. (H.) l. separatus</i> Couzijn, 1981
<i>H. (H.) l. spartanicus</i> Couzijn, 1981
<i>H. (H.) longimanus angustimanus</i> (Thorell 1877)
<i>H. (H.) l. belitungensis</i> Couzijn, 1981
<i>H. (H.) l. borneensis</i> (Thorell, 1877)
<i>H. (H.) l. humilis</i> (Simon, 1877)
<i>H. (H.) l. longimanus</i> (Herbst, 1800)
<i>H. (H.) l. marmoratus</i> Couzijn, 1981
<i>H. (H.) l. paris</i> Couzijn, 1981
<i>H. (H.) l. tarawakaensis</i> Couzijn, 1981
<i>H. (H.) l. bengkalitensis</i> Couzijn, 1981
<i>H. (H.) madoerensis</i> Couzijn, 1981
<i>H. (H.) petersii luzonensis</i> Couzijn, 1981
<i>H. (H.) p. mindanaensis</i> Couzijn, 1981
<i>H. (H.) p. petersii</i> (Thorell, 1876)
<i>H. (H.) spinifer solitarius</i> Couzijn, 1981
<i>H. (H.) s. spinifer</i> Ehrenberg, 1828
<i>Heterometrus</i> (<i>Javanimetrus</i>) <i>c. cyaneus</i> (C.L. Koch, 1836)
<i>H. (J.) c. insulanus</i> Couzijn, 1981
<i>H. (J.) c. sumatrensis</i> Couzijn, 1981
<i>Heterometrus</i> (<i>Srilankametrus</i>) <i>i. indus</i> (DeGeer, 1778)
<i>H. (S.) i. laevitensus</i> Couzijn, 1981

**Heterometrus thorellii* (Pocock, 1897) was misidentified as *Heterometrus (H.) bengalensis* (C.L. Koch, 1841), which was, in turn, misidentified as *Heterometrus (Chersonesometrus) fastigosus* Couzijn, 1981.

the adult stage (Appendices 2, 3): total length, 1 (0.5%); chelicerae, 4 (2%); carapace, 30 (16%); pedipalps, 61 (32.5%); coxosternum, 1 (0.5%); legs, 33 (18%); genital operculum, 1 (0.5%); hemispermatophore, 3 (1.5%); pectines, 5 (3%); tergites, 5 (3%); sternites, 4 (2%); metasoma, 29 (16%); telson, 5 (3%); ecology and behavior, 3 (1.5%). The breakdown of morphological characters by character system is as follows: surface macrosculpture, 70 (38%); macrosetae, 32 (17%); colour patterns, 19 (10%); shape and size, 29 (16%); surface topography, 13 (7%); trichobothria, 9 (5%); genitalia, 4 (2%); ungues, 3 (2%); sutures, 3 (2%); dentition, 1 (1%). Fifty-eight (31%) characters were sexually dimorphic with 30 and 28 characters scored only for males and females, respectively. Male and female characters were scored with question marks for *S. barberi* and *S. couzijni*, respectively known only from the adult female and adult male, and partially scored for *C. nathanorum*, only known from immatures. Morphological terminology follows previous papers on Scorpionidae by the second author (e.g., Prendini, 2000b, 2001a; Prendini et al., 2003; Tahir and Prendini, 2014; Prendini, 2016; Prendini and Loria, 2020).

Forty-seven (25%) multistate characters were treated as unordered/non-additive (Fitch, 1971) in the phylogenetic analyses to avoid *a priori* hypotheses of character state transformation. Fifty-two (28%) characters were uninformative. Morphological characters were optimized on the preferred phylogenetic hypothesis using WinClada ver. 1.00.08.

DNA sequencing

Two loci in the nuclear genome and three loci in the mitochondrial genome were amplified and sequenced at the AMNH Sackler Institute for Comparative Genomics. The nuclear small subunit ribosomal RNA (18S rDNA, hereafter "18S") and the D3 region of the large subunit ribosomal RNA (28S rDNA, hereafter "28S"), and the mitochondrial small subunit ribosomal RNA (12S rDNA, hereafter "12S"), large subunit ribosomal RNA (16S rDNA, hereafter "16S"), and Cytochrome *c* Oxidase subunit I (hereafter "COI"), were selected as this combination of loci is highly informative for resolving relationships from species to family level in a range of scorpion taxa (Prendini et al., 2003; González-Santillán and Prendini, 2015; Santibáñez-López et al., 2014; Ceccarelli et al., 2016; Ojanguren-Affilastro et al., 2016; Esposito et al., 2018, Esposito and Prendini, 2019).

DNA was extracted from leg muscle tissue using the Qiagen (Germantown, MD) DNeasy Blood and Tissue Kit, using the manufacturer's protocol. The polymerase chain reaction (PCR) was performed in thermocyclers (Eppendorf EP and Eppendorf Pro, Hamburg, Germany) using a variety of optimization protocols, primer combinations (Table 3) and reagents, including Taq (Fisher Scientific, Hampton, NH), TopTaq (Qiagen), REDTaq (Sigma-Aldrich, St. Louis, MO), REDTaq ReadyMix (Sigma-Aldrich), and Illustra PuReTaq Ready-To-Go PCR beads (GE Healthcare, Chicago, IL). The adjuvants bovine serum albumin (BSA), dimethyl sulphoxide (DMSO) and MgCl₂ were added when necessary to assist amplification. The COI and 18S loci were sequenced in two and three fragments, respectively, whereas all other loci were sequenced in a single fragment. PCR clean-up was performed using Agencourt Ampure (Beckman Coulter, Brea, CA) on a Biomek NX robot (Beckman Coulter). Sanger dideoxy sequencing was conducted using standard protocols for the BigDye Terminator ver. 3.1 cycle sequencing kit (Life Technologies, Carlsbad, CA) and prepared for an ABI 3730XL automated DNA sequencer (Perkin Elmer, Waltham, CA) using Agencourt CleanSEQ (Beckman Coulter) on the Biomek NX robot. Each locus was sequenced in both the forward and reverse directions in order to verify results. Double-stranded sequences were edited, primer sequences removed, and assembled into consensus sequences using Sequencher ver. 5.4.6 (Gene Codes Corp., Ann Arbor, MI), and deposited in GenBank (<https://www.ncbi.nlm.nih.gov/genbank>; Table 4).

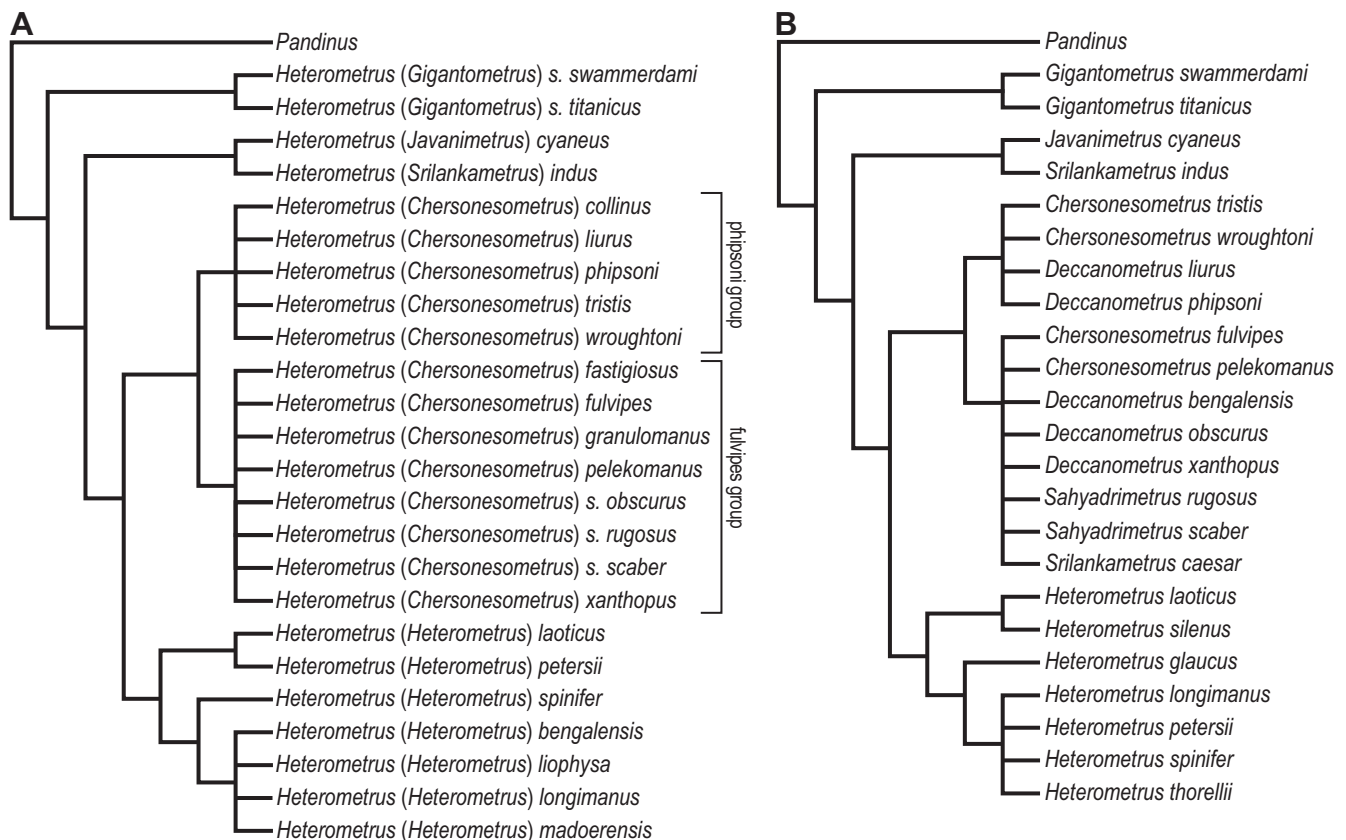


Fig. 6. Couzijn's (1981: 80, fig. 21) dendrogram portraying hypothetical affinities between the subgenera, species groups and species of *Heterometrus* Ehrenberg, 1828 (Scorpionidae: Heterometrinae Simon, 1879). A. Couzijn's (1981) classification, in which *Heterometrus thorellii* (Pocock, 1897) was misidentified as *Heterometrus* (*H.*) *bengalensis* (C.L. Koch, 1841), which was, in turn, misidentified as *Heterometrus* (*Chersonesometrus*) *fastigosus* Couzijn, 1981, and *Srilankametrus caesar* (C.L. Koch, 1841) was misidentified as *Heterometrus* (*Chersonesometrus*) *granulomanus* Couzijn, 1981. B. Reinterpreted according to the revised classification of Prendini and Loria (2020), with four synonyms removed: *Palamnaeus phipsoni collinus* Pocock, 1900 = *Deccanometrus phipsoni* (Pocock, 1893); *Heterometrus liophysa* var. *madoerensis* Kopstein, 1921 = *Heterometrus glaucus* (Thorell, 1876); *H. (C.) fastigosus* = *Deccanometrus bengalensis* (C.L. Koch, 1841); *Heterometrus* (*Chersonesometrus*) *granulomanus* Couzijn, 1981 = *Srilankametrus caesar* (C.L. Koch, 1841).

Sequence lengths were calculated using the “gc_calculator.py” script in Biopython (Cock et al., 2009). Sequences of the 18S, 28S and COI loci were length-invariant, respectively comprising 1761 nucleotide base-pairs (bp), 514 bp and 1078 bp in all terminals, except *N. hierichonticus*, which was 515 bp for 28S (Table 5). Sequences of the 12S locus varied from 328–335 bp with a mode of 333 bp and the 16S locus from 481–487 bp with a mode of 481 bp.

Sequence alignment and statistics

DNA sequence alignment was performed with MAFFT ver. 7.429 by auto-aligning with the “leavegappyregion” option selected (Katoh et al., 2002; Katoh and Standley, 2013). The L-INS-i method, an iterative refinement method that uses a local pairwise alignment with an affine gap cost, was selected as the best alignment for all loci (Katoh and Toh, 2008).

DNA sequence alignments were checked using Mesquite ver. 3.5 (Maddison and Maddison, 2019). The length of the aligned sequences was 1761 bp for 18S, 515 bp for 28S, 342 bp for 12S, 492 bp for 16S, and 1078 bp for COI, summing to a total of 4188 bp for the concatenated dataset (Table 5). Gaps representing insertion/deletion events were present in the aligned 12S (20 gaps), 16S (11 gaps) and 28S (1 gap).

Average percentage GC content, number of variable positions and parsimony informative sites were calculated for all aligned loci in MEGA ver. 7 (Kumar et al., 2016). AT content was higher in the 12S, 16S and COI loci whereas GC content was higher in the 18S and 28S loci, typical of arthropod and nuclear genomes (DeSalle et al., 1987; Simon et al., 1994; Wahlberg and Zimmermann, 2000; Prendini et al., 2003). The numbers of variable positions and parsimony-informative sites were highest for COI, followed by 16S, 12S, 28S and 18S (Table 5). When considered as a percentage of the alignment length of each locus, however, the numbers of variable positions and parsimony-informative sites were highest for 12S, followed by 16S, COI, 28S and 18S, as in other scorpion taxa (Prendini et al., 2003; González-Santillán and Prendini, 2015; Santibáñez-López et al., 2014; Ojanguren-Affilastro et al., 2016; Esposito et al., 2018). Also as expected for a protein-coding gene, the third codon position of the COI was the most informative, containing 77% and 82% of the variable positions and parsimony-informative sites in the locus, followed by the first codon position, containing 19% and 15%, respectively. The number of haplotypes for each locus, calculated using *DNAcol-lapser* (<http://users-birc.au.dk/biopv/php/fabox/dnacollapser.php>) in FaBox (Villesen, 2007), was highest in COI, followed by 12S, 16S, 28S and 18S.

Table 2

Genera and species of Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879) with known countries and Indian states of occurrence, based on Prendini and Loria (2020)

Species	Distribution
<i>Chersonesometrus bastawadei</i> (Prendini and Loria, 2020)	India (Karnataka)
<i>C. beccaloniae</i> (Kovářík, 2004)	India (Tamil Nadu)
<i>C. fulvipes</i> (C.L. Koch, 1837)	India (Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, National Capital Territory, Rajasthan, Uttar Pradesh), Pakistan
<i>C. hendersoni</i> (Prendini and Loria, 2020)	India (Karnataka)
<i>C. madraspatensis</i> (Pocock, 1900)	India (Andhra Pradesh, Chhattisgarh, Karnataka, Odisha, Tamil Nadu, Telangana, Puducherry)
<i>C. nathanorum</i> (Prendini and Loria, 2020)	India (Tamil Nadu)
<i>C. pelekomanus</i> (Couzijn, 1981)	India (Karnataka)
<i>C. shivashankari</i> (Prendini and Loria, 2020)	India (Andhra Pradesh, Karnataka)
<i>C. tristis</i> (Henderson, 1919)	India (Andhra Pradesh, Karnataka, Tamil Nadu)
<i>C. wroughtoni</i> (Pocock, 1899)	India (Karnataka, Maharashtra)
<i>Deccanometrus bengalensis</i> (C.L. Koch, 1841)	Bangladesh, China (Tibet Autonomous Region), India (Assam, Haryana, Himachal Pradesh, Jharkhand, Madhya Pradesh, Odisha, Rajasthan, Uttar Pradesh, Uttarakhand, West Bengal), Nepal
<i>D. latimanus</i> (Pocock, 1894)	Pakistan
<i>D. liurus</i> (Pocock, 1897)	India (Madhya Pradesh)
<i>D. obscurus</i> (Couzijn, 1981)	India (Gujarat, Maharashtra)
<i>D. phipsoni</i> (Pocock, 1893)	India (Andhra Pradesh, Chhattisgarh, Karnataka, Madhya Pradesh, Maharashtra, Tamil Nadu)
<i>D. ubicki</i> (Kovářík, 2004)	India (Puducherry)
<i>D. xanthopus</i> (Pocock, 1897)	India (Maharashtra, Telangana)
<i>Gigantometrus swammerdami</i> (Simon, 1872)	India (Andhra Pradesh, Chhattisgarh, Jharkhand, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Odisha, Puducherry, Tamil Nadu, Telangana, West Bengal)
<i>G. titanicus</i> (Couzijn, 1981)	Sri Lanka
<i>Heterometrus glaucus</i> (Thorell, 1876)	India (Nicobar Islands), Indonesia (Sumatra and offshore islands)
<i>H. laevigatus</i> (Thorell, 1876)	Myanmar, Thailand
<i>H. laoticus</i> Couzijn, 1981	Cambodia, Laos, Thailand
<i>H. longimanus</i> (Herbst, 1800)	Brunei, India (Andaman Islands), Indonesia, Malaysia, Philippines, Singapore
<i>H. petersii</i> (Thorell, 1876)	Malaysia (Penang), Singapore
<i>H. silenus</i> (Simon, 1884)	Cambodia, Vietnam
<i>H. spinifer</i> (Ehrenberg, 1828)	Malaysia (peninsular), Thailand
<i>H. thorellii</i> (Pocock, 1897)	Myanmar
<i>Javanimetrus cyaneus</i> (C.L. Koch, 1836)	India (Nicobar Islands), Indonesia, Malaysia, Philippines, Thailand
<i>Sahyadrimetrus barberi</i> (Pocock, 1900)	India (Kerala, Tamil Nadu)
<i>S. kanarensis</i> (Pocock, 1900)	India (Goa, Karnataka)
<i>S. mathewi</i> (Prendini and Loria, 2020)	India (Kerala)
<i>S. rugosus</i> (Couzijn, 1981)	India (Karnataka, Kerala, Tamil Nadu)
<i>S. scaber</i> (Thorell, 1876)	India (Goa, Karnataka, Kerala, Puducherry, Tamil Nadu)
<i>S. tikaderi</i> (Prendini and Loria, 2020)	India (Tamil Nadu)
<i>Srilankametrus caesar</i> (C.L. Koch, 1841)	India (Puducherry)
<i>S. couzijni</i> (Prendini and Loria, 2020)	India (Tamil Nadu)
<i>S. gravimanus</i> (Pocock, 1894)	Sri Lanka
<i>S. indus</i> (DeGeer, 1778)	Sri Lanka
<i>S. pococki</i> (Prendini and Loria, 2020)	Sri Lanka
<i>S. serratus</i> (Pocock, 1900)	Sri Lanka
<i>S. yaleensis</i> Kovářík et al., 2019	Sri Lanka

Phylogenetic analyses

The morphological character matrix and the concatenated dataset of aligned nuclear and mitochondrial DNA sequences (hereafter, “the molecular dataset”) were analyzed simultaneously using Maximum Likelihood (ML), Bayesian Inference (BI), and parsimony with equal weighting and implied weighting (Goloboff, 1993), applying five values for the concavity constant, k , ranging from strong to mild ($k = 1, 3, 10, 60, 100$). Additionally, to provide an independent comparison, the morphological and molecular datasets were analyzed

separately using parsimony with equal weighting and implied weighting with $k = 1, 3, 10, 60, 100$, and the molecular dataset was analyzed separately using ML and BI.

Parsimony analyses were performed with TNT ver. 1.1 (Goloboff et al., 2003, 2008). Gaps were treated as missing data and uninformative characters deactivated with the xinct command. A script from Dimitrov et al. (2012, 2013), modified by Santibáñez-López et al. (2014), which includes tree drifting, mixed sectorial search and tree fusing was used for the tree search: hold 10000; rseed1; xm: noverb nokeep; rat: it 0 up 4 down 4 au 0 num 36 give 99 equa; dri:

it 10 fit 1.00 rfi 0.20 aut 0 num 36 give 99 xfa 3.00 equa; sec: mins 45 maxs 45 self 43 incr 75 minf 10 god 75 drift 6 glob 5 dglob 10 rou 3 xss 10- 14 + 2 noxev noeq; tf: rou 5 minf 3 best ke nochoo swap; xm: level 10 nochk rep 50 fuse 3 dri 10 rss css noxss mult nodump conse 5 conf 75 nogive notarg upda autoc 3 xmix; xm; xmult.

A jackknife analysis was performed to evaluate nodal support using another script modified from Dimitrov et al. (2012): hold 10 000; rseed1; mult: noratchet repl 10 tbr hold 1; resample jak repl 1000 freq from 0 [mult]. A bootstrap analysis was performed for comparison with 1000 pseudoreplicates. Partitioned Bremer support was calculated with implied weighting and a low value for the concavity constant ($k = 1$) using the TNT script “pbsup.run” (Baker and DeSalle, 1997; Table 6). Tree length, consistency indices (CI), retention indices (RI), fit and adjusted homoplasy were also calculated (Table 7).

The ML analysis was conducted using RAxML-HPG ver. 8.2.12 on the CIPRES Gateway (<https://www.phylo.org>; Stamatakis, 2006, 2014). The dataset was partitioned by loci, and a MULTIGAMMA model implemented with an Mk model (Lewis, 2001) for the morphological partition. A second ML analysis, in which an ascertainment bias was applied to the morphological partition (ML-ASC) was performed for comparison. A rapid bootstrap analysis with 1000 replicates (Stamatakis, 2014) was used to search for the ML tree, following Ojanguen-Affilastro et al. (2016).

Twenty-four nucleotide substitution models were tested for each locus using JModeltest ver. 2.1.6 (Guindon and Gascuel, 2003; Darriba et al., 2012) on the CIPRES gateway. Models were selected using the Akaike Information Criterion (AIC) and a gamma distribution assumed for the morphological dataset (Lewis, 2001; Table 5). Using these models, BI was conducted in MrBayes v. 3.2.7 (Ronquist and Huelsenbeck, 2003) with unlinked parameters for all partitions and the following settings: mcmc; nchains = 4; print frequency = 1000; sample frequency = 1000; diagnosing frequency = 1000; burnin = 0.25; savebrlens = yes. The analysis was

terminated after 25 million generations when the standard deviation of the split frequencies was below 0.01.

Ecomorphotypes and ancestral state reconstruction

An analysis of 10 morphological and behavioral characters (Table 8; Figs. 2–5), considered potentially adaptive, was conducted to understand how dispersal into new and different habitats may have driven morphological diversification in Asian scorpionids. Five of these characters concern colouration of the carapace, tergites, sternites, metasoma, pedipalps and legs. It is well known that scorpions occurring in humid and forested habitats are mostly or entirely dark, typically blackish or brownish in colour, whereas scorpions occurring in arid habitats with open vegetation are paler, typically brownish or yellowish in colour, often with contrasting patterns, such as paler legs, pedipalp segments and telson (Prendini, 2001b, 2012). The adaptive function of these differences in colouration is unknown but they are presumed to confer some protection from vertebrate predation, by providing camouflage when scorpions are active on the surface at night. Dark colouration would presumably be advantageous in the darkness of the forest floor and/or in areas with dark, humic substrates, whereas mixed or contrasting colouration would be advantageous for breaking up the body outline in habitats with open vegetation and a heterogeneous substrate (e.g., savanna), and pale colouration would be advantageous in areas with sparse vegetation and pale, sandy substrates (e.g., desert).

Two characters concern the morphology of the tarsi, used for locomotion and burrow construction (Prendini, 2001b). Scorpions occurring in three-dimensional, humid habitats, such as forests, typically possess longer, more gracile legs enabling a more errant lifestyle, and a well developed pseudonychia, which appears to assist with climbing tree trunks and rock walls to seek out prey and retreats. Scorpions occurring in arid habitats typically possess

Table 3

Primers used to generate DNA sequences from two nuclear and three mitochondrial loci for phylogenetic analysis of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879)

Loci	Primer	Sequence (5'–3')	Reference
18S rDNA	18S1F	TACCTGGTTGATCCTGCCAGTAG	Giribet et al. (1996)
	18S5R	CTTGGCAAATGCTTTTCGC	Giribet et al. (1996)
	18S3F	GTTTCGATTCCGGAGAGGGA	Giribet et al. (1996)
	18Sbi	GAGTCTCGTTCGTTATCGGA	Wheeler et al. (1993)
	18Sa2.0	ATGGTTGCAAAGCTGAAAC	Wheeler et al. (1993)
	18S9R	GATCCTTCCGCAGGTTACCTAC	Giribet et al. (1996)
28S rDNA	28SA	GACCCGTCTTGAAGCACG	Nunn et al. (1996)
	28SBout	CCCACAGCGCCAGTTCTGCTTACC	Prendini and Wheeler (2005)
12S rDNA	12SAI	AAACTAGGATTAGATACCCTATTAT	Kocher et al. (1989)
	12SBI	AAGAGCGACGGCGCATGTGT	Kocher et al. (1989)
16S rDNA	16SA	CTCCGGTTTGAACCTCAGATCA	Simon et al. (1991)
	16SB	CGCCTGTTTATCAAAAACAT	Simon et al. (1991)
	16Sai	GTGCAAAGGTAGCATAATCA	Gantenbein et al. (1999)
Cytochrome Oxidase <i>c</i> Subunit I Oxidase I	Nancy	CCCGGTAATAATATAAACTTC	Simon et al. (1991)
	EXTB	CCTATTGAWARAACATARTGAAAATG	Simon et al. (1991)
	EXTA	GAAAGTTTATATTTTAATTTTACCTGG	Simon et al. (1991)
	HCO	TAAACTTCAGGGTGACCAAAAAATCA	Folmer et al. (1994)
	LCO1490	GGTCAACAAATCATAAAGATATTGG	Folmer et al. (1994)
	HCOoutout	GTAAATATATGRTGDGCTC	Prendini et al. (2003)
	CI-J01718	GGNGGATTGGAAATTGRTTRGTTC	Simon et al. (1991)
	CI-N02776	GGATAATCAGAATANCGNCGAGG	Simon et al. (1991)
	COImodF	ATCATAAGGATATTGGGACTATGT	Bryson et al. (2013)
	CruzR	CATACCCAAAGARCCAAAAGG	Valdez-Cruz et al. (2004)

Table 4
Tissue samples and GenBank accession codes for DNA sequences of the 18S rDNA (18S), 28S rDNA (28S), 12S rDNA (12S), 16S rDNA (16S), and Cytochrome *c* Oxidase Subunit I (COI) loci used for phylogenetic analysis of the genera and species of Asian forest scorpions (Scorpionidae: Heterometrinidae Simon, 1879)

Family	Species	AMCC	Country	18S	28S	12S	16S	COI
Diplocentridae	<i>Nebo hierichonticus</i> (Simon, 1872)	LP 11138	Israel	MT417898	MT418125	MT417773	MT418351	MT418015
Scorpionidae	<i>Opisthophthalmus capensis</i> (Herbst, 1800)	LP 1046	South Africa	MT417899	AY156532	AY156547	AY156562	AY156577
	<i>Pandinus imperator</i> (C.L. Koch, 1841)	LP 1601	Ghana	MT417900	AY156537	AY156552	AY156567	AY156582
	<i>Scorpio fuscus</i> (Ehrenberg, 1829)	LP 1556	Israel	MT417901	AY156539	AY156554	AY156569	AY156584
	<i>Chersonesometrus bastavadei</i> Prendini and Loria, 2020	LP 16785	India	MT417902	MT418126	MT417774	MT418352	MT418016
		LP 16787		MT417903	MT418127	MT417775	MT418353	MT418017
	<i>Chersonesometrus beccaloniae</i> (Kovářík, 2004)	LP 13133		MT417904	MT418128	MT417776	MT418354	MT418018
		LP 13134		MT417905	MT418129	MT417777	MT418355	MT418019
	<i>Chersonesometrus fulvipes</i> (C.L. Koch, 1837)	LP 8268		MT417911	MT418135	MT417783	MT418361	MT418025
		LP 13125		MT417906	MT418130	MT417778	MT418356	MT418020
		LP 13981		MT417907	MT418131	MT417779	MT418357	MT418021
		LP 13990		MT417908	MT418132	MT417780	MT418358	MT418022
		LP 13997		MT417909	MT418133	MT417781	MT418359	MT418023
		LP 14031		MT417910	MT418134	MT417782	MT418360	MT418024
	<i>Chersonesometrus hendersoni</i> Prendini and Loria, 2020	LP 16776		MT417912	MT418136	MT417784	MT418362	MT418026
		LP 16782		MT417913	MT418137	MT417785	MT418363	MT418027
	<i>Chersonesometrus madraspatensis</i> (Pocock, 1900)	LP 1533		MT417915	AY156527	AY156542	AY156557	AY156572
		LP 4324		MT417916	MT418139	MT417787	MT418365	MT418029
		LP 13127		MT417914	MT418138	MT417786	MT418364	MT418028
	<i>Chersonesometrus pelekomanus</i> (Couzijn, 1981)	LP 16773		MT417917	MT418140	MT417788	MT418366	MT418030
		LP 16774		MT417918	MT418141	MT417789	MT418367	MT418031
	<i>Chersonesometrus tristis</i> (Henderson, 1919)	LP 2276		MT417921	MT418144	MT417792	MT418370	MT418034
		LP 13135		MT417919	MT418142	MT417790	MT418368	MT418032
		LP 13136		MT417920	MT418143	MT417791	MT418369	MT418033
	<i>Deccanometrus bengalensis</i> (C.L. Koch, 1841)	LP 14238		MT417922	MT418145	MT417793	MT418371	MT418035
	<i>Deccanometrus latimanus</i> (Pocock, 1894)	LP 11568	Pakistan	MT417923	MT418146	MT417794	MT418372	MT418036
		LP 11569		MT417924	MT418147	MT417795	MT418373	MT418037
		LP 11601		MT417925	MT418148	MT417796	MT418374	MT418038

Table 4
(Continued)

Family	Species	AMCC	Country	18S	28S	12S	16S	COI
Scorpionidae	<i>Deccanometrus obscurus</i> (Couzjin, 1981)	LP 6225	India	MT417929	MT418152	MT417800	MT418378	MT418042
		LP 13974		MT417926	MT418149	MT417797	MT418375	MT418039
		LP 16800		MT417927	MT418150	MT417798	MT418376	MT418040
	<i>Deccanometrus xanthopus</i> (Pocock, 1897)	LP 16817		MT417928	MT418151	MT417799	MT418377	MT418041
		LP 13126		MT417930	MT418153	MT417801	MT418379	MT418043
		LP 13971		MT417931	MT418154	MT417802	MT418380	MT418044
	<i>Gigantometrus swammerdami</i> (Simon, 1872)	LP 16814		MT417932	MT418155	MT417803	MT418381	MT418045
		LP 1607		MT417935	AY156530	AY156545	AY156560	AY156575
		LP 13128		MT417933	MT418156	MT417804	MT418382	MT418046
	<i>Gigantometrus titanicus</i> (Couzjin, 1981)	LP 13129	Sri Lanka	MT417934	MT418157	MT417805	MT418383	MT418047
		LP 12287		MT417936	MT418158	MT417806	MT418384	MT418048
		LP 12288		MT417937	MT418159	MT417807	MT418385	MT418049
	<i>Heterometrus glaucus</i> (Thorell, 1876)	LP 13922	Indonesia	MT417938	MT418160	MT417808	MT418386	MT418050
		LP 13928		MT417939	MT418161	MT417809	MT418387	MT418051
		LP 1789		MT417945	MT418167	MT417815	MT418393	MT418057
	<i>Heterometrus laevigatus</i> (Thorell, 1876)	LP 12335	Thailand	MT417940	MT418162	MT417810	MT418388	MT418052
		LP 12337		MT417941	MT418163	MT417811	MT418389	MT418053
		LP 13045		MT417942	MT418164	MT417812	MT418390	MT418054
	<i>Heterometrus laoticus</i> Couzjin, 1981	LP 13047	Laos	MT417943	MT418165	MT417813	MT418391	MT418055
		LP 13048		MT417944	MT418166	MT417814	MT418392	MT418056
		LP 11330		MT417946	MT418168	MT417816	MT418394	MT418058
		LP 11331	Thailand	MT417947	MT418169	MT417817	MT418395	MT418059
		LP 11352		MT417948	MT418170	MT417818	MT418396	MT418060
		LP 12336		MT417949	MT418171	MT417819	MT418397	MT418061
	<i>Heterometrus longinanus</i> (Herbst, 1800)	LP 13046	Laos	MT417950	MT418172	MT417820	MT418398	MT418062
		LP 15584		MT417951	MT418173	MT417821	MT418399	MT418063
		LP 16489		MT417952	MT418174	MT417822	MT418400	MT418064
		LP 11948	Malaysia	MT417953	MT418175	MT417823	MT418401	MT418065
		LP 12114		MT417954	MT418176	MT417824	MT418402	MT418066
		LP 12115		MT417955	MT418177	MT417825	MT418403	MT418067
		LP 12116		MT417956	MT418178	MT417826	MT418404	MT418068
		LP 12139		MT417957	MT418179	MT417827	MT418405	MT418069

Table 4
(Continued)

Family	Species	AMCC	Country	18S	28S	12S	16S	COI
Scorpionidae	<i>Heterometrus petersii</i> (Thorell, 1876)	LP 1604	Singapore	MT417958	AY156529	AY156544	AY156559	AY156574
		LP 8269		MT417960	MT418181	MT417829	MT418407	MT418071
		LP 16458		MT417959	MT418180	MT417828	MT418406	MT418070
	<i>Heterometrus silenus</i> (Simon, 1884)	LP 1603	Vietnam	MT417965	AY156528	AY156543	AY156558	AY156573
		LP 1888		MT417966	MT418186	MT417834	MT418412	MT418076
		LP 11248		MT417961	MT418182	MT417830	MT418408	MT418072
	<i>Heterometrus spinifer</i> (Ehrenberg, 1828)	LP 11252	Cambodia	MT417962	MT418183	MT417831	MT418409	MT418073
		LP 11353		MT417963	MT418184	MT417832	MT418410	MT418074
		LP 15590		MT417964	MT418185	MT417833	MT418411	MT418075
		LP 1906		MT417972	MT418192	MT417840	MT418418	MT418082
		LP 10581		MT417967	MT418187	MT417835	MT418413	MT418077
		LP 11920		MT417968	MT418188	MT417836	MT418414	MT418078
	<i>Heterometrus thorellii</i> (Pocock, 1892)	LP 11949	Myanmar	MT417969	MT418189	MT417837	MT418415	MT418079
		LP 11950		MT417970	MT418190	MT417838	MT418416	MT418080
		LP 11974		MT417971	MT418191	MT417839	MT418417	MT418081
		LP 2704		MT417980	MT418200	MT417848	MT418426	MT418090
		LP 13049		MT417973	MT418193	MT417841	MT418419	MT418083
<i>Javanimetrus cyaneus</i> (C.L. Koch, 1836)		LP 13050		MT417974	MT418194	MT417842	MT418420	MT418084
		LP 13051		MT417975	MT418195	MT417843	MT418421	MT418085
		LP 13052		MT417976	MT418196	MT417844	MT418422	MT418086
		LP 13153		MT417977	MT418197	MT417845	MT418423	MT418087
		LP 13155		MT417978	MT418198	MT417846	MT418424	MT418088
		LP 15604		MT417979	MT418199	MT417847	MT418425	MT418089
		LP 2309	Indonesia	MT417984	MT418204	MT417852	MT418430	MT418094
		LP 8270		MT417985	MT418205	MT417853	MT418431	MT418095
		LP 12112		MT417981	MT418201	MT417849	MT418427	MT418091
		LP 12113		MT417982	MT418202	MT417850	MT418428	MT418092
		LP 12117		MT417983	MT418203	MT417851	MT418429	MT418093

Table 5

Statistics for nuclear loci, 18S rDNA and 28S rDNA, and mitochondrial loci, 12S rDNA, 16S rDNA and Cytochrome *c* Oxidase subunit I (COI) in phylogenetic analysis of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879): number of haplotypes; length (nucleotide base-pairs) of unaligned and aligned sequences; average percent GC content; number of variable positions (VP); number of parsimony-informative sites (PI); and model selected for Bayesian Inference using the Akaike Information Criterion (AIC)

Locus	Haplotypes	Unaligned	Aligned	GC	VP	PI	AIC
18S rDNA	8	1761	1761	51	28	5	GTR + I
28S rDNA	13	514–515	515	58	30	9	GTR + I + G
12S rDNA	99	328–335	342	26	190	165	HKY + I + G
16S rDNA	96	481–487	492	31	242	197	HKY + I + G
COI	106	1078	1078	38	435	393	GTR + I + G
COI 1 st Codon	–	–	359	46	81	57	–
COI 2 nd Codon	–	–	359	42	19	13	–
COI 3 rd Codon	–	–	360	26	335	323	–

shorter, more robust legs and a reduced pseudonychium, assisting with burrow excavation in the hard substrate and generally associated with a more sedentary lifestyle. Large median ocelli are also common among forest scorpions, presumably enhancing light sensitivity in the darkness, a character not required in habitats with less vegetation cover.

Burrow site selection is also associated with habitat type. Whereas scorpions which burrow under stones or other surface debris may be found in any habitat, scorpions which burrow in open ground (i.e., without logs, stones or other surface debris) are often associated with arid habitats, suggesting the ability to excavate burrows in open ground may have been a precursor for scorpion diversification in barren, desert environments where surface debris is relatively scarce (Prendini, 2001b; Prendini et al., 2003).

The states of each character were assigned a score inversely associated with the mean monthly precipitation and vegetation cover of the habitat such that states associated with densely vegetated, mesic habitats received a lower score than states associated with sparsely vegetated, arid habitats. Following the approach of Ojanguren-Affilastro et al. (2016), the scores for each species were then summed to produce an ensemble index of adaptation to habitat aridity (scale = 1–10) for each species, in turn used to classify the species into three ecomorphotypes along a continuum from wet, evergreen forest (score = 1) to dry savanna, scrubland (score = 10).

In order to quantify the aridity of the habitat, remote sensing data for mean monthly precipitation and normalized difference vegetation index (NDVI) were downloaded from WorldClim ver. 2 (<http://worldclim.org/version2>) and NASA Earth Observatory (<https://earthobservatory.nasa.gov/global-maps>), respectively, and associated with georeferences of the collection sites of the terminal taxa using the sample extraction method in ArcMap ver. 10.1 (Appendix 4). The mean values for each ecomorphotype were then calculated from the mean values for the terminals.

The dataset was reduced to one terminal per species for this analysis, bringing the number of terminals to 45, including all 41 Asian scorpionid species and the four outgroup taxa. Ten ingroup species were represented only by morphological data. DNA sequences for the reduced dataset were aligned using MAFFT ver. 7 and the parameters described above. The reduced morphological and molecular datasets were analyzed simultaneously with ML using RAXML-HP ver. 8 and the parameters described above. The ML tree recovered was converted into an ultrametric tree using the commands makeChronosCalib and chronos in the R ver. 3.5.3 (R Core Team Packages) program *ape* (Paradis et al., 2004). The ultrametric tree with the lowest log-likelihood score was selected for subsequent R analyses, achieved by setting the rate-smoothing parameter lambda to zero and using a relaxed substitution model.

Ancestral state reconstruction of the habitat aridity index was conducted on the ultrametric tree using ML with the R program *phytools* (Revell, 2012). Log likelihoods and ancestral rates were calculated for nodes using three models: all rates equal (ER), symmetric (SYM), and all rates different (ARD). Absolute values of log-likelihoods were mapped onto the nodes of the ultrametric tree. Chi-squared tests were performed. A parsimony-based, node-driven method of ancestral reconstruction (Ramírez and Michalik, 2014; Ojanguren-Affilastro et al., 2016) was tested for comparison.

Ancestral state reconstruction was also performed individually on each character contributing to the habitat aridity index using the methods described above. Pagel's (1999) lambda (λ) was calculated to test for the phylogenetic signal in each character using the fitDiscrete function in the R program *geiger* (Harmon et al., 2007).

Results

Separate analyses

All separate analyses of the morphological matrix with parsimony recovered the Asian scorpionids, i.e., Heterometrinae, as monophyletic and sister to *Pandinus*, except the analysis with implied weighting and $k = 1$, in which the Asian scorpionids were paraphyletic with respect to *Pandinus* (Table 9). The Asian scorpionid genera, *Gigantometrus*, *Heterometrus*, and *Javanimetrus*, as defined by Prendini and Loria (2020), were each consistently monophyletic, whereas *Deccanometrus* and *Sahyadrimetrus* were consistently paraphyletic. *Srilankametrus* was monophyletic in the analyses with implied weighting and $k = 3$ or 10, but paraphyletic in the analyses with equal weighting or implied weighting and $k = 1$, 60, or 100. *Chersone-sometrus* was consistently monophyletic except in the analysis with equal weighting, in which it was paraphyletic with respect to *Gigantometrus*.

All separate analyses of the molecular data with parsimony also recovered Heterometrinae monophyletic, except the analyses with equal weighting or implied weighting and $k = 60$ and 100. *Gigantometrus*, *Heterometrus*, *Javanimetrus*, *Sahyadrimetrus* and *Srilankametrus*

Table 6

Nodal support values for clades recovered in phylogenetic analysis of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879), based on separate and simultaneous (Simul) analyses of the morphological (Mor) and molecular (Mol, concatenated aligned DNA sequences of two nuclear and three mitochondrial gene markers) datasets using Bayesian inference (BI), maximum likelihood (ML) and parsimony (P) with implied weighting and low value of the concavity constant ($k = 1$). Nodal support was evaluated using posterior probabilities (PP), bootstrap (B) or partitioned Bremer support (PBr). Grey cells indicate clades with low support, no effect (0) or negative support

			Heterometrinae	Chersonesometrus	Deccanometrus	Gigantometrus	Heterometrus	Javanimetrus	Sahyadrimetrus	Srilankametrus	(C + G)	(D + C + G)	(J + Sa)	(J + D + C + G)	(Sa + D + C + G)	(Sa + J + D + C + G)	(Sr + Sa + J + D + C + G)
BI	PP	Simul	1	1	0.96	1	1	1	1	1	1	1	-	0.61	1	1	1
		Mol	1	-	-	1	1	1	0.99	1	1	1	-	1	-	1	1
ML	B	Simul	100	73	66	100	100	100	68	99	99	92	-	-	68	77	100
		Mol	96	-	<50	100	100	100	56	99	100	91	-	62	-	62	100
P	B	Simul	89	100	<50	100	100	100	<50	100	100	55	<50	-	-	<50	100
		J	95	92	57	100	100	100	49	99	99	74	39	-	-	47	99
	PBr	18S	0	-	-	0	0	0	0	0	-	0	0	-	-	0	0
		28S	1.7	-	-	-0.3	-0.3	-0.3	1.7	1.7	-	-0.3	1.7	-	-	1.7	-0.3
		12S	1.2	-	-	0.2	-4.8	-4.8	1.2	1.2	-	-4.8	1.2	-	-	1.2	4.8
		16S	-4.8	-	-	14.2	-2.8	-2.8	-4.8	-4.8	-	-2.8	-4.8	-	-	-4.8	-2.8
		COI	2.2	-	-	2.2	-5.8	-5.8	2.2	2.2	-	-5.8	2.2	-	-	2.2	-5.8
		Mor	20	-	-	-4.0	24	16	3	14	-	14	3	-	-	3	17

were consistently monophyletic, whereas *Chersonesometrus* was consistently paraphyletic with respect to *Gigantometrus*. *Deccanometrus* was monophyletic in the analyses with equal weighting or implied weighting and $k = 60$ or 100 , but paraphyletic in the analyses with implied weighting and $k = 1, 3$, or 10 (Table 9).

Separate analysis of the molecular dataset with ML recovered a tree with optimization likelihood of -30943.40 . The Asian scorpionids formed a monophyletic group, sister to *Scorpio*, with high bootstrap support (96%). All Asian scorpionid genera were monophyletic except for *Chersonesometrus*, again rendered paraphyletic by *Gigantometrus*. All Asian scorpionid genera recovered as monophyletic received high bootstrap support ($\geq 99\%$; Table 6) except *Deccanometrus* and *Sahyadrimetrus*, which were less well supported (44% and 56%, respectively). Relationships among some genera and major clades were highly supported ($\geq 91\%$), e.g., the clade comprising all Asian scorpionid genera except *Heterometrus*; the clade comprising *Chersonesometrus*, *Deccanometrus*, and *Gigantometrus*; and the clade comprising *Chersonesometrus*

and *Gigantometrus* (Table 6). The clade comprising *Chersonesometrus*, *Deccanometrus*, *Gigantometrus*, and *Javanimetrus*, and the clade comprising those genera and *Sahyadrimetrus* received moderate support (62%). The tree obtained by separate analysis of the molecular dataset with BI was largely congruent with the ML tree except that *Deccanometrus* was also paraphyletic. Most relationships among the genera and clades received high posterior probability values in the BI analysis of the molecular dataset (≥ 0.99 ; Table 6)

Simultaneous analyses

Simultaneous analyses of the morphological and molecular datasets with parsimony and equal weighting or implied weighting consistently recovered Heterometrinae as monophyletic and sister to *Pandinus*. All Asian scorpionid genera were consistently monophyletic, except for *Chersonesometrus*, which was paraphyletic in the analysis with equal weighting, and *Deccanometrus*, which was paraphyletic in the analyses with equal weighting or implied weighting and $k = 10$,

Table 7

Tree statistics for phylogenetic analysis of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879)

	Weight	MPTs	Length	CI	RI	Fit	AH
Morphology	IW: $k = 1$	1	580	0.319	0.885	63.30	68.70
	IW: $k = 3$	6	562	0.329	0.891	84.46	47.54
	IW: $k = 10$	3	551	0.336	0.894	108.01	23.99
	IW: $k = 60$	1	550	0.336	0.894	126.45	5.55
	IW: $k = 100$	1	550	0.336	0.894	128.55	3.45
	EW	408	550	0.336	0.894	81.77–83.76	–
Molecular	IW: $k = 1$	1	5402	0.246	0.788	250.22	518.78
	IW: $k = 3$	1	5316	0.250	0.793	386.15	382.85
	IW: $k = 10$	1	5309	0.250	0.793	552.13	216.87
	IW: $k = 60$	1	5300	0.251	0.794	711.68	57.32
	IW: $k = 100$	1	5299	0.251	0.794	732.71	36.29
	EW	12	5299	0.251	0.794	383.99–384.19	–
Simultaneous	IW: $k = 1$	3	6000	0.252	0.802	305.59	595.41
	IW: $k = 3$	3	5947	0.255	0.805	464.52	436.48
	IW: $k = 10$	3	5930	0.255	0.805	656.36	244.64
	IW: $k = 60$	3	5925	0.256	0.806	837.06	63.94
	IW: $k = 100$	3	5925	0.256	0.806	860.58	40.42
	EW	66	5925	0.256	0.806	462.88–463.64	–

Length, consistency index (CI), retention index (RI), fit and adjusted homoplasy (AH) of most parsimonious trees (MPTs) obtained by separate and simultaneous analyses of the morphological and molecular (concatenated aligned DNA sequences of two nuclear and three mitochondrial gene markers) datasets, with equal weighting (EW) and implied weighting (IW), applying five values of the concavity constant (k).

60, or 100. The analyses with equal weighting or implied weighting and $k = 1$ or 3 placed *Javanimetrus* sister to *Sahyadrimetrus* (Fig. 7c) whereas the analyses with implied weighting and $k = 10$, 60 or 100 placed *Javanimetrus* sister to a clade comprising *Chersonesometrus*, *Deccanometrus* and *Gigantometrus*.

All Asian scorpionid genera received high bootstrap support (100%) except for *Deccanometrus* and *Sahyadrimetrus* ($\leq 50\%$). *Chersonesometrus*, *Gigantometrus*, *Heterometrus*, *Javanimetrus* and *Srilankametrus* also received high jackknife support ($\geq 92\%$), whereas *Deccanometrus* and *Sahyadrimetrus* received lower support (57% and 49%, respectively). The clade comprising all Asian scorpionid genera except *Heterometrus* received high bootstrap and jackknife support (100% and 99%, respectively), as did the clade comprising *Chersonesometrus* and *Gigantometrus* (100% and 99%, respectively). The clade comprising *Chersonesometrus*, *Deccanometrus*, and *Gigantometrus* received low bootstrap (55%) and moderate jackknife (74%) support. The clade comprising *Javanimetrus* and *Sahyadrimetrus* also received low bootstrap and jackknife support ($\leq 50\%$). Partitioned Bremer support values were negative or had no effect for some clades (Table 6).

Simultaneous analysis of the morphological and molecular datasets with ML recovered a tree with optimization likelihood of -34881.09 (Figs. 7a, 8). Heterometrinae again formed a monophyletic group, sister to *Pandinus*, with high bootstrap support (100%), and all Asian scorpionid genera were monophyletic. *Gigantometrus*, *Heterometrus*, *Javanimetrus* and

Srilankametrus each received high bootstrap support ($\geq 99\%$), whereas *Chersonesometrus*, *Deccanometrus* and *Sahyadrimetrus* received moderate support (73%, 66% and 68%, respectively). Relationships among the Asian scorpionid genera obtained by the ML analysis were congruent with those obtained by the parsimony analyses, except for the positions of *Javanimetrus* and *Sahyadrimetrus* (Table 9), and mostly well supported (Fig. 7a). For example, the clade comprising all Asian scorpionid genera except *Heterometrus*, the clade comprising *Chersonesometrus* and *Gigantometrus*, and the clade comprising *Chersonesometrus*, *Deccanometrus*, *Gigantometrus*, *Javanimetrus*, and *Sahyadrimetrus* each received high bootstrap support ($\geq 77\%$). The tree obtained by simultaneous analysis of the morphological and molecular datasets with ML-ASC was congruent with the ML tree except for the position of *Javanimetrus*, which was placed sister to *Sahyadrimetrus* (Table 9).

The tree obtained by simultaneous analysis with BI was mostly congruent with the ML trees (Fig. 7b). The Asian scorpionids again formed a monophyletic group, sister to *Pandinus*, all Asian scorpionid genera were monophyletic, and the relationships among them resembled the ML trees except for the positions of *Javanimetrus* and *Sahyadrimetrus* (Table 9; Fig. 7b). All genera received high posterior probabilities (≥ 0.96). Support for relationships among the genera and major clades was also high (≥ 0.99), except for the clade comprising *Chersonesometrus*, *Deccanometrus*, *Gigantometrus*, and *Javanimetrus*, which received moderate support (0.61).

Table 8

Distribution of ten characters concerning morphology and burrow architecture used to reconstruct ancestral ecomorphotypes for the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879)

Taxon	Characters	Score	Taxon	Characters	Score
<i>Nebo hierichonticus</i>	11112 11100	[9]	<i>Heterometrus glaucus</i>	00010 00010	[2]
<i>Opisthophthalmus capensis</i>	12010 11110	[8]	<i>Heterometrus laevigatus</i>	00000 00010	[1]
<i>Scorpio fuscus</i>	11100 11110	[7]	<i>Heterometrus laoticus</i>	00000 00010	[1]
<i>Pandinus imperator</i>	10010 01110	[5]	<i>Heterometrus longimanus</i>	00010 00010	[2]
<i>Deccanometrus bengalensis</i>	10010 01100	[4]	<i>Heterometrus petersii</i>	00010 00010	[2]
<i>Deccanometrus latimanus</i>	11001 11111	[8]	<i>Heterometrus silenus</i>	00000 00010	[1]
<i>Deccanometrus liurus</i>	10011 11100	[6]	<i>Heterometrus spinifer</i>	00010 00010	[2]
<i>Deccanometrus obscurus</i>	10010 01100	[4]	<i>Heterometrus thorelli</i>	00000 00010	[1]
<i>Deccanometrus phipsoni</i>	10010 01100	[4]	<i>Javanimetrus cyaneus</i>	10000 00110	[3]
<i>Deccanometrus ubicki</i>	10010 11100	[5]	<i>Sahyadrimetrus barberi</i>	10010 00110	[4]
<i>Deccanometrus xanthopus</i>	11111 11111	[10]	<i>Sahyadrimetrus kanarensis</i>	10010 00110	[4]
<i>Chersonesometrus bastawadei</i>	10010 11100	[5]	<i>Sahyadrimetrus mathewi</i>	10010 00110	[4]
<i>Chersonesometrus beccaloniae</i>	10110 01100	[5]	<i>Sahyadrimetrus rugosus</i>	10010 01100	[4]
<i>Chersonesometrus fulvipes</i>	10010 11110	[6]	<i>Sahyadrimetrus scaber</i>	10010 00110	[4]
<i>Chersonesometrus hendersoni</i>	10110 11100	[6]	<i>Sahyadrimetrus tikaderi</i>	10010 00110	[4]
<i>Chersonesometrus madraspatensis</i>	10010 11110	[6]	<i>Srilankametrus caesar</i>	10000 01111	[5]
<i>Chersonesometrus nathanorum</i>	10110 01100	[5]	<i>Srilankametrus couziji</i>	10000 01111	[5]
<i>Chersonesometrus pelekomanus</i>	10110 01100	[5]	<i>Srilankametrus gravimanus</i>	10000 01111	[5]
<i>Chersonesometrus shivashankari</i>	10011 11110	[7]	<i>Srilankametrus indus</i>	10000 01110	[4]
<i>Chersonesometrus tristis</i>	10110 01100	[5]	<i>Srilankametrus pococki</i>	10000 01110	[4]
<i>Chersonesometrus wroughtoni</i>	10110 01100	[5]	<i>Srilankametrus serratus</i>	10000 01111	[5]
<i>Gigantometrus swammerdami</i>	10010 10110	[5]	<i>Srilankametrus yaleensis</i>	10000 01111	[5]
<i>Gigantometrus titanicus</i>	10010 00110	[4]			

Terminology follows Appendix 3. Each character state was assigned a score inversely associated with vegetation cover and precipitation of the habitat such that states associated with densely vegetated, humid habitats received a lower score than states associated with sparsely vegetated, arid habitats. The scores for each species were then summed to produce an ensemble index of adaptation to habitat aridity (scale = 1–10), allowing species to be classified into three ecomorphotypes along a continuum from wet, evergreen forest (score = 1) to dry savanna, scrubland (score = 10), and ancestral state reconstruction of ecomorphotypes to be performed on the phylogeny.

1. Carapace, median ocelli size: large, distance between ocelli < ocellus width [0]; small, distance between ocelli ≥ ocellus width [+1].
2. Carapace, tergites and metasoma colouration: dark, black [0]; brown, reddish-brown or yellowish brown [+1]; pale [+2].
3. Sternites III–VII, infuscation: dark (infuscate) [0]; pale (immaculate) [+1].
4. Telson, colouration: as dark as metasomal segment V [0]; paler than metasomal segment V [+1].
5. Pedipalps, infuscation: femur, patella and chela [0]; femur and patella only [+1]; none [+2].
6. Legs, colouration: very dark or heavily infuscate [0]; pale (though perhaps very lightly infuscate) [+1].
7. Basitarsi length: long [0]; short [+1].
8. Telotarsi, pseudonychium: prominent, acuminate [0]; reduced, rounded [+1].
9. Burrows constructed in the open: absent [0]; present [+1].
10. Burrows constructed under stones: present [0]; absent [+1].

Preferred hypothesis

Comparison of the separate and simultaneous analyses of the morphological and molecular datasets revealed several trends (Table 9). Heterometrinae were monophyletic in 80% of the analyses. *Gigantometrus*, *Heterometrus*, and *Javanimetrus* were monophyletic in all analyses, whereas *Sahyadrimetrus* was monophyletic in all except separate analyses of the morphological dataset and *Srilankametrus* monophyletic in all except separate analyses of the morphological dataset with equal weighting or implied weighting and $k = 1, 60$ or 100 . *Chersonesometrus* and *Deccanometrus* were monophyletic in 60% and 40% of the analyses, respectively. However, both genera were monophyletic in 90% and 60% of the simultaneous analyses, respectively. Weak support for the monophyly of *Deccanometrus* may be explained by the

absence of molecular data for three species, representing 40% of the species diversity of the genus.

Although relationships among the genera and clades were less congruent among the analyses (Table 9), a clade comprising *Chersonesometrus* and *Gigantometrus* was consistently recovered (Table 9). Relationships obtained by separate analyses of the molecular dataset were generally congruent with those obtained by the simultaneous analyses, and included two major clades of Asian scorpionids, i.e., a Southeast Asian clade comprising the species of *Heterometrus*, and a predominantly South Asian clade comprising all genera and species from the Indian subcontinent and Sri Lanka, including the monotypic Southeast Asian *Javanimetrus*. However, the analyses recovered three alternative hypotheses for the position of *Javanimetrus* (Table 9): sister to a clade comprising the Indian genera,

Table 9

Nodal stability for selected clades and taxa in phylogenetic analysis of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879), based on separate and simultaneous (Simul) analyses of the morphological (Mor) and molecular (concatenated aligned DNA sequences of two nuclear and three mitochondrial gene markers, Mol) datasets using Maximum Likelihood (ML) including ascertainment bias with Lewis correction (ML-ASC), Bayesian Inference (BI) and parsimony with equal weighting (EW) and implied weighting (IW), applying five values of the concavity constant (k)

		Heterometrinae	Chersonesometrus	Deccanometrus	Gigantometrus	Heterometrus	Javanimetrus	Sahyadriimetrus	Srilankametrus	(C + G)	(D (C + G))	(J + Sa)	(J + D + C + G)	(Sa + D + C + G)	(H + J)	(Sr (Sa + J + D + C + G))	(H (Sr (Sa + J + D + C + G)))
Simul	ML																
	ML - ASC																
	BI																
	EW																
	IW ($k = 1$)																
	IW ($k = 3$)																
	IW ($k = 10$)																
	IW ($k = 60$)																
Mol	IW ($k = 100$)																
	ML																
	BI																
	EW																
	IW ($k = 1$)																
	IW ($k = 3$)																
	IW ($k = 10$)																
	IW ($k = 60$)																
Mor	IW ($k = 100$)																
	EW																
	IW ($k = 1$)																
	IW ($k = 3$)																
	IW ($k = 10$)																
	IW ($k = 60$)																
	IW ($k = 100$)																

Highlighted cells indicate clades and taxa recovered under the various parameters applied (classification follows Prendini and Loria, 2020). Abbreviations for genera as follows: C: *Chersonesometrus*; D: *Deccanometrus*; G: *Gigantometrus*; H: *Heterometrus*; J: *Javanimetrus*; Sa: *Sahyadriimetrus*; Sr: *Srilankametrus*.

Chersonesometrus, *Deccanometrus*, *Gigantometrus*, and *Sahyadrimetrus* (Fig. 7a); sister to a clade comprising *Chersonesometrus*, *Deccanometrus*, and *Gigantometrus*, to the exclusion of *Sahyadrimetrus* and *Srilankametrus* (Fig. 7b); sister to *Sahyadrimetrus* (Fig. 7c).

The tree recovered by simultaneous analysis of the morphological and molecular datasets with ML, is presented as the preferred hypothesis for the phylogeny of the Asian forest scorpions (Figs. 7a, 8) based on an assessment of the morphological character optimizations, and the higher bootstrap support values for the relationships among genera and clades.

Ecomorphotypes and ancestral state reconstruction

The habitat aridity adaptation index enabled three ecomorphotypes to be associated with habitats along a continuum of progressively decreasing precipitation and vegetation cover from wet (score = 1) to dry (score = 10): 1–4, wet silvicolous ecomorphotype inhabiting evergreen forest; 5–6, dry silvicolous ecomorphotype inhabiting deciduous forest; 7–10, savannicolous ecomorphotype inhabiting savanna or scrubland (Figs. 2–5). The wet silvicolous ecomorphotype was the most common, observed in 21 (51%) of the Asian scorpionid species, followed by the dry silvicolous ecomorphotype, observed in 17 (42%), and the savannicolous ecomorphotype, observed in only three (7%). The mean monthly precipitation and normalized difference vegetation index (NDVI) calculated for each ecomorphotype confirmed the prediction that the habitat of the savannicolous ecomorphotype is more arid than the dry silvicolous ecomorphotype which is, in turn, more arid than the wet silvicolous ecomorphotype (Fig. 9b, c).

Most Asian scorpionid genera exhibited one or two ecomorphotypes among the species. All species of *Heterometrus* and *Sahyadrimetrus*, one species of *Gigantometrus*, and the monotypic *Javanimetrus*, were wet silvicolous, whereas all except one species of *Chersonesometrus*, and the other species of *Gigantometrus* were dry silvicolous. Five species of *Srilankametrus* were dry silvicolous, whereas two were wet silvicolous. *Deccanometrus* exhibited the greatest variation with all three ecomorphotypes observed among its species. Three species were wet silvicolous, two, dry silvicolous, and two, savannicolous. One species of *Chersonesometrus* was also savannicolous. Among the outgroup taxa, *P. imperator* was dry silvicolous whereas *N. hierichonticus*, *O. capensis* and *S. fuscus* were savannicolous (all three species inhabit Mediterranean scrubland).

Ancestral state reconstruction of the habitat aridity index resulted in log-likelihood scores of -48.22852 , -54.03222 and -75.34216 for the ER, SYM and ARD models, respectively. The ARD model was preferred over the ER and SYM models based on a chi-

squared test ($P < 0.01$) although results were congruent as the ecomorphotype of the ancestor of each genus was consistently recovered across all three reconstructions. The ancestor of all Asian scorpionids was recovered as wet silvicolous (Fig. 9a) with high probability ($> 90\%$). The ancestors of *Deccanometrus*, *Heterometrus*, *Javanimetrus*, *Sahyadrimetrus* and *Srilankametrus* were also recovered as wet silvicolous whereas the ancestor of *Chersonesometrus* and *Gigantometrus* was recovered as dry silvicolous in all models. Ancestral state reconstruction using the parsimony-based, node-driven method recovered the same results except that the ancestor of *Deccanometrus* was dry silvicolous and the ancestor of *Gigantometrus*, wet silvicolous.

Ancestral state reconstructions of the characters (Fig. 10) contributing to the habitat aridity adaptation index were congruent across the ER, SYM and ARD models, although the ARD model was preferred for all characters on account of the lowest log likelihood. Pagel's (1999) lambda and chi-squared tests also indicated a strong phylogenetic signal ($\lambda \geq 0.96$) and statistical significance ($P < 0.01$) for all characters.

Discussion

Couzijn's (1978, 1981) classification revalidated and updated

Couzijn (1978, 1981) presented a phenetic cluster analysis of the species of *Heterometrus*, the first quantitative analysis of morphological variation in the Asian forest scorpions, based on 131 characters, 75 of which were biometric (Fig. 6). Initially, Couzijn (1978) recognized three subgenera and five species groups: the exclusively Indian *Chersonesometrus*, containing the fulvipes and phipsoni species groups of India; the monotypic *Gigantometrus* of India and Sri Lanka; and the nominotypical subgenus, *Heterometrus*, containing the indus species group of India and Sri Lanka, and the cyaneus and spinifer species groups of Southeast Asia. Couzijn (1981) subsequently revised his previous classification by replacing the cyaneus and indus groups with subgenera *Javanimetrus* and *Srilankametrus*, respectively, thus restricting the nominotypical subgenus to the former spinifer group of Southeast Asia, and resulting in five subgenera and two species groups (Table 1).

Couzijn (1981) also proposed a scheme of relationships among the subgenera based on his analysis, according to which the African scorpionid genus, *Pandinus*, was the closest relative of the Asian scorpionids, of which *Gigantometrus* was suggested to be the most basal, sister to a monophyletic group comprising two clades, one comprising *Chersonesometrus* and

Heterometrus, the other comprising *Javanimetrus* and *Srilankametrus* (Fig. 6).

Prendini (2000a) and Prendini et al. (2003) tested the monophyly and position of *Heterometrus* in Scorpionoidea and Scorpionidae, respectively, with phylogenetic analyses of morphology and multilocus DNA sequence data for exemplar species. *Heterometrus* was consistently recovered as monophyletic and sister to *Pandinus*, as proposed by Couzijn (1981). Additionally, Prendini et al. (2003) recovered two clades within *Heterometrus*, comprising the exemplar species from South and Southeast Asia, respectively. Although the sister-group arrangement of the South Asian *H. fulvipes* (C.L. Koch, 1837) and *H. swammerdami* Simon, 1872 recovered in the analyses appeared to falsify Couzijn's (1981) hypothesis that *H. swammerdami* was the basal-most member of *Heterometrus*, Prendini et al. (2003: 209, 218) noted that “a detailed phylogenetic analysis of the species of *Heterometrus* is required to confirm this finding” and “a computational cladistic analysis of relationships among the species of *Heterometrus*... should help to clarify the validity of the five subgenera currently recognised.”

Kovářík (2004: 1) ignored the work of Prendini et al. (2003) and disagreed with many of the taxonomic decisions of Couzijn (1981) stating:

... although Couzijn's extensive character set gives the impression of a definitive work, at a closer examination the inclusion of many of his characters and the weight given to many others invite considerable doubt... Most of his morphometric characters are abandoned because of their substantial intraspecific variation, and his opinion is concurred with only as to the importance of the form and surface of the chela of pedipalps (granulation, punctation, carinae) and to some extent the granulation of the carapace.

While problems with some of Couzijn's (1978, 1981) decisions have been identified, e.g., by Tikader and Bastawade (1983) and Prendini et al. (2003), Kovářík's (2004: 1) characterization of his work is, at best, misrepresented. Couzijn (1981: 10–16, 26, 27) did not neglect the characters established by Pocock (1900 and citations therein), such as sexual dimorphism, the colour of the legs, or development of the metasomal carinae, as is evident, e.g., from his Table 2 (characters 45, 111–113, 118, 119, 128), and figures 8 and 9. Rather, Couzijn (1978, 1981) recognized that “a source or taxonomic difficulty” such as the Asian forest scorpions requires a detailed, quantitative approach considering all available evidence in a rigorous and objective manner.

In contrast, Kovářík (2004: 2, 4) synonymized Couzijn's (1981) subgenera based on a rather simplistic assessment of a handful of characters, devoid of analysis, which boils down to an appeal to authority:

Couzijn (1978, 1981) described four subgenera but gave no characters that would permit to unequivocally distinguish them from the nominotypical subgenus. The invalidity of his subgenera is also clear from the below listed synonymizations of species across them, and I therefore regard all of them as synonyms. The only subgenus that could possibly be upheld is *Gigantometrus*, which comprises closely related species easily distinguishable from others. However, the genus *Heterometrus* contains more of such groups (e.g. *H. madraspatensis* and *H. gravimanus*, or *H. petersii* and *H. laoticus*), and in my opinion, they have the character of species groups rather than of subgenera. I have found two characters in *Heterometrus* that could serve to justify subgeneric divisions. One of them is sexual dimorphism, which Couzijn for some reason did not include in his analysis. However, after examination of most of the species it has become clear that this character evolved in the genus more than once, because it differentiates otherwise closely related species that definitely cannot be placed in separate subgenera. As examples can serve *H. liophysa*, *H. longimanus*, *H. spinifer* and *H. cimrmani* sp. n., whose females and juveniles are virtually indistinguishable, and the species can be reliably identified only on males. The other character is the genital operculum which comes in two basic shapes, but they cross some of the species group boundaries. I therefore believe that subgeneric divisions within the genus *Heterometrus* are not justified.

In a subsequent critique of the systematics of Asian forest scorpions, Tahir and Prendini (2014: 3) reiterated the need for a “quantitative cladistic analysis of relationships among the species of *Heterometrus*... to reassess Couzijn's (1981) five subgenera, which were rather arbitrarily synonymized.” The recommendation was again ignored when Rossi (2016: 19) reinstated subgenus *Gigantometrus* solely on the grounds that it can be “easily differentiated by the larger chela.”

The phylogenetic analyses presented here vindicate Couzijn's (1978, 1981) subdivision of the Asian forest scorpions, with one exception. As defined by Couzijn (1981), *Chersonesometrus*, and its *fulvipes* and *phipsoni* species groups were rendered paraphyletic by *Gigantometrus* and *Javanimetrus*, justifying the creation of *Deccanometrus* and *Sahyadrimetrus* by Prendini and Loria (2020). Additionally, Couzijn's (1981) scheme of relationships among the subgenera and species groups of Asian forest scorpions (Fig. 6) was falsified, as neither the basal position of *Gigantometrus* nor the clades comprising *Chersonesometrus* and *Heterometrus* or *Javanimetrus* and *Srilankametrus* were recovered. Considering the complexity of the taxa and the technology and methods available at the time, however, Couzijn's (1978, 1981) efforts were a singular accomplishment.

Finally, it should be noted that the concept that the Asian scorpionid taxa comprise more than one genus is not new. For example, Pocock (1892: 39) considered the Southeast Asian taxa, then assigned to *Palamnaeus*

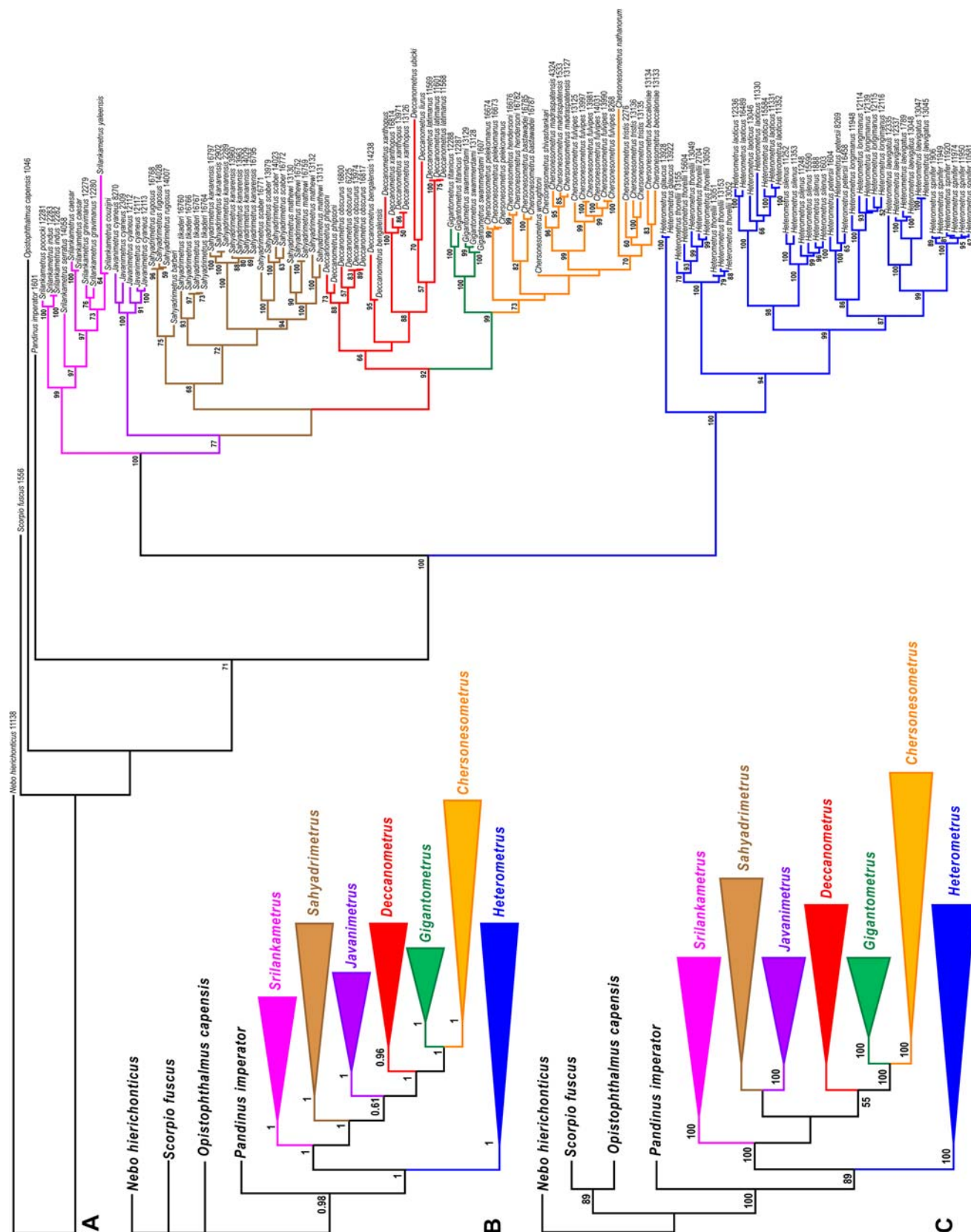


Fig. 7. Phylogeny of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879) obtained by simultaneous phylogenetic analysis of 186 morphological characters and 4188 aligned DNA nucleotides from three mitochondrial and two nuclear gene loci. A. Preferred phylogeny obtained using maximum likelihood. B. Majority rule (50%) Bayesian consensus tree with collapsed nodes for genera. C. Strict consensus of three most-parsimonious trees obtained with implied weighting and $k = 1$, with collapsed nodes for genera. Numbers adjacent to nodes are bootstrap support values ($> 50\%$) (A, C) or posterior probabilities (> 0.5) (B).

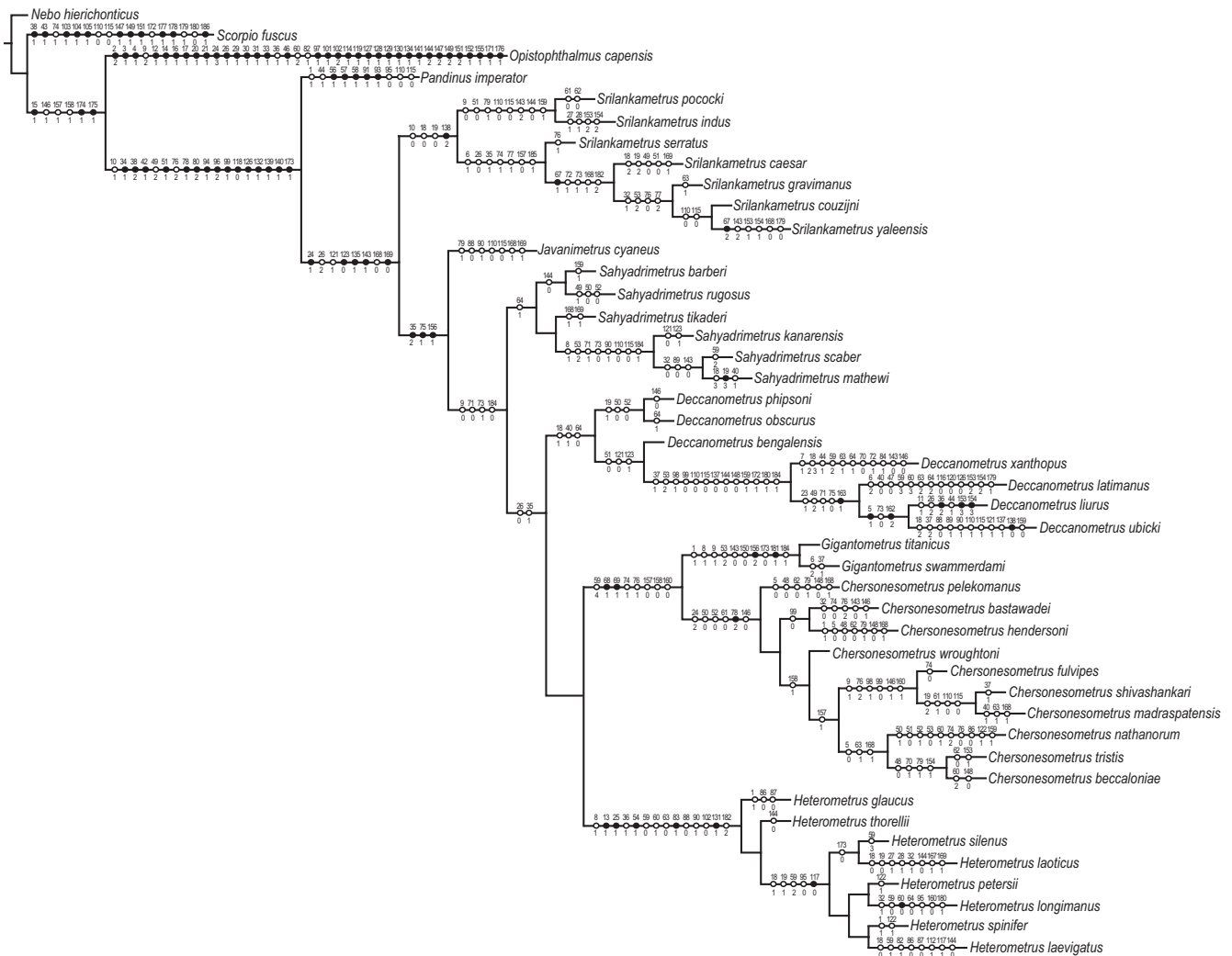


Fig. 8. Preferred phylogeny of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879) with unambiguous morphological synapomorphies optimized. Black circles indicate uniquely derived apomorphic states, white circles parallel derivations of apomorphic states. Numbers above circles indicate characters, numbers below indicate states (Appendix 3).

Thorell, 1876, distinct from the Indian taxa, which he retained in *Scorpio* L., 1758, following Thorell (1876).

Redefinition of *Heterometrus*, a genus widespread in Southeast Asia

Couzijn's (1978, 1981) concept of the nominotypical subgenus of *Heterometrus* (Table 1), comprising *H. glaucus* (Thorell, 1876) [as *Heterometrus liophysa* (Thorell, 1888)], *Heterometrus laoticus* Couzijn, 1981, *H. longimanus*, *Heterometrus silenus* (Simon, 1884) [as *Heterometrus petersii* (Thorell, 1876)], *H. spinifer* [including *Heterometrus laevigatus* (Thorell, 1876)], and *Heterometrus thorellii* (Pocock, 1892) [as *H. bengalensis*] agrees fully with the phylogenetic analyses presented here, which included 43 terminals representing all eight species recognized by Prendini and Loria (2020; Table 2) with at least two terminals each. All separate

and simultaneous analyses of the morphological and molecular datasets, using parsimony, ML or BI, recovered *Heterometrus* as monophyletic and sister to all other Asian scorpionid genera with high support, as well as the monophyly of each species. Material from Penang and Singapore, assessed to be conspecific with the holotype of *Palamnaeus petersii*, also from Singapore, based on morphological similarity, was genetically more closely related to *H. longimanus* than to morphologically indistinguishable material of *H. spinifer* from the Malaysian mainland and Tioman, leading to the conclusion that *H. petersii* is a cryptic species.

Heterometrus was supported in the preferred phylogenetic hypothesis (Fig. 8) by 30 molecular synapomorphies and 14 morphological synapomorphies. The genus is widespread across Southeast Asia, from the Andaman and Nicobar Islands, throughout Indochina, the Thai-Malay Peninsula and offshore islands (Ko Chang, Ko Lanta, Ko

Muk, Ko Samui, Langkawi, the Myeik archipelago, Penang, the Phi Phi Islands, Phuket, Phú Quoc Island, Tioman, Tinggi and Singapore), the Riau archipelago, Sumatra, Weh, Simeulue, Babi, Nias, the Batu archipelago, Siberut, the Mentawai archipelago, Bangka, Belitung, Java, Madura, Bali, Borneo, Miang-Besar, and the Philippine islands of Balabac, Mindanao, Luzon, Palawan, Panay, and Tawi Tawi in the Sulu archipelago, to Wallace's Line. It has been recorded from the following countries: Brunei, Cambodia, India, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand and Vietnam (Fig. 1f). The distribution of *Heterometrus* overlaps that of *Javanimetrus* in Borneo, Luzon, Nias, Sumatra, and Thailand.

The scheme of relationships among the species of *Heterometrus* presented in Fig. 8 was consistently recovered with high support. This topology suggests that *Heterometrus* dispersed into Southeast Asia from India via Sumatra and Myanmar (Loria and Prendini, 2020).

Elevation of Javanimetrus, a monotypic genus from Southeast Asia

Couzijn (1978) proposed the cyaneus group of subgenus *Heterometrus* to accommodate *H. cyaneus* (C.L. Koch, 1836), which was separated from other Asian scorpionids by the presence of an acarinate and granulo-reticulate pedipalp chela, in addition to 16 biometric characters. Couzijn (1981) subsequently created subgenus *Javanimetrus* to include the species and its two former subspecies, and proposed that it was the sister group of *Srilankametrus*, based on increased denticulation of the dorsosubmedian and dorsolateral carinae of the metasoma, a reduced number of pectinal teeth, and weak sexual dimorphism of the pedipalps. Kovařík (2004) synonymized subgenus *Javanimetrus* with *Heterometrus*, as discussed above.

Five terminals of *J. cyaneus* were included in the phylogenetic analyses presented here. *Javanimetrus* was consistently monophyletic in the analyses and supported in the preferred phylogenetic hypothesis (Fig. 8) by 58 molecular synapomorphies and seven morphological synapomorphies.

The phylogenetic position of *Javanimetrus* varied among the analyses presented. *Javanimetrus* was placed sister to a clade comprising the Indian genera, *Chersonesometrus*, *Deccanometrus*, *Gigantometrus*, and *Sahyadrimetrus*, in the ML analysis (Fig. 7a), a relationship also recovered by the separate analysis of the molecular dataset with parsimony and equal weighting or implied weighting and $k = 60$ or 100. It was placed sister to a clade comprising *Chersonesometrus*, *Deccanometrus*, and *Gigantometrus* (Fig. 7b) in the simultaneous analysis with BI; the simultaneous analysis using parsimony with implied weighting and $k = 10, 60$, or 100; and the

separate analyses of the molecular dataset with ML, BI, and parsimony with implied weighting and $k = 1, 3$ or 10. It was placed sister to *Sahyadrimetrus* (Fig. 7c) in the simultaneous analyses with ML-ASC or parsimony with equal weighting or implied weighting and $k = 1$ or 3, as well as in the separate analysis of the morphological data with implied weighting and $k = 1$, in which *Sahyadrimetrus* was paraphyletic.

The unique morphology of *Javanimetrus*, which more closely resembles the scorpionids of South Asia than those of Southeast Asia, together with its phylogenetic position, justify its revalidation and elevation to the rank of genus by Prendini and Loria (2020).

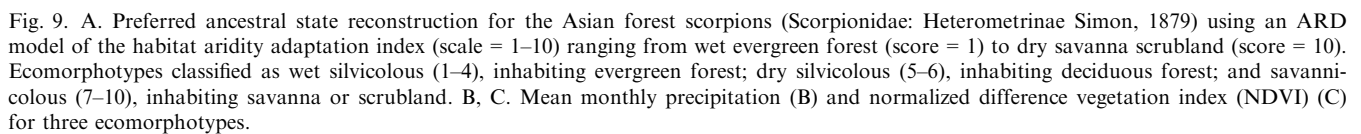
Aside from *Heterometrus*, *Javanimetrus* is the only Asian scorpionid present in Southeast Asia. Its range is much more restricted than that of *Heterometrus*, however, with records in Sumatra and the neighboring islands of Nias, Borneo, Java, Madura and Bali, as well as a single locality each in the Nicobar Islands, southern Thailand, the offshore island of Ko Muk, and Luzon (Fig. 1e). *Javanimetrus* does not occur in Indochina. Its disjunct distribution is presumed to be relictual.

Elevation of Srilankametrus, endemic to Sri Lanka and southern India

Couzijn (1978) proposed the indus group of subgenus *Heterometrus* to accommodate *H. indus* (DeGeer, 1778), based on 16 biometric characters. Couzijn (1981) subsequently created subgenus *Srilankametrus* to include the species and its subspecies, and proposed it as the sister group of *Javanimetrus*, based on increased denticulation of the dorsosubmedian and dorsolateral carinae of the metasoma, a reduced number of pectinal teeth, and weak sexual dimorphism of the pedipalps. Kovařík (2004) synonymized subgenus *Srilankametrus* with *Heterometrus*, as discussed above.

The phylogenetic analyses presented here included 10 terminals representing the seven species of *Srilankametrus* recognized by Prendini and Loria (2020), three of which were represented by at least two terminals each. *Srilankametrus* was monophyletic with high support in all except the separate analyses of the morphological dataset with equal weighting or implied weighting and $k = 1, 60$, or 100. *Srilankametrus* was supported by 32 molecular synapomorphies and four morphological synapomorphies in the preferred phylogenetic hypothesis (Fig. 8) and was consistently placed sister to a clade comprising all other Asian scorpionid genera, except *Heterometrus*. Its phylogenetic position and unique morphology justify its revalidation and elevation to the rank of genus by Prendini and Loria (2020).

Srilankametrus indus and *S. pococki* Prendini and Loria, 2020 formed a monophyletic group, placed sister to the other species of *Srilankametrus*, in all



Srilankametrus caesar and *S. couzijni* are endemic to southern India whereas the other five species of the genus are endemic to Sri Lanka. The distribution of *Srilankametrus* overlaps the distributions of *Chersonometrus* and *Deccanometrus* in southern India, and of *Gigantometrus* in southern India and Sri Lanka (Fig. 1d).

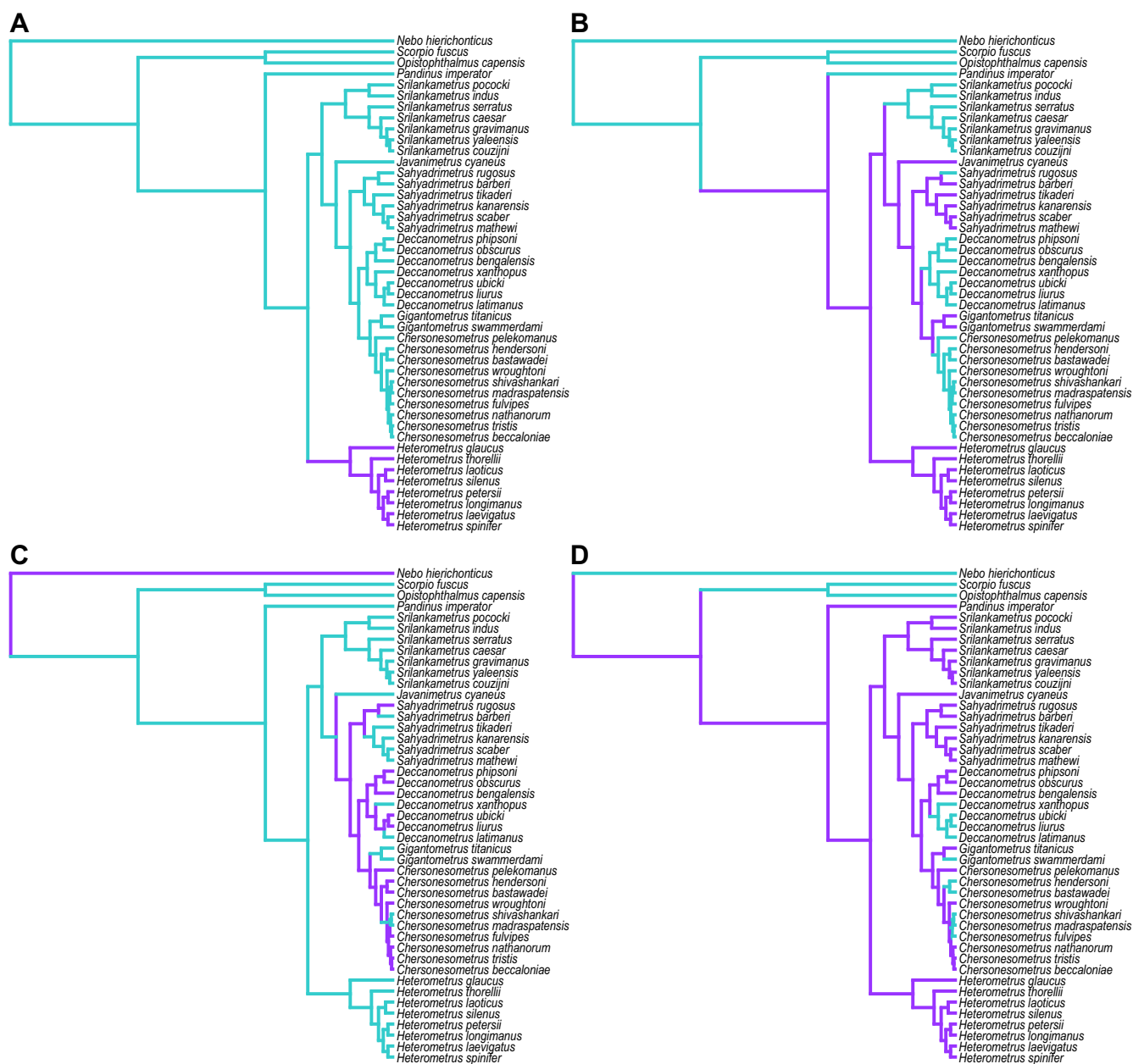


Fig. 10. Ancestral state reconstruction of selected characters contributing to the habitat aridity adaptation index for the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879), using the preferred ARD model. A. Carapace median ocelli size: large, distance between ocelli less than ocellus width (purple) or small, distance between ocelli equal to or greater than ocellus width (blue). Telotarsi pseudonychium: reduced, rounded (blue) or prominent, acuminate (purple). B. Basitarsi length: short (blue) or long (purple). C. Burrows constructed in open: present (blue) or absent (purple). D. Leg colour: predominately pale (blue) or very dark, heavily infuscated (purple).

Elevation of *Gigantometrus*, the largest Asian scorpion, from India and Sri Lanka

Couzijn (1978, 1981) created subgenus *Gigantometrus* to accommodate the largest Asian scorpion species, *G. swammerdami* and its subspecies, considered distinct from all other Asian scorpionids due to

its putatively primitive characters and apparent resemblance to *Pandinus*, notably the acarinate, granular pedipalp chela, as well as its Indian-Sri Lankan distribution, considered characteristic of the relictual fauna of Gondwana. For similar reasons, Couzijn (1981) considered *Gigantometrus* basal to all other Asian scorpionids.

As noted above, Kovařík (2004) apparently recognized the uniqueness of *Gigantometrus* in suggesting it was the only subgenus that could potentially be upheld yet proceeded to synonymize it. Recently, Rossi (2016: 19) reinstated it with the following argument: “I also restore the subgenus *Gigantometrus* stat. n., synonymized without any justification by Kovařík (2004), which... [is] easily differentiated by the larger chela.” Unfortunately, Rossi’s (2016) argument for the revalidation of *Gigantometrus* scarcely differs from Kovařík’s (2004) argument for its synonymy. Each represents an appeal to authority, far removed from “the method of polythetic analysis” applied by Couzijn (1978, 1980).

Five terminals of *Gigantometrus*, representing two species from southern India and Sri Lanka, were included in the phylogenetic analyses presented here. *Gigantometrus* was consistently monophyletic in the analyses and supported in the preferred phylogenetic hypothesis (Fig. 8) by 18 molecular synapomorphies and 10 morphological synapomorphies. All analyses consistently placed *Gigantometrus* monophyletic with *Chersonesometrus*, confirming the exemplar-based analyses of Prendini et al. (2003), and falsifying Couzijn’s (1981) hypothesis that *Gigantometrus* is the basalmost lineage of Asian scorpionids (Fig. 6). The phylogenetic position and unique morphology of *Gigantometrus* justify its elevation to the rank of genus by Prendini and Loria (2020).

Gigantometrus is distributed across southern and eastern India and Sri Lanka (Fig. 1c). Its distribution overlaps those of *Chersonesometrus* and *Deccanometrus* in India, and *Srilankametrus* in southern India and Sri Lanka.

The terminals of *Gigantometrus* from India and Sri Lanka formed two clades in the analyses presented here, supporting Couzijn’s (1981) hypothesis that the populations in Sri Lanka, assigned to the subspecies, *H. (G.) swammerdami titanicus* Couzijn, 1981, and subsequently elevated to the rank of species by Lourenço and Huber (2002), but synonymized with *H. swammerdami* by Kovařík (2004), are genetically as well as morphologically distinct from those in India. This finding justifies the revalidation of *Gigantometrus titanicus* by Prendini and Loria (2020).

Elevation and redefinition of Chersonesometrus, endemic to central and southern India

Based on his phenetic analysis, Couzijn (1978) created subgenus *Chersonesometrus* to accommodate all scorpionid species endemic to the Indian subcontinent, except *G. swammerdami*, and designated *Heterometrus (C.) fulvipes* as type species. Couzijn (1978, 1981) assigned the morphological diversity in *Chersonesometrus* to two groups (Table 1): the phipsoni group, occurring mostly in western and southern India, and the fulvipes group, distributed throughout central and

southern India. Couzijn (1981) considered *Chersonesometrus* so morphologically diverse that no single trait could be used to characterize all its component species, and placed it sister to the nominotypical subgenus, *Heterometrus*. Kovařík (2004) synonymized *Chersonesometrus* with *Heterometrus*, however.

As defined by Couzijn (1981), *Chersonesometrus* and its fulvipes and phipsoni species groups were consistently rendered paraphyletic by *Gigantometrus* and *Javanimetrus* in the analyses presented here, justifying its redefinition and the creation of two new genera by Prendini and Loria (2020). In order to restore the monophyly of *Chersonesometrus*, *Deccanometrus* was created to accommodate *H. bengalensis*, *H. latimanus* (Pocock, 1894), *H. liurus*, *H. obscurus* (Couzijn, 1981), *H. phipsoni*, *H. ubicki* and *H. xanthopus*, whereas *Sahyadrimetrus* was created to accommodate *H. barberi*, *H. kanarensis* (Pocock, 1900), *H. rugosus* (Couzijn, 1981), *H. scaber* (Thorell, 1876) and two new species. The remaining 10 species accommodated in *Chersonesometrus*, as redefined by Prendini and Loria (2020; Table 2), and represented in the analyses by 24 terminals, consistently formed a monophyletic group, placed sister to *Gigantometrus*, thus refuting Couzijn’s (1981) hypothesis that *Chersonesometrus* is sister to *Heterometrus* (Fig. 6). *Chersonesometrus* was supported in the preferred phylogenetic hypothesis (Fig. 8) by six morphological synapomorphies. Its phylogenetic position and unique morphology justify its revalidation and elevation to the rank of genus by Prendini and Loria (2020).

Chersonesometrus occurs across central and southern India (Fig. 1a), where its distribution overlaps those of *Deccanometrus* and *Gigantometrus*. The type species, *C. fulvipes*, recorded in many states of central and northern India, and possibly also the extreme east of Pakistan (Tahir and Prendini, 2014), is the most widely distributed whereas the other nine species are known from fewer localities, primarily on the Deccan Plateau.

The simultaneous analyses of the morphological and molecular datasets consistently recovered two clades among the species of *Chersonesometrus*, one comprising *C. bastawadei* Prendini and Loria, 2020 and *C. hendersoni* Prendini and Loria, 2020, and another comprising *C. fulvipes*, *C. madraspatensis* (Pocock, 1900) and *C. shivashankari*. The positions of other taxa were more variable.

Relationships among the terminals sampled from across the distributions of *C. fulvipes* and *C. madraspatensis* supported the decision to retain the latter. Pocock’s (1900) subspecies, *H. fulvipes madraspatensis*, based mainly on a difference in the shape of the male pedipalps, was synonymized with *H. (C.) fulvipes* by Couzijn (1981) but revalidated and elevated to the rank of species by Tikader and Bastawade (1983), a decision upheld by Fet (2000) and Kovařík (2004).

Sahyadrimetrus, a genus endemic to the Western Ghats

The Western Ghats or Sahyadri of India is a biodiversity hotspot containing many endemic taxa (Menon and Bawa, 1997; Gunawardene et al., 2007). Ten (24%) Asian scorpionid species have been recorded in the area, seven of which are endemic. Prendini and Loria (2020) created a new genus, *Sahyadrimetrus*, to accommodate six species (Table 2), three of which, *S. barberi*, *S. kanarensis* and *S. scaber*, were previously assigned to subgenus *Chersonesometrus* by different authors (Couzijn, 1981; Tikader and Bastawade, 1983; Fet, 2000; Sureshan et al., 2007), whereas *S. rugosus* (as its junior synonym, *H. keralaensis* Tikader and Bastawade, 1983) was assigned to subgenus *Heterometrus* based on putative morphological similarities with *H. longimanus* (Tikader and Bastawade, 1983; Fet, 2000; Sureshan et al., 2007) which proved to be convergent.

The phylogenetic analyses presented here included 24 terminals representing the six species of *Sahyadrimetrus*, all except one of which were represented by at least two terminals each (molecular data were absent, and the morphology incompletely scored for *S. barberi*, known only from the holotype female). *Sahyadrimetrus* was consistently monophyletic in the simultaneous analyses of the morphological and molecular datasets and the separate analyses of the molecular dataset. The genus was supported in the preferred phylogenetic hypothesis (Fig. 8) by 17 molecular synapomorphies and one morphological synapomorphy.

Whereas all simultaneous analyses consistently recovered two monophyletic groups within *Sahyadrimetrus*, one comprising *S. barberi* and *S. rugosus*, the other comprising *S. kanarensis*, *S. mathewi* Prendini and Loria, 2020, *S. scaber* and *S. tikaderi* Prendini and Loria, 2020, the position of *Sahyadrimetrus* varied. *Sahyadrimetrus* was placed sister to *Javanimetrus* in the simultaneous analyses with ML-ASC and parsimony with equal weighting or implied weighting and $k = 1$ or 3, and the separate analysis of the morphological dataset with parsimony and implied weighting with $k = 1$, but sister to a clade including *Chersonesometrus*, *Deccanometrus*, *Gigantometrus*, and *Javanimetrus* in the simultaneous analyses using BI or parsimony with implied weighting and $k = 10$, 60, or 100, as well as in the separate molecular analyses using ML, BI or parsimony with implied weighting and $k = 1$, 3, or 10. *Sahyadrimetrus* was placed sister to a clade comprising *Chersonesometrus*, *Deccanometrus*, and *Gigantometrus* in the simultaneous analysis with ML and the separate analyses of the molecular dataset using parsimony with equal weighting or implied weighting with $k = 60$ or 100.

Sahyadrimetrus occurs from low to high elevations in the Western Ghats is allopatric or parapatric with *Chersonesometrus*, *Deccanometrus*, and *Gigantometrus* (Fig. 1d). *Sahyadrimetrus mathewi* and *S. scaber* are more widespread than the other three species, especially *S. barberi* and *S. rugosus* which appear to be somewhat restricted. Phylogeographic structure, coincident with morphological differences, among terminals formerly assigned to *S. scaber*, from lower and higher elevations to the north and south of the Bharathapuzha River and Palakkad (Palghat) Gap, a disjunction in the Western Ghats between the Nilgiri Hills and Anaimalai Hills, justified the recognition of two new species, *S. mathewi* and *S. tikaderi*.

Deccanometrus, a genus widespread across the Indian Subcontinent

Prendini and Loria (2020) created a new genus, *Deccanometrus*, to accommodate seven Indian species (Table 2), four of which, *D. bengalensis*, *D. liurus*, *D. phipsoni*, and *D. xanthopus* were previously assigned to subgenus *Chersonesometrus* (Couzijn, 1981; Tikader and Bastawade, 1983; Fet, 2000; Table 1), whereas one, *D. latimanus*, was mistakenly assigned to *Srilankametrus* (Tikader and Bastawade, 1983; Fet, 2000).

The phylogenetic analyses presented here included 17 terminals representing all seven species of *Deccanometrus*, five of which were each represented by at least two terminals (molecular data were absent for *D. liurus*, *D. phipsoni*, and *D. ubicki*). *Deccanometrus* was consistently monophyletic in the simultaneous analyses of the morphological and molecular datasets using ML, ML-ASC, BI or parsimony with implied weighting and $k = 1$ or 3. The genus was supported in the preferred phylogenetic hypothesis (Fig. 8) by nine molecular synapomorphies and three morphological synapomorphies. *Deccanometrus* was also monophyletic in the separate analyses of the molecular dataset using ML or parsimony with equal weighting or implied weighting with $k = 60$ or 100.

All simultaneous analyses of the morphological and molecular datasets and separate analyses of the molecular dataset using ML, BI, or parsimony with equal weighting or implied weighting and $k = 3$, 10, 60, or 100, consistently placed *Deccanometrus* with the clade comprising *Chersonesometrus* and *Gigantometrus*. All simultaneous analyses of the morphological and molecular dataset consistently recovered a clade comprising *D. latimanus*, *D. liurus*, *D. ubicki* and *D. xanthopus*. However, relationships among the other species varied among the analyses.

Deccanometrus was monophyletic in 60% of the simultaneous analyses. Its paraphyly in the remaining analyses may be due to the absence of molecular data

for almost half its species. For example, *D. bengalensis*, *D. obscurus*, and *D. phipsoni* formed a monophyletic group with the clade comprising *Chersonesometrus* and *Gigantometrus*, to the exclusion of the other four species of *Deccanometrus*, which formed a separate clade, in the simultaneous analyses using parsimony with implied weighting and $k = 10$, 60, or 100; *D. obscurus* and *D. phipsoni* formed a monophyletic clade with *Chersonesometrus* and *Gigantometrus* in the simultaneous analysis using parsimony with equal weighting; and *D. latimanus* formed a monophyletic clade with *Chersonesometrus* and *Gigantometrus* in the separate analyses of the molecular dataset with BI or parsimony with implied weighting and $k = 3$ or 10. *Deccanometrus* was rendered paraphyletic by *Javanimetrus* in the separate analysis of the molecular dataset with parsimony and implied weighting with $k = 1$.

The distribution of *Deccanometrus* overlaps those of its sister genera, *Chersonesometrus* and *Gigantometrus* (Fig. 1b). Three species of the genus are widespread but allopatric: *D. bengalensis* extends from the plains of northern India to the Himalayan foothills; *D. phipsoni* inhabits the Eastern Ghats and the central part of the Western Ghats; and *D. xanthopus* inhabits the Deccan Plateau. Three other species are more range-restricted: *D. latimanus* inhabits northern Pakistan; *D. liurus* is known from two localities in the Central Highlands; *D. obscurus* inhabits the northern part of the Western Ghats, and *D. ubicki* has only been recorded from a single locality near the southeastern coast.

Suprageneric classification of Scorpionidae

Although not the direct focus of this study, relationships among the non-Asian scorpionids recovered by the analyses presented here (Figs. 7, 8) necessitate comments on recent changes to the suprageneric classification of Scorpionidae self-published, without peer review by Rossi (2016). In addition to “reinstating” three scorpionoid taxa well justified two decades ago with phylogenetic analyses and diagnoses by Prendini (2000a), revalidated by Prendini and Wheeler (2005), and in widespread use since—i.e., Diplocentridae, Nebinae Kraepelin, 1905 and Urodacidae Pocock, 1893—Rossi (2016: 16–22) proposed new taxa “without the need of more complex, and always controversial, phylogenetic analyses.” Two subfamilies were recognized in Urodacidae, including a new, monotypic subfamily, created to accommodate the monotypic genus, *Aops* Volschenk and Prendini, 2008, the point of which is unclear.

Rossi (2016) proposed the following changes within Scorpionidae (Table 10): two subfamilies and six tribes (three new) were recognized, all except one tribe accommodating a single genus; Rugodentinae Bastawade et al., 2005 and *Rugodentus* Bastawade et al.,

2005 were respectively synonymized with Pandininae Thorell, 1876 and *Heterometrus*, resulting in a junior secondary homonym and a *nomen novum*; *Protophthalmus* Lawrence, 1969 was revalidated from synonymy with *Opisthophthalmus*, and *Protophthalmus holmi* Lawrence, 1968 and *Protophthalmus jenseni* Lamoral, 1972, erroneously claimed to be new combinations (both were originally described as species of *Protophthalmus*); three former subgenera of *Pandinurus* Fet, 2000, i.e., *Pandiborellius* Rossi, 2015, *Pandicaporiacous* Rossi, 2015, and *Pandipalpus* Rossi, 2015, were elevated to the rank of genus; former subgenus *Gigantometrus* of *Heterometrus* was revalidated; and a new monotypic subgenus, *Dunlopandinoides* Rossi, 2016, was created within *Pandinoides* Fet, 2000.

Aside from the low information content of monotypic taxa (Platnick, 1976), there are several problems with Rossi's (2016) proposed classification centering around the failure to follow best practices in terms of scientific rigor and repeatability, and the absence of a body of evidence to support the changes, concerns that are also applicable to the classification presented by Kovařík et al. (2017) and similar contributions in self-published or self-edited books and online journals lacking rigorous peer review or offering, at best, crony review (Wüster et al., 2001; Prendini and Wheeler, 2005; Williams et al., 2006; Jäch, 2007a, b; Wallach et al., 2009; Kaiser et al., 2013; Kaiser, 2013, 2014; Measey, 2013; Reynolds et al., 2014; Denzer et al., 2015; Rhodin et al., 2015; Garnett and Christidis, 2017; Schutze et al., 2017; Esposito et al., 2018; Troncoso-Palacios et al., 2019). Specifically, the changes presented were either unsubstantiated by or violate the results of phylogenetic analysis which, despite the opinion of Rossi (2016), is almost universally regarded by professional scientists as the basis for modern biological classification insofar as it constitutes the evidence for monophyly, the fundamental requirement for upholding any group in a Linnean hierarchy. For example, subfamily Scorpioninae and tribes Opisthophthalmi Rossi, 2016 and Pandinini Thorell, 1876, as construed by Rossi (2016; Table 10), were demonstrably paraphyletic based on phylogenetic analyses of morphology and/or multilocus DNA sequences by Lamoral (1978, 1979), Prendini (2000a), Prendini et al. (2003), and the present contribution.

Whereas the monophyletic group comprising *Heterometrus*, *Pandinus* and allied genera posited by Couzijn (1981) and others, i.e., Pandininae sensu Rossi (2016), was recovered in all phylogenetic analyses to date, e.g., Prendini (2000a), Prendini et al. (2003) and the present study, only one analysis (Prendini, 2000a) placed *Opisthophthalmus* and *Scorpio* in a monophyletic group. However, that analysis formed part of a larger study of relationships within superfamily Scorpionoidea, based on exemplar species, and did not

Table 10

Rossi's (2016) self-published suprageneric classification of Scorpionidae Latreille, 1802

Family Scorpionidae Latreille, 1802
Subfamily Scorpioninae Latreille, 1802
Tribe Scorpionini Latreille, 1802
<i>Scorpio</i> Linnaeus, 1758
Tribe Opisthophthalmini Rossi, 2016
<i>Opisthophthalmus</i> C.L. Koch, 1837
Tribe Protophthalmini Rossi, 2016
<i>Protophthalmus</i> Lawrence, 1969
Subfamily Pandininae Thorell, 1876
Tribe Pandinini Thorell, 1876
<i>Pandiborellius</i> Rossi, 2015
<i>Pandiborellius</i> (<i>P.</i>) Rossi, 2015
<i>Pandiborellius</i> (<i>Pandipavesius</i>) Rossi, 2015
<i>Pandicaporiaccous</i> Rossi, 2015
<i>Pandinoides</i> Fet, 2000
<i>Pandinoides</i> (<i>P.</i>) Fet, 2000
<i>Pandinoides</i> (<i>Dunlopandinoides</i>) Rossi, 2016
<i>Pandinopsis</i> Vachon, 1974
<i>Pandinurus</i> Fet, 2000
<i>Pandinurus</i> (<i>P.</i>) Fet, 2000
<i>Pandinurus</i> (<i>Pandinoriens</i>) Rossi, 2015
<i>Pandinus</i> Thorell, 1876
<i>Pandipalpus</i> Rossi, 2015
Tribe Pandinopsini Rossi, 2016
<i>Pandinops</i> Birula, 1913
Tribe Heterometrinini Simon, 1879
<i>Heterometrus</i> Ehrenberg, 1828
<i>Heterometrus</i> (<i>H.</i>) Ehrenberg, 1828
<i>Heterometrus</i> (<i>Gigantometrus</i>) Couzijn, 1978

Table 11

Revised suprageneric classification of Scorpionidae Latreille, 1802 by Prendini and Loria (2020) with species counts in parentheses

Family Scorpionidae Latreille, 1802 (161)
Subfamily Heterometrinae Simon, 1879 (41)
<i>Chersonesometrus</i> Couzijn, 1978 (10)
<i>Deccanometrus</i> Prendini and Loria, 2020 (7)
<i>Gigantometrus</i> Couzijn, 1978 (2)
<i>Heterometrus</i> Ehrenberg, 1828 (8)
<i>Javanimetrus</i> Couzijn, 1981 (1)
<i>Sahyadrimetrus</i> Prendini and Loria, 2020 (6)
<i>Srilankametrus</i> Couzijn, 1981 (7)
Subfamily Opisthophthalmini Rossi, 2016 (59)
<i>Opisthophthalmus</i> C.L. Koch, 1837 (59)
Subfamily Pandininae Thorell, 1876 (44)
<i>Pandiborellius</i> Rossi, 2015 (10)
<i>Pandinoides</i> Fet, 2000 (3)
<i>Pandinops</i> Birula, 1913 (9)
<i>Pandinopsis</i> Vachon, 1974 (1)
<i>Pandinurus</i> Fet, 2000 (15)
<i>Pandinus</i> Thorell, 1876 (4)
<i>Pandipalpus</i> Rossi, 2015 (2)
Subfamily Scorpioninae Latreille, 1802 (17)
<i>Scorpio</i> Linnaeus, 1758 (17)

include molecular data. In contrast, the analyses of Prendini et al. (2003), based on multilocus DNA sequence data and morphology, for a larger sample of scorpionid exemplars, failed to recover the clade

comprising *Opisthophthalmus* and *Scorpio*, most of the analyses instead placing *Scorpio* sister to the clade comprising *Heterometrus* and *Pandinus*, to the exclusion of *Opisthophthalmus*. Results of the analyses presented here also consistently failed to place *Opisthophthalmus* and *Scorpio* in a monophyletic group. The evidence overwhelmingly suggests that Scorpioninae, as defined by Rossi (2016), is paraphyletic, necessitating the elevation of Opisthophthalmini to the rank of subfamily by Prendini and Loria (2020; Table 11).

All phylogenetic analyses to date recovered the monophyly of *Opisthophthalmus* and placed *Opisthophthalmus holmi* (Lawrence, 1969) terminally within it (Lamoral, 1978, 1979; Prendini, 2000a; Prendini et al., 2003). Analyses based on morphology and multilocus DNA sequences supported the opinion of Lamoral (1979: 719) “that *Protophthalmus* is congeneric with *Opisthophthalmus* and that *holmi* and *jensenii* represent an extreme sister group separated from other species by a series of derived character states.” Pending evidence to the contrary, both *Opisthophthalmus* and Opisthophthalmini were rendered paraphyletic by *Protophthalmus* and Protophthalmini, justifying their synonymy by Prendini and Loria (2020).

Unpublished data, indicating that *Pandipalpus* Rossi, 2015, *Pandinoides*, and *Pandinops* Birula, 1913, form a monophyletic group, to the exclusion of other genera of Pandininae, imply that Pandinini, as defined by Rossi (2016), is also paraphyletic, justifying the synonymy of Pandinopsini Rossi, 2016 by Prendini and Loria (2020). Recent changes to the generic classification of Pandininae, including the creation of new genera and subgenera based on trivial morphological differences, are critiqued elsewhere (Prendini, 2016; Prendini and Loria, 2020).

Rossi (2016) recognized the Asian scorpionids as a tribe of Pandininae for reasons unspecified. The two lineages are sister taxa, with similar levels of taxonomic and morphological diversity, justifying the elevation of Heterometrinini, the monophyly of which was also consistently recovered in the analyses presented here, to the rank of subfamily by Prendini and Loria (2020).

Finally, Rossi's (2016) synonymy of Rugodentinae and *Rugodentus* rendered Scorpionidae, Pandininae Thorell, 1876, Heterometrinini Simon, 1879, and *Heterometrus* paraphyletic, justifying the revalidation and elevation of Rugodentinae to the rank of family, and the revalidation of *Rugodentus*, also discussed further by Prendini and Loria (2020).

Dispersal and evolution of ecomorphotypes in Asian forest scorpions

The present study provides insight into the evolution of ecomorphotypes in Asian Scorpionidae, which

complements a biogeographical investigation of the origins, dispersal and diversification of Heterometrinae, presented elsewhere (Loria and Prendini, 2020). According to the reconstruction presented here, the ancestor of Asian scorpionids was wet silvicolous and inhabited humid, evergreen forests (Fig. 9). This result agrees favorably with the suggestion by Prendini et al. (2003) that the ancestor of Scorpionidae probably inhabited tropical rainforests because *Pandinus* and the basalmost Asian scorpionids, as well as several scorpionoid outgroup taxa, including Heteroscorpionidae Kraepelin, 1905, most Hormuridae Laurie, 1896, and the basal species of Urodacidae, Diplocentridae, and Bothriuridae Simon, 1880 inhabit forests, mesic savanna or caves.

Asian scorpionids separated from their African relatives when India rifted from Africa (Couzijn, 1981; Prendini et al., 2003) but the precise timing of this divergence is a matter of debate. Santibáñez-López et al. (2017) suggested the divergence between Heterometrinae and *Pandinus* occurred at the Jurassic-Cretaceous boundary, 145 million years ago (Ma). Sharma et al. (2018) suggested superfamily Scorpionoidea diversified at the end of the Cretaceous, roughly 90 Ma, implying Heterometrinae and *Pandinus* diverged more recently. Santibáñez-López et al. (2018) presented an even earlier date, placing this divergence within the Paleogene, less than 30 Ma. An analysis based on a more comprehensive taxon sample suggests the divergence occurred in the mid-Mesozoic, ca. 115 Ma, however (Loria and Prendini, 2020).

During India's journey to Eurasia, the Cretaceous-Tertiary (KT) mass extinction followed the Chicxulub meteorite impact in Mexico and Deccan volcanism on the Indian subcontinent (Wilfe et al., 2003; Keller, 2014). Due to environmental changes caused by these events, range contraction confined many taxa to refugia in the Western Ghats of India and the Central Highlands of Sri Lanka (Boussuyt and Milikovitsh, 2001; Conti et al., 2002; Joshi and Karanth, 2013). The reconstruction presented here suggests the ancestors of *Sahyadrimetrus* and *Srilankametrus*, which respectively inhabit the Western Ghats and the Central Highlands, retained the ancestral wet silvicolous ecomorphotype for Asian scorpionids, implying that the humid, evergreen forests of these mountain ranges remained more stable and were less adversely affected by the KT mass extinction, consistent with the refugium hypothesis (Loria and Prendini, 2020).

Once volcanism subsided, recolonization of the rest of the Indian mainland allowed the ancestors of more derived genera such as *Chersonesometrus* and *Deccanometrus* to colonize and diversify into new and often drier habitats, including deciduous forest and savanna on the Deccan Plateau, the Indo-Gangetic Plains, and the Himalayan foothills (Loria and Prendini, 2020).

Dispersal into these habitats selected for characters associated with dry silvicolous and savannicolous ecomorphotypes including paler and/or contrasting colouration, shorter, more heavily armed leg tarsi, better suited for burrowing in hard, dry substrata, and burrow construction in open ground (Fig. 10). This pattern of ecomorphotype evolution is consistent with other genera of Scorpionidae, in which arid-adapted species are usually more derived than mesic-adapted species (Prendini, 2000a, 2001b). For example, diversification of the scorpionid taxa *Opisthophthalmus*, *Scorpio*, and Pandininae is attributed to the aridification of the African continent (Prendini et al., 2003).

Interestingly, evolution and diversification into other, equally mesic habitats, appears to have occurred in at least one clade of Asian scorpionids. Upon India's collision with Eurasia, the ancestors of *Heterometrus* dispersed into new tropical rainforest habitats from which they had previously been absent, evidently acquiring several adaptive characters in the process. For example, enlargement of the ocelli, a synapomorphy for *Heterometrus* (Figs. 3a, 10), may have facilitated radiation into tropical rainforests by improving the ability to detect light through the dense canopy. Other synapomorphies of *Heterometrus*, such as longer pedipalps, longer, less heavily armed leg basitarsi, better suited to climbing and roaming than burrowing, and the development a prominent pseudonygium on the telotarsi, the latter often associated with a corticolous ecomorphotype (Prendini, 2001b; Figs. 3d, 10), may have permitted these species to take better advantage of temporary shelters such as under peeling bark and in tree holes, well above ground level, in addition to burrowing.

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Conflict of Interest

None declared.

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Note added in Proof. *Pandinurus intermedius* (Borelli, 1919) and *Pandinus lowei* Kovařík, 2012, synonymized by Prendini and Loria (2020) based on unpublished genomic and morphological evidence (in preparation), were revalidated two weeks later in: Kovařík, F., Lowe, G. and Elmi, H. Sh. A. (2020) Scorpions of the Horn of Africa (Arachnida: Scorpiones). Part XXV. Description of *Pandinurus awalei* sp. n. and the male of *Pandiborellius somalilandus* (Kovařík, 2012), with remarks on recent synonymies (Scorpionidae: Pandiniinae). *Euscorpius*, 322, 1–21. Prendini and Loria's (2020) decisions were supported by ca. 4200 nucleotide base-pairs of DNA sequence data from five gene loci. The material analyzed genetically in support of the synonymy of *P. intermedius* comprised eight samples from five localities in Ethiopia, spanning the localities from which material was previously identified as *Pandinurus citernii* (Borelli, 1919) and *P. intermedius* by Kovařík et al. (2017, 2020), and matching the descriptions of those species provided therein, as well as three additional localities near the Somali type locality of both species, from the Kenyan side of the border. The material analyzed in support of the synonymy of *P. lowei* comprised nine samples, three from the type locality and its vicinity in the Upemba National Park of the Democratic Republic of Congo, matching the original description and material examined from the type locality, at the MNHN (ex MRAC 114034), and six others, three each from Malawi and Tanzania, spanning the distribution of *Pandipalpus viatoris* (Pocock, 1890), and matching type and nontype material examined by Prendini (2016). Consistent with the evidence of low genetic divergence among the samples in question, the morphological differences cited in support of *P. intermedius* and *P. lowei* do not hold up when a larger sample of material is examined. Consequently, both synonyms are upheld: *Pandinus lowei* Kovařík, 2012 = *Pandipalpus viatoris* (Pocock, 1890), syn. nov.; *Pandinus intermedius* Borelli, 1919 = *Pandinurus citernii* (Borelli, 1919), syn. nov.

Appendix 1

Terminal taxa, specimens and tissue samples used for phylogenetic analysis of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879). Depositories for material examined are abbreviated as follows: American Museum of Natural History (AMNH), New York, U.S.A.; Natural History Museum (BMNH), London, U.K.; Bombay Natural History Society (BNHS), Mumbai, India; California Academy of Sciences (CAS), San Francisco, U.S.A.; Field Museum of Natural History (FMNH), Chicago, U.S.A.; Institute of Fundamental Studies (IFS), Kandy, Sri Lanka; Lee Kong Chian Natural History Museum (LKC), National University of Singapore; Museo Civico di Storia Naturale “Giacomo Doria”, Genoa (MCSNG), Italy; Museum of Comparative Zoology (MCZ), Harvard University, Cambridge, MA; Muséum National d’Histoire Naturelle (MNHN), Paris, France; Musée Royal d’Histoire Naturelle de Belgique (MRHNB), Brussels, Belgium; Naturhistorisches Museum Wien (NHMW), Vienna, Austria; Naturhistoriska Museet, Göteborg (NMG), Sweden; Naturhistoriska Riksmuseet (NRS), including the Julio Ferrer Collection (JF), Stockholm, Sweden; Naturalis, Rijksmuseum van Natuurlijke Historie (RMNH), Leiden, The Netherlands; Forschungsinstitut und Naturmuseum Senckenberg, Frankfurt (SMF), Germany; U.S. National Museum of Natural History (USNM), Smithsonian Institution, Washington, DC; Western Australian Museum (WAM), Perth; Museum für Naturkunde, Berlin (ZMB), Germany; Zoologisches Institut und Zoologisches Museum, Universität Hamburg (ZMH), Germany; Natural History Museum of Denmark, Copenhagen (ZMUC). Tissue samples are deposited in the Ambrose Monell Collection for Molecular and Microbial Research (AMCC) at the AMNH.

Outgroup

Nebo hierichonticus (Simon, 1872): ISRAEL: *Haifa Distr.*: Mount Carmel National Park: Ha’Agam campsite, upper Nahal Oren, 32°43’N 35°01’E, 219 m, 22.viii.2011, L. Prendini, T.L. Bird and E. Gefen., 6 ♂, 5 ♀, 3 subad. ♂, 2 subad. ♀, 2 juv. ♂, 5 juv. ♀ (AMNH), 1 juv. ♂, 1 juv. ♀ (AMCC [LP 11138]); Palestine, Afule, 25.ix.1934, Dr. F.R.S. Shaw, 5 ♂, 2 ♀ (FMNH); Mt. Carmel, 20.vii.1960, 1 subad. ♂ (CAS), 12.xi.1960, Dr. M.R. Warburg, 1 juv. ♂ (CAS).

Opisthophthalmus capensis (Herbst, 1800): SOUTH AFRICA: Western Cape Prov.: *Cape Distr.*: Oranjezicht, Cape Town, Table Mountain (northern slope) [33°56’S 18°25’E], 2.viii.1997, L. Prendini and E. Scott, 1 ♂, 1 ♀ (CAS), 1 juv. ♂ (AMCC 100811 [LP 1046]).

Scorpio fuscus (Ehrenberg, 1829): ISRAEL: *Haifa Distr.*: Mount Carmel, Ramat Ha’Nadiv [32°33’N 34°57’E], vi.1998, Y. Lubin, 1 subad. ♂ (AMCC 101705 [LP 1556]); Bet-Guvzin, 27.iii.1967, N. Roman, 1 ♂ (AMNH); Jerusalem, 25.vii.1961, P. Amitai, ♀ (AMNH); Palestine, Afule, 25.ix.1934, Dr. F.R.S. Shaw, 5 ♀, 2 subad. ♂, 2 juv. ♀ (FMNH). LEBANON: Beirut, campus of American University of Beirut, 23.x.1961, A.J. Sam, 2 ♀, 1 juv. ♀ (CAS).

Pandinus imperator (C.L. Koch, 1841): GHANA: xi.1997, D. Taylor, 1 ♂ (AMCC 101703 [LP 1601]); Cape Coast [05°06’N 01°14’W], Univ. Cape Coast Collection, Dr. J. Boggs, 1 ♂, 1 ♀, 5 juv. (AMNH). LIBERIA: *Nimba Co.*: Oldtown Gobonwea, 225 mi. from Monrovia, 40 mi. E of Mt. Nimba [07°33’N 08°37’W], C.D. Miller III, 2 ♂, 2 ♀ (AMNH), 4 ♂, 3 ♀ (USNM).

Ingroup

Chersonesometrus bastawadei Prendini and Loria, 2020: INDIA: Karnataka: *Bagalkot Distr.*: Badami, Bhootnath Temple, Badami–Mahakoota Path, 15°55’N 75°41’E, 607 m, 23.xi.2009, R. Datta, 1 subad. ♂ paratype (AMCC [LP 16785]); Naganath Temple, NW of

Navilahole, 15°54’N 75°44’E, 541 m, 23.xi.2009, R. Datta, holotype ♀, paratypes: 7 ♀, 5 subad. ♂, 1 subad. ♀, 11 juv. ♂, 4 juv. ♀ (AMNH), 2 juv. ♂, 1 juv. ♀ (AMCC [LP 16787]).

Chersonesometrus beccaloniae (Kovářík, 2004): INDIA: Tamil Nadu: *Nilgiris Distr.*: Ootacamund [Ooty, 11°25’N 76°42’E], 1 ♂ (ZMH). Vibudhi Malai Murugan Temple, Bokkapuram, 11°33’N 76°39’E, 1023 m, 27.xi.2004, S. Basi, 1 ♀, 1 subad. ♂, 7 juv. ♂, 6 juv. ♀ (AMNH), 2 juv. ♂ (AMCC [LP 13133]). *Vellore Distr.*: Javadu Hills, S of Madanur on Highway 46 (Ambur–Vellore), 12°51’N 78°49’E, 306 m, 30.xi.2004, S. Basi, 1 ♀ (AMNH); Yelagiri Hills, Mangalam Road, Yelagiri, 12°35’N 78°39’E, 1000 m, 29.xi.2004, S. Basi, 1 ♀, 1 juv. ♂, 3 juv. ♀, 1 ♀ carapace and sinistral pedipalp segments (AMNH), 1 juv. ♂, 1 juv. ♀ (AMCC [LP 13134]).

Chersonesometrus fulvipes (C.L. Koch, 1837): INDIA: Haryana: *Faridabad Distr.*: Aravalli Hills, NNE of Mangar, 28°23’N 77°11’E, 252 m, 14.xi.2006, P. Menon, 1 ♀, 6 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 13990]). Madhya Pradesh: *Hoshangabad Distr.*: Satpura, Pipariya Road 19, between Matkuli and Pipariya, 22°39’N 78°23’E, 364 m, 6.xii.2006, P. Menon, 1 ♀ (AMNH), 1 juv. ♂ (AMCC [LP 14031]); Raini Pani, 22°37’N 78°07’E, 385 m, 4.xii.2006, P. Menon, 1 ♀, 1 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 13997]); Raini Pani road, 22°37’N 78°08’E, 369 m, 5.xii.2006, P. Menon, 3 ♀, 1 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 13981]). Maharashtra: *Chandrapur Distr.*: Chanda [Chandrapur, 19°58’N 79°18’E], Central Provinces, *Heterometrus (Chersonesometrus) fulvipes*: neotype [= *Palamnaeus fulvipes bombayensis* holotype]: 1 ♀ (BMNH 1913.1.6.2 ex BNHS). *Satara Distr.*: Khambatki Ghat, S of Khandala, on M.B.T. Road (Highway 47), 18°01’N 74°01’E, 928 m, 17.xi.2004, S. Basi, 2 pedipalp chelae (AMNH), 1 juv. ♀ (AMCC [LP 13125]), 937 m, 30.xi.2006, P. Menon, 2 ♂, 1 ♀, 9 juv. ♂, 5 juv. ♀ (AMNH), 1 juv. ♂, 1 juv. ♀ (AMCC [LP 13957]), 902 m, 29.xi.2009, R. Datta, 2 ♂, 1 ♀ (AMNH), 2 juv. ♂, 1 juv. ♀ (AMCC [LP 16812]). Pakistan [alleged], iii.2008, ex pet trade, 1 ♂ (AMNH [LP 8268]), [pedipalp chela] (AMCC [LP 8268]).

Chersonesometrus hendersoni Prendini and Loria, 2020: INDIA: Karnataka: *Ballari Distr.*: Sanapur Lake, Sanapura, 15°22’N 76°26’E, 477–478 m, 22.xi.2009, R. Datta, holotype ♀, paratypes: 1 subad. ♀, 7 juv. ♂, 6 juv. ♀ (AMNH), 2 juv. ♀ (AMCC [LP 16776]); Tungabhadra River, Narayanpet, 15°20’N 76°25’E, 448 m, 22.xi.2009, R. Datta, paratypes: 2 ♀, 2 subad. ♂, 1 subad. ♀, 6 juv. ♂, 1 juv. ♀ (AMNH), 2 juv. ♂ (AMCC [LP 16782]).

Chersonesometrus madraspatensis (Pocock, 1900): INDIA: vi.1999, ex F. Somma, 1 ♀ (AMCC 101695 [LP 1533]). Andhra Pradesh: *Chittoor Distr.*: Kothakota, on Chittoor–Tirupati Road 61, 13°23’N 79°06’E, 372 m, 30.xii.2004, S. Basi, 1 subad. ♂ (AMCC [LP 13127]). Tamil Nadu: *Chennai Distr.*: *P. f. madraspatensis*: lectotype ♂, paralectotype ♂ (BMNH), Madras [Chennai, 13°05’N 80°17’E], Henderson. *Kancheepuram Distr.*: Nemmeli [Nemmeli, 12°43’N 80°13’E], 45 km from Chennai, on eastern coastline, 30.v.2005, H.A. Vadivelu, A. and R. Ali, and F. Paul, 4 ♂, 4 ♀, 1 juv. ♂, 1 juv. ♀ (AMCC [LP 4324]).

Chersonesometrus nathanorum Prendini and Loria, 2020: INDIA: Tamil Nadu: *Theni Distr.*: 1 subad. ♂ holotype, 1 juv. ♀ paratype (MNHN RS 3145), India, Monts Palni, 8 km NE Ganguvarpatti village [Ganguvarpatti, 10°12’N 77°44’E], 20.ii.1943, Dr. K. Lindberg, se pente en montant [upwards slope].

Chersonesometrus pelekomanus (Couzijn, 1981): INDIA: Nilgiris, Deccan, holotype ♂, 1 ♂ (SMF RII/1088/19). Bombay [Bombay Presidency], Deccan, paratype ♀ (SMF RII/329), 1 subad. ♀, 10 juv. ♂, 8 juv. ♀ (SMF RII/329). Karnataka: *Ballari Distr.*: Jarmali, NW of Jarmali Durgamma Temple, 14°50’N 76°28’E, 675 m, 22.xi.2009, R. Datta, 1 juv. ♂ (AMCC [LP 16774]), 1 juv. ♂ (AMNH). *Chitradurga Distr.*: Chitradurga, Jogi Matti Road, 14°11’N 76°24’E, 1000 m, 21.xi.2009, R. Datta, 1 ♂, 2 ♀, 1 subad. ♀, 19 juv. ♂, 15 juv. ♀ (AMNH), 1 juv. ♂, 2 juv. ♀ (AMCC [LP 16773]).

Chersonesometrus shivashankari Prendini and Loria, 2020: INDIA: Andhra Pradesh: *Anantapur Distr.*: Wagra Karoor/Wagra Karur/

Wagra Karour [Vajrakarur, 15°02'N 77°23'E], Chaper, 1 ♂ (MNHN RS 0083 [Simon coll. 7015]), 1893, Chaper, 1 ♂, 1 ♀, 2 juv. ♂ paratypes (MNHN RS 0093). Karnataka: *Bellary Distr.*: Bellahunshi [Bellahunshi, 15°10'N 76°16'E], 30.viii.1912, Fletcher, holotype ♂, 1 juv. ♂, 2 juv. ♀ paratypes (BMNH); near road from Hampasagar [Hampasagara/Hampasagaram, 15°07'N 76°01'E] to Hadagalli [Hadagali, 15°01'N 75°56'E], 1.ix.1912, Y.R., 1 ♀, 4 juv. ♀ paratypes (BMNH); Hospet [15°16'N 76°24'E], 30.viii.1912, Fletcher, 2 juv. ♀ paratypes (BMNH).

Chersonesometrus tristis (Henderson, 1919): INDIA: Andhra Pradesh: *Chittoor Distr.*: Ithepalli/Agarala, NW of, 13°35'N 79°15'E, 340 m, 1.xii.2004, S. Basi, 2 ♂ [1 ♂ remains], 1 subad. ♂, 1 juv. ♂ (AMNH), 1 juv. ♂ (AMCC [LP 13135]). *Kadapa Distr.*: Highway 58, between Rajukunta and Rapur, 14°11'N 79°25'E, 289 m, 12.ii.2004, S. Basi, 1 juv. ♂ (AMCC [LP 13136]), 14°12'N 79°28'E, 148 m, 12.ii.2004, S. Basi, 1 juv. ♀ (AMNH). Karnataka: *Mandya Distr.*: Maddur [Madduru, 12°35'N 77°03'E], iii.2003, 1 ♀ (AMNH), 1 juv. ♂ (AMCC [LP 2275]), 1 juv. ♀ (AMCC [LP 2276]). *Mysore Distr.*: Mysore [Mysuru, 12°18'N 76°38'E], M. Seiter, 1 ♂, 1 ♀, 1 juv. (NHMW 28529).

Chersonesometrus wroughtoni (Pocock, 1899): INDIA: Karnataka: *Belgaum Distr.*: Belgaum [15°52'N 74°30'E], W.A. Talbot, lectotype ♀, paralectotypes: 3 ♀, 3 juv. ♂, 1 juv. ♀ (BMNH 1897.9.17.1-4.11.20 [BNHS 1163, 1165, 1168, 1169, 1171]). Maharashtra: *Kolhapur Distr.*: Gadingal [Gadhinglaj, 16°14'N 74°21'E], Kolhapur, Wray, paralectotype ♀ (BMNH 1897.9.5.2 [BNHS 1027]).

Deccanometrus bengalensis (C.L. Koch, 1841): CHINA: Tibet Autonomous Region: *Ngari Prefecture*: Tibet, south region of Pulan [Purang/Burang Xian, 30°15'N 81°10'E], low valley of Kongque He River, near border with Nepal, vii.1931, *Heterometrus tibetanus*: holotype ♂ (MNHN RS 8608), 1 ♂ paratype (MNHN RS 8726). INDIA: Assam: *Heterometrus (Chersonesometrus) fastigosus*: holotype ♂, 1 ♀, 1 subad. ♂ paratypes (SMF II/8886/228). Odisha: Rangamatia Village, near Baripada, 21°55'N 86°45'E, T. Kroes, 1 ♂, 1 ♀ (AMNH), 3 juv. (AMCC [LP 14238]). Uttarakhand: *Dehradun Distr.*: Dehra Dun [30°19'N 78°02'E], 1 ♂ (BMNH 1897.9.17.5 [BNHS 1236]), F. Gleadow, 1 ♂ (BMNH 1896.10.20.70 [BNHS 843]), Gamble, 2 ♂, 1 ♀ (BMNH 1896.2.2.1-2). *Pauri Garhwal Distr.*: Munala [Mundala, 29°48'N 78°31'E], A.N. Cassia, 1 juv. ♂, 1 juv. ♀ (BMNH 1896.9.26.85.86 [BNHS 764]). West Bengal: Bengal, 1 ♂, 1 ♀ (BMNH 1860.15); Bengal, Lam, *Buthus bengalensis* lectotype [= *Heterometrus (Chersonesometrus) fastigosus* paratype]: 1 ♂ (ZMB 70/2), paralectotypes [= *H. (C.) fastigosus* paratypes]: 1 ♂ (ZMB 70/1), 3 ♂ (ZMB 70/4, 5, 7), 1 ♀ (ZMB 70/3), 1 subad. ♀ (ZMB 70/6). *Kolkata Distr.*: Calcutta [Kolkata, 22°34'N 88°22'E], Barabank?, B.V. College, rec. P.K. Sina, 17.viii.1965, ex Stahnke coll., 1 ♀ (CAS 9071134 [ASU 60-271/75-0321]). *Paschim Medinipur Distr.*: Kharqpur [Kharagpur, 22°20'N 87°20'E], 30 m, 20.vi.1960, L. Murmu, in yard, 1 ♂ (CAS 9071140), 17.ii.1961, C.L. Kau, 1 ♀ (CAS 9071133).

Deccanometrus latimanus (Pocock, 1894): India, holotype ♀ (BMNH 1879.7.3.1 old 1859/88). PAKISTAN: Khyber Pakhtoon Khawa: *Boner Distr.*: Daggar Tehsil, Daggar, 34°31'N 72°28'E, 27.ix.2012, M. Shahid, M. Ahsan and A. Khan, 1 ♀ (AMNH), 1 subad. ♂ (AMCC [LP 11569]), x.2012, M. Azhar, 2 ♂, 4 ♀, 4 subad. ♂, 2 subad. ♀ (AMNH), 1 subad. ♀ (AMCC [LP 11601]); Boner Tehsil, Daggar, 3 km W of Rega village, 33°51'N 72°48'E, 2257 ft., 13.ix.2012, M. Ahsan, 1 ♂, 1 ♀ (AMNH). *Peshawar Distr.*: Peshawar Tehsil, Peshawar, 15 km W on road to Landi Kotal, 34°01'N 71°35'E, 1158 ft., 24.ix.2012, M. Shahid, M. Ahsan and A. Khan, 3 ♀ (AMNH), 1 subad. ♀ (AMCC [LP 11568]).

Deccanometrus liurus (Pocock, 1897): INDIA: Madhya Pradesh: *Gwalior Distr.*: Gwalior [26°13'N 78°11'E], Dane (Medical Officer of Makarajah Scindia), *Palamnaeus liurus*: lectotype ♀ [BNHS 995], paralectotypes: 2 ♀ [BNHS 992, 944], 1 subad. ♂ [BNHS 987], 4 juv. ♂ (BMNH 1896.12.15.20-25).

Deccanometrus obscurus (Couzijn, 1981): INDIA: Maharashtra: *Mumbai City Distr.*: Bombay [Mumbai, 18°59'N 72°50'E], 6.iii.1967, S.R. Sane, ex Stahnke coll., 4 ♂ (CAS 9070410 [ASU 67-351], 9070414 [ASU 67-317, 322], 9070415 [ASU 67-301]), 8 ♀ (CAS 9070408 [ASU 67-311], 9070412 [ASU 67-313], 9070413 [ASU 67-309, 315], 9070414 [ASU 67-302, 312], 9070415 [ASU 67-304, 306]). *Mumbai Suburban Distr.*: Sanjay Ghandi Park, Kanheri caves [19°12'N 72°54'E], Bombay [Mumbai], 24.v.2006, S. Huber, 1 ♀ (AMNH), 1 juv. ♂ (AMCC [LP 6225]). *Pune Distr.*: Bhimashankar, Path to Hanuman Temple, 19°04'N 73°32'E, 900 m, 1.xii.2006, P. Menon, 2 ♂, 2 ♀, 1 subad. ♂, 1 subad. ♀, 8 juv. ♂, 14 juv. ♀ (AMNH), 1 juv. ♂, 2 juv. ♀ (AMCC [LP 13974]); Bhimashankar, near Maa Anjani Temple, 19°04'N 73°32'E, 942–956 m, 30.xi.2009, R. Datta, 2 ♀ (AMNH), 1 juv. ♂ (AMCC [LP 16817]). *Raigad Distr.*: Matheran [18°59'N 73°16'E], near Bombay [Mumbai], xi.1896, M. Maindron, *Heterometrus (Chersonesometrus) scaber obscurus*: holotype ♀ (MNHN RS 0107). *Sindhudurg Distr.*: Amboli, ca. 300 m S, 15°58'N 74°00'E, 699 m, 26.xi.2009, R. Datta, 2 ♂, 1 ♀, 1 subad. ♀, 1 juv. ♂, 1 juv. ♀, 1 ♀ dextral pedipalp patella and chela (AMNH), 1 juv. ♂, 1 juv. ♀ (AMCC [LP 16800]).

Deccanometrus phipsoni (Pocock, 1893): INDIA: Madras [Madras Presidency], *Scorpio phipsoni*: lectotype ♂ (BMNH 1879.7.3.2-3). Andhra Pradesh: *Prakasam Distr.*: Podili [15°36'N 79°37'E], 16.i.1967, D.E. Johnson, ex Stahnke coll., 1 ♂ (CAS 9070409 [ASU 67-104]). Chhattisgarh: “Indes anglaises (Provinces centrales)” [Central Provinces of British India], 1915, G. Babault, 2 ♂, 6 ♀, 1 subad. ♂, 2 subad. ♀, 1 juv. ♂ (MNHN RS 0080), Chilpy, 19.iii.1914, G. Babault, 24 juv. (MNHN RS 0103); Rajadhar [21°43'N 82°00'E], 1915, G. Babault, 1 juv. ♀ (MNHN RS 0111). *Bilaspur Distr.*: Billespoor [Bilaspur, 22°05'N 82°08'E], northern India, iv.1871, M.M. Carleton, 1 ♀ (MCZ 88165). Odisha: *Koraput Distr.*: Pottangi [18°34'N 82°58'E], 13 mi. SE, 550 m, 4.ii.1962, E.S. Ross and D.Q. Cavanaugh, 1 subad. ♂ (CAS 9070417). Tamil Nadu: *Nilgiris Distr.*: Nilgiri Hills [11°25'N 76°30'E], W.M. Daly, *Palamnaeus phipsoni collinus*: holotype ♀, 1 ♀, 2 juv. ♂ paratypes (BMNH 1894.8.21.4-7). *Salem Distr.*: Shevaroy Hills [11°47'N 78°12'E], *Scorpio phipsoni*: 1 subad. ♂ paralectotype (BMNH 1879.7.3.2-3); Yercauda, Sheveroy Hills [11°47'N 78°12'E], J.R. Henderson, 1 ♀, 1 subad. ♂, 2 juv. ♂, 2 juv. ♀ (BMNH 1894.10.24.9-12); Yercaud [11°46'N 78°14'E], 4000 ft., 6.iii.1962, E.S. Ross and D.Q. Cavanaugh, 1 subad. ♂, 1 juv. ♂, 1 juv. ♀ (CAS 9070416).

Deccanometrus ubicki (Kovářík, 2004): INDIA: Puducherry: *Karikal Distr.*: Karikal [Karikal], Madras, viii.1962, P.S. Nathan, ex Stahnke coll., paratype ♀ (CAS 9071138 [ASU 63-1114]), same data, except: ix.1962, paratype ♀ (CAS 9071135 [ASU 63-1122]), same data, except: x.1962, 2 ♀ paratypes (CAS), same data, except: vi.1963, paratype ♂ (CAS 9081136 [ASU 63-1138]), same data, except: vii.1963, 3 ♀, 1 subad. ♀ paratypes (CAS 90711339 [ASU 63-1142/46/47]); Karikal territory, Kurumbargum [Kurumbagaram, 10°59'N 79°46'E] P.O. [Post Office], Tanjore Distr., i.1954, P.S. Nathan, ex Stahnke coll., paratype ♀ (CAS 9071137 [ASU 63-1114]). Karum, Kurum Bagaram [Kurumbagaram, 10°59'N 79°46'E], 7.ii.1947, P.S. Nathan, 1 subad. ♀ (AMNH).

Deccanometrus xanthopus (Pocock, 1897): INDIA: Maharashtra: *Pune Distr.*: Mayureshwar, ca. 2 km NW of Supe, 18°21'N 74°22'E, 644–646 m, 29.xi.2009, R. Datta, 1 ♂, 1 ♀ (AMNH), same data, except: 676 m, 1 juv. ♀ (AMCC [LP 16814]). *Satara Distr.*: Kadao Tal (Satara) [Khatav, 17°39'N 74°22'E], S Dekhan, Bombay [Bombay Presidency], A.D. Wilkins, lectotype ♂, paralectotypes: 1 ♀, 2 subad. ♂ [BNHS 707], 1 subad. ♀ [BNHS 703] (BMNH 1896.9.26.88-91 [BNHS 700, 703]); Khambatki Ghat, S of Khandala, on M.B.T. Road (Highway 47), 18°01'N 74°01'E, 928 m, 17.xi.2004, S. Basi, 1 ♀ (AMNH), 1 juv. ♂ (AMCC [LP 13126]), 937 m, 30.xi.2006, P. Menon, 3 ♀, 1 subad. ♀, 1 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 13971]), 902 m, 29.xi.2009, R. Datta, 1 ♂, 4 ♀, 1 juv. ♂ (AMNH), 1 juv. ♂ (AMCC [LP 16813]); Kadao Tal (Satara) [Khatav, 17°39'N

74°22'E], S Dekhan, Bombay [Bombay Presidency], A.D. Wilkins, *Palamnaeus xanthopus*: lectotype ♂, paralectotypes: 1 ♀, 2 subad. ♂ [BNHS 707], 1 subad. ♀ [BNHS 703] (BMNH 1896.9.26.88-91 [BNHS 700, 703]).

Gigantometrus swammerdami (Simon, 1872): 16.iv.1875, G.N. Westring, *Pandinus asper*: 1 subad. ♂ holotype (NMG 92). “Java”, Dupont, *Pandinus kochii*: 2 ♀ syntypes (ZMB 62). “India Orientalis” [East Indies], *Heterometrus swammerdami*: holotype ♀ (MNHN RS 0110 [Simon coll. 1500]), 1 subad. ♂, 1 subad. ♀, 1 juv. ♀ (ZMB 2284). INDIA: ex pet trade, 1 ♂ (AMNH), [leg] (AMCC 101700 [LP 1607]). Andhra Pradesh: Chittoor Distr.: Kothakota, on Chittoor–Tirupati Road 61, 13°23'N 79°06'E, 372 m, 30.xi.2004, S. Basi, 1 subad. ♂ (AMCC [LP 13128]); Puttur Chittoor [Puttūr, 12°46'N 75°13'E], E. Chell, xi.1952, 1 ♂ (USNM). Maharashtra: Balaghat Distr.: Baihar [22°06'N 80°33'E], 1.ii.1949, R.L. Fleming, 1 ♀ (FMNH 086 257); Mukki-Balagash [Mukki-Balaghat, 22°09'N 80°41'E], 27.viii.1957, G.A. Frhr. V. Maydell, 1 ♀ (ZMH). Tamil Nadu: Coimbatore Distr.: Coimbatore [11°00'N 76°58'E], W. Brady, *Palamnaeus swammerdami flavimanus*: lectotype ♀, paralectotype ♂ (BMNH 1899.11.2.292-3); Coimbatore [11°00'N 76°58'E], vi.1954, P.S. Nathan, ex Stahnke coll., 2 ♂ [ASU 1454.1, 2], 1 subad. ♂ [ASU 1454.5] (CAS 9070406), 2.vii.1954, P.S. Nathan, ex Stahnke coll., 1 ♂ (CAS 9070418 [ASU 1451]), viii.1957, P.S. Nathan, ex Stahnke coll., 1 ♂ [ASU 58-78], 1 ♀ [ASU 58-75] (CAS 9070402). Kanchipuram Distr.: Echankaranai, 12°41'N 80°02'E, 43 m, 2–3.xii.2004, S. Basi, 1 ♂, 2 subad. pedipalp chelae (AMNH), 1 subad. ♂ (AMCC [LP 13129]). Ramanthapuram Distr.: Ramnad [Ramanthapuram, 09°22'N 78°50'E], *Scorpio lucidipes*: holotype ♂ (MNHN RS 0112 [Simon coll. 1542]).

Gigantometrus titanicus (Couzijn, 1981): SRI LANKA: Ceylon, E.C. Fernando, 1 ♂ (FMNH 086 264). Central Prov.: Kandy Distr.: Hasalaka, SE of Rathna Ella Falls, 07°22'N 80°56'E, 200 m, 3.ii.2014, L. Prendini and P. Horsley, 1 ♂, 1 ♀ (AMNH), 1 juv. ♀ (AMCC [LP 12288]). North Central Prov.: Polonnaruwa Distr.: Giritale, roadside, 07°59'N 80°55'E, 173 m, 27–29.i.2014, L. Prendini and P. Horsley, 1 ♀ (AMNH); Polonnaruwa [07°56'N 81°00'E], 29.i.2014, local villagers, 1 ♂, 1 ♀, 2 subad. ♂, 4 subad. ♀, 2 juv. ♂, 3 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 12287]). Northern Prov.: Jaffna Distr.: Jaffnapatam [Jaffna, 09°40'N 80°02'E], Dr. A.E.S. Twynam, 1 ♀ (BMNH). North Western Prov.: Puttalam Distr.: Chilaw [07°34'N 79°48'E], near sea-level, 13.v.1965, C.M. Bogert, 1 ♂ (AMNH). Western Prov.: Colombo Distr.: Colombo [06°56'N 79°52'E], 1900, E. de la Croix, *Heterometrus (Gigantometrus) swammerdami titanicus*: holotype ♀ (MNHN RS 0084). Gampaha Distr.: Negombo [07°12'N 79°52'E], west coast, N of Colombo, 26.iv.1974, Dr. J. Balazuc, *H. (G.) s. titanicus*: paratype ♂ (MNHN RS 7759).

Heterometrus glaucus Thorell, 1876: “Bengal”, 7.viii.1868, G. Litman, *Palamnaeus costimanus* var. *γ glaucus*: lectotype ♀ (NMG 88). INDONESIA: Aceh Prov.: North Nias Reg.: Nias [Nias Island, 01°05'N 97°35'E], Sumatra, P. Beyer, 1 ♀ (SMF 5322); Nias, Dr. Schrieber, 1 ♂, 1 ♀ (1885.29), 1 ♂ (BMNH 1885.54); Pulo [Pulau] Nias Island, 24.iii.1903, W.L. Abbott, 1 ♀ (USNM 57-593); Nias Island, Alasa Village, 01°12'N 97°25'E, 136 m, 3.xi.2016, S.F. Loria and M. Roppo, 3 ♀, 1 subad. ♀ (AMNH), 1 juv. ♂, 1 juv. ♀ (AMCC [LP 13922]); Afulu Retreat, 01°15'N 97°14'E, ca. 18 m, 2.xi.2016, S.F. Loria and M. Roppo, 1 juv. ♂, 1 juv. ♀ (AMCC [LP 13928]). Sabang City Reg.: Poeloe Weh [Pulau Weh, 05°49'N 95°18'E], 1919, P. Buitendijk, *Heterometrus (H.) liophysa separatus*: holotype ♂, paratype ♀ (RMNH 185). Simeulue Reg.: Poeloe Babi [Babi Island, 02°05'N 96°39'E], W coast of Sumatra, ii.1908, P. Buitendijk, *Heterometrus (H.) liophysa spartanicus*: 1 subad. ♂ holotype (RMNH 187). East Java Prov.: Madoera [Madura, 07°00'S 113°20'E], v.1910, P. Buitendijk, *Heterometrus liophysa* var. *madoerensis*: 1 juv. ♀ holotype (RMNH 188). West Sumatra Prov.: Mentawai Islands Reg.: Mentawai Islands, Siberut [Siberot Island, 01°20'S 98°55'E],

Heterometrus laevifrons: 1 juv. ♂ holotype, 1 juv. ♀ paratype (SMF 8883/225). Padang Pariaman Reg.: Padang, Sumatra, J. van den Hoeven *Heterometrus (H.) liophysa*: neotype ♂ (RMNH 78).

Heterometrus laevigatus (Thorell, 1876): “Nova Hollandia [Australia], Melbourne”, 11.ix.1860, G.E. Bothen, *Palamnaeus laevigatus*: 1 subad. ♀ holotype (NMG 90). THAILAND: Chumphon Prov.: Phato Distr.: Khao Kai Jae Waterfall, along main road/trail leading towards waterfall, 09°55'N 98°57'E, 26.iv.2014, S.F. Loria and P. Horsley, 1 ♀ (AMNH), 1 juv. ♂ (AMCC [LP 12337]). Krabi Prov.: Krabi Distr.: Khao Phanom Bencha, around bungalows, 08°14'N 98°55'E, 126 m, 25–26.xi.2014, S.F. Loria and M.M. Locke, 1 juv. ♀ (AMCC [LP 13048]). Phattalung Prov.: Si Banphot Distr.: Khao Pu-Khao Ya, 07°41'N 99°52'E, 186 m, 24.xi.2014, S.F. Loria and M.M. Locke, 3 ♀, 1 subad. ♂, 1 juv. ♂ (AMNH), 1 juv. ♀ (AMCC [LP 13047]). Phetchaburi Prov.: Kaeng Krachan Distr.: Kaeng Krachan, around lake, near road, 12°53'N 99°38'E, 100 m, 24.iv.2014, S.F. Loria and P. Horsley, 3 ♂, 2 juv. ♂, 2 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 12335]). Satun Prov.: Khuan Don Distr.: Thale Ban, along road and trails, 06°43'N 100°10'E, 163 m, 23.xi.2014, S.F. Loria and M.M. Locke, 1 ♂, 1 ♀, 1 subad. ♂, 1 subad. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 13045]). Myanmar [alleged], ex pet trade, 2002, 1 ♀ (AMNH [LP 1789]), 1 juv. ♂ (AMCC [LP 1789]). Surat Thani Prov.: Ko Samui Distr.: Koh Samui, Na Mueang-1 Waterfall, around grounds of waterfall, 09°28'N 99°59'E, 30 m, 15.xii.2018, S.F. Loria, 2 ♂, 1 ♀ (AMNH).

Heterometrus laoticus Couzijn, 1981: CAMBODIA: Siem Reap Prov.: Bakorng Distr.: Kantrieng Commune, Tatrow Village, 13°20'N 104°02'E, > 10 m, vii.1999, B.L. Stuart, 1 ♀, 1 subad. ♀, 1 juv. ♂ (FMNH 086 259 [408]). Siem Reap, Angkor Wat, 13°24'N 103°52'E, 36 m, 15.vii.2017, P. Jäger, 1 ♂ (SMF). LAOS: 1884–1899, Neis, holotype ♂, paratype ♀ (MNHN RS 0030). Khammouane Prov.: Boualapha Distr.: between Boualapha [17°19'N 105°43'E] and Ban Nong Ping, 3.v.2012, P. Jäger, 1 ♀ (SMF), 1 ♀ (AMCC [LP 11330]). Gnommalath Distr.: Tham Nam Lot (Lod) near Ban Naden village, 17°30'N 105°23'E, 260 m, 13–14.vi.2012, L. Prendini, S.F. Loria, and P. Kanyavong, 3 ♂, 2 subad. ♂, 1 subad. ♀, 2 juv. ♀, two pedipalp chelae (AMNH), 1 juv. ♀ (AMCC [LP 11352]), 19.v.2018, S.F. Loria, L. Ngo, S.T. Kangnavong and P.T. Thon Thondet, 1 ♂ (AMNH). Thakhek Distr.: Thakhek, 17°24'N 104°48'E, 15.vi.2012, P. Kanyavong, in garden, 1 subad. ♂ (AMNH); Thakek [17°24'N 104°50'E], 9–13 road km E, 25–27.iv.2012, P. Jäger, 1 subad. ♂ (SMF), 1 subad. ♂ (AMCC [LP 11331]). Luang Prabang Prov.: Pak Seng Distr.: Pak-Xeng town, on Pak Xeng–Sam Soun Road, 20°10'N 102°39'E, 13.v.2018, S.F. Loria, L. Ngo and S.T. Kangnavong, 1 ♂, 1 ♀ (AMNH), 1 ♀ pedipalp chela (AMCC [LP 15584]). THAILAND: Kanchanaburi Prov.: Sai Yok Distr.: Sai Yok, 14°26'N 98°51'E, 88 m, 22.iv.2014, S.F. Loria and P. Horsley, 1 ♀, 1 juv. ♂ (AMNH), 1 ♀, 1 juv. ♀ (AMCC [LP 12336]). Ratchaburi Prov.: Phueng Distr.: Suan Phueng Nature Education Park, 13°31'N 99°14'E, 21.xi.2018, 5th ASA Congress Attendees, 1 juv. ♂ (AMCC [LP 16489]). Ubon Ratchathani Prov.: Khong Chiam Distr.: Kaeng Tana, trail to Pra Cave and Phaphueng Rock Garden, 15°18'N 105°29'E, 112 m, 8.xii.2014, S.F. Loria and M.M. Locke, 1 ♂ (AMNH), [tissue] (AMCC [LP 13046]).

Heterometrus longimanus (Herbst, 1800): East Indies, 26.x.1843, J. Hall, *Palamnaeus angustimanus*: holotype ♂ (NMG 87). “Bengal”, 7.viii.1868, G. Litman, *Palamnaeus costimanus* var. *γ glaucus*: paralectotype ♀ (NMG 88). INDONESIA: Bangka-Belitung Islands Prov.: Bangka [Pulau Bangka, 02°15'S 106°00'E], Vosmaer and Budding, *Heterometrus (H.) longimanus belitungensis*: holotype ♂ (RMNH 786), paratype ♀ (RMNH 86). Riau Prov.: Riouw [Riau], A.H.G. Blokzeijl, *Heterometrus (H.) longimanus paris*: holotype ♂, paratype ♀ (RMNH 83). Bengkulu Reg.: Bengkulu [Bengkalis Island, 01°27'N 102°24'E], M. Maindron, *Heterometrus (H.) longimanus bengkalitensis*: holotype ♂, paratype ♀ (MNHN RS 0048). West Sumatra Prov.: Padang Panjang Reg.: Ajer Mantior [Ajermantsjur/Ajer Mantjoer, near Padang Panjang, 00°28'S 100°24'E], Sumatra, viii.1878, O. Beccari,

Palamnaeus liophysa: syntypes: 1 juv. ♂, 1 juv. ♀ (MCSNG). Solok Reg.: Gunung Talang [00°59'S 100°41'E], 1000 m, 1926, V. Duivenbode and E. Jacobson, *Heterometrus* (*H.*) *longimanus*: neotype ♂ (RMNH 291). MALAYSIA: Sabah: Sandakan Div.: Kinabatangan Distr., Kampung Bilit, Myne Resort, near Kinabatangan River, 05°29'N 118°14'E, 20–66 m, 22–23.vii.2013, S.F. Loria and J. Huff, 1 ♂, 1 subad. ♂, 1 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 12114]); Kinabatangan Distr., northern Borneo, SE end of Dewhurst Bay [05°35'N 118°35'E], 6.v–15.vi.1950, R.F. Inger and D.D. Davis, 1 ♀ (FMNH 086 267). Sarawak: Sarawak, Borneo, Doria and Beccari, *Palamnaeus costimanus* var. *β borneensis*: 1 ♂, 1 ♀, 1 juv. ♂ syntypes (MCSNG). Kuching Div.: Kubah National Park, Matang, 4.2 km from front gate, 01°36'N 110°12'E, 657 m, 5–6.viii.2013, J. Huff and A. Ang, 1 ♂ (AMNH), 1 juv. ♀ (AMCC [LP 12116]); Kuching, 8 Jalan Ong Kwan Hin, 01°33'N 110°21'E, 29 m, 6.viii.2013, J. Huff and A. Ang, 1 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 12139]); Kuching, in overgrown residential area between Taman Bunga Teratai and inner recesses of Lintang Park, within central city area, 01°32'N 110°21'E, 1–20.viii.2010, A. Ang, 5 ♂, 5 ♀ (AMNH); residential property off Jalan Mendu, 2.5 km from Kuching City Centre, 01°33'N 110°22'E, 27.xi.2010, A. Ang, 5 ♂, 5 ♀ (AMNH). Miri Div.: Gunung Mulu National Park, trails around park headquarters, 04°03'N 114°49'E, 58 m, 6.v.2014, S.F. Loria and P. Horsley, 1 ♀, 1 subad. ♂, 1 subad. ♀, 1 juv. ♂ (AMNH), trails near research headquarters, 04°03'N 114°49'E, 40 m, 29.vii–5.viii.2013, J. Huff and A. Ang, 1 ♂, 1 subad. ♀, 2 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 12115]). Selangor: Kuala Selangor Distr.: Bukit Malawati, Kuala Selangor, 03°20'N 101°15'E, 8–12 m, 9.vi.2013, L. Prendini and S.F. Loria, 1 ♀, 1 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 11948]). PHILIPPINES: National Capital Region: Manille [Manila, 14°36'N 120°59'E], Laglaise, M.N., *Pandinus humilis*: 3 juv. ♀ syntypes (MNHN RS 0021 [Simon coll. 2499]). Palawan Prov.: Balabac, Dalawan Bay [07°54'N 117°05'E], 9.x.1961, Noona Danish Expedition 1961–1962, *Heterometrus* (*H.*) *longimanus marmoratus*: holotype ♂, paratype ♀ (ZMUC 24639). Tawi Tawi Prov.: Tarawakan [05°09'N 119°53'E], Tawi Tawi Island, Philippines (Sulu Islands), 7.xi.1961, Noona Danish Expedition 1961–1962, *Heterometrus* (*H.*) *longimanus tarawakanensis*: holotype ♀, 1 subad. ♂ paratype (ZMUC 24641).

Heterometrus petersii (Thorell, 1876): iii.2008, ex pet trade, 1 subad. ♂ (AMNH [LP 8269]), pedipalp chela (AMCC [LP 8269]). MALAYSIA: Penang: Penang [Penang Island, 05°24'N 100°14'E], i.1862, M.D. Parker, 1 ♂ (MCZ); Pulo Penang [Pulau Penang (Penang Island)], vii.1860, W.H.A. Putnam, 1 ♂, 1 ♀ (MCZ); Penang Island, Bukit Bendera/Penang Hill, Moniot Road, 05°25'N 100°15'E, and grounds of Monkey Cup Cafe, 05°25'N 100°16'E, 16–27.x.2017, L. Esposito and S.F. Loria, 3 ♂, 2 ♀ (CAS 9080783), 1 ♂ [leg] (AMCC [LP 16458]), 1 ♀ [leg] (AMCC [LP 16459]). SINGAPORE: “India Orientalis, ex Singapore”, 22.iv.1852, J.F. Lijbeck, *Palamnaeus petersii*: holotype ♀ (NMG 91). Singapore [01°22'N 103°48'E], 1889, Dr. Hamlin, 1 ♂ (MCZ 14970), 1921, 1 subad. ♂ (LKC ZRC-ENT 7964), xii.1921, 1 subad. ♂ (LKC ZRC-ENT 7960), 1 subad. ♀ (LKC ZRC-ENT 7963), xi.2004, captive bred, 1 subad. ♀, 2 juv. ♀ (LKC ZRC-ARA 677); 8th Mile Old Upper Thomson Road/6th Mile Thomson Road [01°23'N 103°49'E], 16–17.ii.1963, K.O. Yee, 1 juv. ♀ (LKC ZRC-ENT 7229 [ZRC-ARA 731]); Gardens [Singapore Botanic Gardens, 01°19'N 103°49'E], 7.viii.1921, 1 ♂ (LKC ZRC-ENT 7961), xii.1921, 1 ♂ (LKC ZRC-ENT 7962), Botanical Gardens, 13.iv.1960, S.C. Keong, 1 ♀ (LKC ZRC-ARA 675), Botanic Gardens, National Orchid Garden, 2010, K. Lin, 1 ♂ (LKC ZRC-ENT 7239); Bukit Kallang, track near [01°22'N 103°49'E], 23.v.1994, C.H. Yeow and J. Choo P.S., 1 subad. ♀ (LKC ZRC-ARA 676); [probably Mandai, 01°25'N 103°47'E], ix.1998, C.K. Wee, 1 ♀ (AMNH [LP 1604]), legs (AMCC 101699 [LP 1604]); Mandai [01°25'N 103°47'E], 1998, C.K. Wee, 1 juv. ♀ (LKC ZRC-ARA 679); SADA Hill [Singapore Air Defence Artillery, 01°26'N 103°50'E], 1 ♂ (LKC ZRC-ENT 7965).

Heterometrus silenius (Simon, 1884): CAMBODIA: Kep Prov.: Kep Distr.: Kep National Park, 10°29'N 104°18'E, 72 m, 24.v.2018, S.F. Loria, T. Chea, and V. Khem, 2 juv. ♀ (AMCC [LP 15590]). PHILIPPINES: Mindanao, Mantano and Rey, *Heterometrus* (*H.*) *petersii mindanaensis*: holotype ♂ (MNHN RS 0023). VIETNAM: Cochinchina, Germain, *Palamnaeus silenius*: lectotype ♂ (MNHN RS 0020), paralectotype ♂ (MNHN RS 8903 ex 0020). Vietnam, x.1997, 1 ♂ (AMCC [LP 1602]), 1 ♀ (AMCC 101697 [LP 1603]), viii.1999, ex pet trade, 1 ♂ (AMNH/AMCC [LP 1888]). Bin Thuan Prov.: Ham Thuan Nam Distr.: Ta Kou Mountain Nature Reserve, Ta Kou Mountain (Nui Ta Kou), trail above pagoda, 10°49'N 107°54'E, 540 m, 22–23.vi.2012, L. Prendini and S.F. Loria, 1 ♂ (AMNH), trail below guesthouse and upper cable station, 10°49'N 107°54'E, 383 m, 22–23.vi.2012, L. Prendini and S.F. Loria, 1 ♂, 2 ♀, 3 juv. ♂, 7 juv. ♀, 1 ♀ pedipalp chela (AMNH), 1 juv. ♀ (AMCC [LP 11353]), 10°49'N 107°53'E, 356 m, 23.vi.2012, L. Prendini and S.F. Loria, 1 ♂, 2 ♀, 1 subad. ♂, 2 juv. ♀ (AMNH). Dong Nai Prov.: Din Quan Distr.: Din Quan [11°09'N 107°16'E], Bienhoa, 18.xii.1932, M. Poilane, 1 ♀ (USNM); Dinh Quan, outskirts of town, 11°16'N 107°25'E, 139 m, 16.ii.2012, L. Prendini and D.P. Phu, 1 ♂, 1 ♀, 1 subad. ♂, 2 juv. ♂, 2 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 11252]). Tan Phu Distr.: Nam Cat Tien National Park, forest trail to Bau Sau [11°28'N 107°20'E], 6–8.v.2001, H.H. Tan et al., 1 ♂ (LKC ZRC-ARA 652); Cat Tien, camp and trail to Big Tree, 11°25'N 107°26'E, 135 m, 16.ii.2012, L. Prendini, 4 ♂, 3 ♀, 1 subad. ♂, 1 subad. ♀, 3 juv. ♂ (AMNH); Kat Tien National Park, *Lagerstroemia* site, 11°26'N 107°25'E, 7.xii.2011, A. Goncharov, 4 juv. ♂ (AMNH, AMCC [LP 11243, 11246]), 8 juv. ♀ (AMNH, AMCC [LP 11242, 11244, 11245, 11247]), *Azelia* site, 11°26'N 107°26'E, 24.xi.2011, A. Goncharov, 2 juv. ♂ (AMNH, AMCC [LP 11250]), 2 juv. ♀ (AMNH, AMCC [LP 11249]), 2 juv. (AMNH, AMCC [LP 11248]).

Heterometrus spinifer (Ehrenberg, 1828): MALAYSIA: Johor: Ledang Distr.: Taman Hutan Lagenda Gunung Ledang (Gunung Ledang Recreational Park), trail around camp, 02°20'N 102°37'E, 43 m, 19.vi.2013, L. Prendini and S.F. Loria, 1 ♂, 2 ♀ (AMNH). Mersing Distr.: Endau-Rompin National Park: Chalet area, 02°32'N 103°25'E, 53 m, 29.vi.2010, M.S. Harvey, K.L. Edward, R. Hashim, and S. Dzarawi, 1 subad. ♂ (WAM Ta209), [leg] (AMCC [LP 10581]); Kuala Jasin camp, trail to Janing Barat Plateau, 02°32'N 103°22'E, 37 m, 16–17.vi.2013, L. Prendini and S.F. Loria, 1 ♀ (AMNH). Kedah: Sult. Kadeh [Kedah], Malacca, 1900, Jhr. A. van de Does de Bijle/Bye, *Heterometrus* (*H.*) *spinifer*: neotype ♂ (RMNH 290). Pahang: Raub Distr.: The ‘Gap’ [03°42'N 101°45'E], Fraser’s Hill, R. West, 10.iv.2001, 1 ♂ (AMNH), [leg] (AMCC [LP 1906]). Rompin Distr.: Endau-Rompin National Park, Sungai Kinchin Base Camp, 02°37'N 103°20'E, 38 m, 18.vi.2013, L. Prendini and S.F. Loria, 1 juv. ♂, 1 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 11950]); Pulau Tioman (Tioman Island): trail between Kampung Genting and Kampung Paya, 02°46'N 104°07'E, 35–41 m, 11.vi.2013, L. Prendini and S.F. Loria, 1 ♀, 1 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 11920]); trail from Air Batang to Salang via Monkey Bay, 02°51'N 104°09'E, 48 m, 13.vi.2013, L. Prendini and S.F. Loria, 1 juv. ♀ (AMCC [LP 11974]); trail from Juara to Tekek, between 02°49'N 104°10'E and 02°49'N 104°11'E, 301–305 m, 15.vi.2013, L. Prendini and S.F. Loria, 1 juv. ♂ (AMNH). Perak: Batang Padang Distr.: Kampung Batu Tujoh [04°15'N 101°19'E], Cameron Highlands, vi.2013, ex Penang Butterfly Farm, 11 ♂, 9 ♀ (AMNH). Selangor: Gombak Distr.: Taman Negeri Selangor (Selangor State Park), 03°18'N 101°42'E, 1314 m, 7.vi.2013, L. Prendini and S.F. Loria, 1 subad. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 11949]).

Heterometrus thorellii (Pocock, 1892): MYANMAR: Irouaddy [Irrawaddy], Vossain, *Heterometrus* (*H.*) *bengalensis*: neotype ♂ (MNHN RS 0045). Bago Reg.: Taungoo Distr.: Bago (East) Region, Pho Kyar Elephant Camp, Yedasho, on trails near bungalows, 19°18'N 96°10'E, 127 m, 20.ix.2014, S.F. Loria and M.M. Locke, 2 ♀, 1 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 13049]). Kayin Reg.:

Hpa-An Distr.: Hpa-An Township, ca. 10 mi. E of Hpa-An Kalauk Htaung Village [16°53'N 97°39'E], v.2018, H.T. Bone, 3 ♂ (AMNH), 1 juv. ♀ (AMCC [LP 15604]). Mandalay Reg.: *Mandalay Distr.*: Mandalay, 22°01'N 96°07'E, 20.ii.2014, M. Roppo, 1 ♂ (AMNH), 1 subad. ♀ (AMCC [LP 13155]). *Myingyan Distr.*: Mt. Popa, Popa Mountain Resort, 20°55'N 95°13'E, 777–815 m 16–17.xi.2014, S.F. Loria and M.M. Locke, 5 ♀, 3 subad. ♂, 3 subad. ♀, 2 juv. ♂, 1 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 13051]). *Nyaung-U Distr.*: Kyauk Kuumin Pagoda, near Nyaung-U, 21°12'N 94°56'E, 58 m, 18.xi.2014, S.F. Loria and M.M. Locke, 1 subad. ♂, 12 juv. ♂, 9 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 13052]); Nyaung-U, on dirt road towards Kyauk Kuumin Pagoda, 21°12'N 94°56'E, 81–101 m, 18.xi.2014, S.F. Loria and M.M. Locke, 1 ♀, 4 juv. ♂, 4 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 13153]). Yangon Reg.: *Yangon North Distr.*: Hlawga National Park, 23 km NNW Yangon, 17°03'N 96°06'E, 16–20.ix.2003, C. Griswold, 1 ♂ (CAS 9019143), [leg] (AMCC [LP 2704]); Hlawga National Park, 17°03'N 96°07'E, 55 m, 22.xi.2014, S.F. Loria and M.M. Locke, 2 ♀, 1 subad. ♂, 3 juv. ♂, 4 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 13053]); Yangon, plantation on outskirts, 17°12'N 96°10'E, 56 m, 22.xi.2014, S.F. Loria and M.M. Locke, 1 ♀, 1 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 13050]).

Javanimetrus cyaneus (C.L. Koch, 1836): INDONESIA: Java, 4 ♀ (AMNH [24849]), xii.2002, 1 ♂ (AMNH), [leg] (AMCC [LP 2309]), iii.2008, ex pet trade, 1 ♀ (AMNH), [pedipalp] (AMCC [LP 8270]). Banten Prov.: *Pandeglang Reg.*: Ujung Kulon National Park: Ujung Jaya, Cibiuk TNUK, 06°47'S 105°31'E, 80–103 m, 17.vii.2013, Loria and Sarino, 1 juv. ♀ (AMCC [LP 12117]); Kertamukti, Mountain Honje, 06°43'S 105°34'E, 74 m, 18.vii.2013, Loria, Sarino and Oman, 2 ♂, 3 ♀, 1 subad. ♀, 3 juv. ♂, 7 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 12113]). East Java Prov.: *Banjuwangi Reg.*: Kalie Baroe/Kali Baru [Kalibaru, 08°16'S 113°59'E], Banjoewangi/Banjuwangi, Java, viii.1902, M.A.J. Fokker, *Heterometrus* (*Javanimetrus*) *c. cyaneus*: neotype ♂ (RMNH 139). Central Madura [07°00'S 113°20'E], x.1903, G.J.A. Steen, *H. (J.) c. insulanus*: holotype ♂, paratype ♀ (RMNH 272). West Java Prov.: *Bogor City Reg.*: Buitenzorg [Bogor, 06°36'S 106°48'E], v.1921, 2 ♂ (AMNH). *Cianjur Reg.*: Ciseureuh, 06°42'S 107°02'E, 1115 m, 4.vii.2013, S.F. Loria, 1 subad. ♂, 1 subad. ♀, 1 juv. ♂ (AMNH), 1 juv. ♀ (AMCC [LP 12112]). West Sumatra Prov.: Padang [00°57'S 100°25'E], Sumatra, Muller, *H. (J.) c. sumatrensis*: holotype ♂, paratype ♀ (RMNH 267). PHILIPPINES: Laguna Prov.: *First Distr.*, *Los Baños Municipality*: Mt. Makiling [14°08'N 121°12'E], Luzon, *Heterometrus* (*H. petersii luzonensis*: holotype ♀, paratype ♀ (SMF RII/8882/24).

Sahyadrimetrus barberi (Pocock, 1900): INDIA: Tamil Nadu: *Tirunelveli Distr.*: Kannikatti [ca. 08°40'N 77°17'E], Tinnevely, evergreen forest in Western Ghats, 2500 ft., C.A. Barber, holotype ♀ (BMNH 1899.9.24.1).

Sahyadrimetrus kanarensis (Pocock, 1900): INDIA: Goa: *Canacona Distr.*: Fort Cabo de Rama [Cabo de Rama, 15°05'N 73°55'E], above coast line, ca. 30 m, 14.i.2003, S. Huber, 1 ♂ (AMCC [LP 2902]), 1 juv. ♂ (AMCC [LP 2289]). *South Goa Distr.*: Agonda, SE on Butterfly Beach Road, 15°02'N 74°00'E, 37 m, 25.xi.2009, R. Datta, 1 subad. ♂, 1 juv. ♂ (AMNH), 1 juv. ♂ (AMCC [LP 16797]); Chaudi, hills N of town, 15°01'N 74°03'E, 10 m, 28.xi.2006, P. Menon, 2 ♂, 1 ♀, 1 subad. ♂, 3 subad. ♀, 3 juv. ♂, 9 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 13960]), 4.xi.2009, R. Datta, 7 ♂, 2 ♀, 10 subad. ♂, 4 subad. ♀, 8 juv. ♂, 10 juv. ♀, 1 ♂ sinistral pedipalp chela, 1 ♀ sinistral and dextral pedipalp chelae (AMNH), 3 juv. ♀ (AMCC [LP 16803]), Cortigão Canacoa [15°01'N 74°01'E], 16.ii.1997, J. Nielsen, 1 juv. ♂ (NRS [JF 53]); Cotigão W.L.S., 14°59'N 74°07'E, 46 m, 25.xi.2009, R. Datta, 2 ♀, 1 subad. ♀, 2 juv. ♂, 1 juv. ♀, 2 ♂ sinistral pedipalp chelae (AMNH), 2 juv. ♂, 1 juv. ♀ (AMCC [LP 16795]); Cotigão, near turnoff to Tirwal, 14°59'N 74°08'E, 98 m, 28.xi.2006, P. Menon, 1 subad. ♀ (AMCC [LP 14029]); Mangal roadside, 15°04'N 74°11'E, 108 m, 27.xi.2006, P. Menon, 1 ♀, 1 subad. ♂, 2 subad. ♀, 2 juv. ♂ (AMNH), 1 juv. ♀ (AMCC [LP 13963]). Karnataka: Kanara, Bombay [Bombay Presidency], T.R.D. Bell,

Palamnaeus scaber kanarensis: lectotype ♂, 2 ♀ paralectotypes (BMNH 1897.11.5.29–30 ex BNHS).

Sahyadrimetrus mathewi Prendini and Loria, 2020: INDIA: Kerala: *Ernakulam Distr.*: near Thattekad, 10°08'N 76°40'E, 39 m, 14.xi.2009, R. Datta, paratypes: 1 ♂, 1 ♀ (AMNH), 1 juv. ♀ (AMCC [LP 16752]), same data, except: 10°08'N 76°41'E, 22–40 m, 15.xi.2009, paratypes: 1 ♂, 1 ♀, 1 subad. ♂, 1 juv. ♂, 1 ♀ sinistral and dextral pedipalp patella and chela (AMNH), 1 juv. ♂, 1 juv. ♀ (AMCC [LP 16759]). *Thiruvananthapuram Distr.*: Kallar–Meenmutty Falls Road, SE of Kallar, 08°43'N 77°08'E, 157 m, 20.xi.2004, S. Basi, Vamanapuram River valley, paratypes: 2 ♂, 5 ♀, 1 subad. ♂, 11 juv. ♂, 10 juv. ♀ (AMNH), 2 juv. ♂, 1 juv. ♀ (AMCC [LP 13130]); Trivandrum [Thiruvananthapuram, 08°31'N 76°56'E], 4.iv.1968, A.P. Mathew, ex Stahnke coll., paratype ♂ (CAS [ASU 68–305]), paratype ♀ (CAS [ASU 68–303]). *Thrissur Distr.*: Attoor, Asurankundu Dam, 10°41'N 76°18'E, 118 m, 25.xi.2004, S. Basi, paratypes: 2 ♂, 2 ♀, 1 juv. ♂, 1 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 13131]); Poomala, Pathazhakundu Dam Road, W side Pathazhakundu Dam, 10°37'N 76°14'E, 136 m, 25.xi.2004, S. Basi, holotype ♂ (AMNH), paratypes: 2 ♂, 2 ♀, 3 subad. ♂, 1 subad. ♀, 2 juv. ♂, 1 juv. ♀ (AMNH), 1 subad. ♀ (AMCC [LP 13132]).

Sahyadrimetrus rugosus (Coutz, 1981): INDIA: Malabar Coast, E. Pougnet, *Heterometrus* (*Chersonesometrus*) *scaber rugosus*: holotype ♀ (MNHN RS 0082 [Simon coll. 17340]). Karnataka: *Kodagu Distr.*: Nalknad Palace Rd, Yevakapadi, 12°14'N 75°38'E, 1115 m, 20.xi.2009, R. Datta, 1 ♀ (AMNH). Kerala: *Wayanad Distr.*: Chaliyar River, above Meenmutty Waterfalls, 11°32'N 76°14'E, 845 m, 24.xi.2006, P. Menon, 1 ♀, 8 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 14007]); Kunnathidavaka, 11°31'N 76°03'E, 838 m, 25.xi.2006, P. Menon, 1 ♀, 1 subad. ♀ (AMNH), 1 subad. ♂ (AMCC [LP 14028]), 893 m, 18–19.xi.2009, R. Datta, 2 ♀, 1 subad. ♂, 3 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 16770]). Pookode Lake, Vythiri, 11°33'N 76°02'E, 748 m, 19.xi.2009, R. Datta, 1 subad. ♂ (AMCC [LP 16768]). Tamil Nadu: *Nilgiris Distr.*: Devala [11°28'N 76°23'E], S India, v.1961, P. Susai Nathan, ex Stahnke coll., 5 ♀ (CAS 9070419 [ASU 61–243], 9070421 [ASU 61–242, 244, 284, 286]).

Sahyadrimetrus scaber (Thorell, 1876): INDIA: Goa: M. Rousseau, 1 ♂ (MNHN RS 0100). Karnataka: *Belagavi Distr.*: Kalsa Waterfall, between Surla and Kankumbi, 15°41'N 74°11'E, 100 m, 28.xi.2006, P. Menon, 1 ♀, 5 juv. ♂, 1 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 14023]). *Dakshina Kannada Distr.*: Kalbale, Virajpet–Byndoor Rd, Guddadka, 12°38'N 75°18'E, 259 m, 20.xi.2009, R. Datta, 1 juv. ♂ (AMCC [LP 16771]). *Shimoga Distr.*: Agumbe, 13°31'N 75°05'E, 629 m, 20.xi.2009, R. Datta, 2 ♀ (AMNH); Jogi Gundi Falls, 13°29'N 75°07'E, 662 m, 21.xi.2009, R. Datta, 1 juv. ♀ (AMCC [LP 16772]). *Udupi Distr.*: Someshwara, Nadpalu, intersection Sitha River with road to Koodlu Theertha Waterfalls, 13°26'N 75°06'E, 135 m, 27.xi.2006, P. Menon, 1 ♂ (AMNH); Someshwara, Sitha River, Seethanadi, 13°29'N 75°01'E, 68 m, 26.xi.2006, P. Menon, 1 ♀ (AMNH), 1 juv. ♀ (AMCC [LP 13979]). Puducherry: *Mahé Distr.*: Malabar, Dussumier, 1 ♂ (MNHN RS 0095); Mahé [11°42'N 75°32'E], *Heterometrus* (*Chersonesometrus*) *s. scaber*: neotype ♀ (MNHN RS 0098); Mahé, Malabar [11°42'N 75°32'E], 1903, M. Maingron, 1 ♀ (MNHN RS 3264).

Srilankametrus tikaderi Prendini and Loria, 2020: INDIA: Tamil Nadu: *Coimbatore Distr.*: Attakatti [10°27'N 76°59'E], 9.vii.1945, Dr. K. Lindberg, holotype ♂, 2 ♀, 6 juv. ♂ paratypes (MNHN RS 3265), 777–1012 m, 17.xi.2009, R. Datta, paratypes: 1 ♂, 1 ♀, 1 subad. ♂, 1 ♀ sinistral pedipalp chela (AMNH), 1 juv. ♀ (AMCC [LP 16766]); Attakatti, ca. 650 m S, 10°26'N 76°59'E, 962 m, 17–18.xi.2009, R. Datta, paratypes: 3 ♂, 3 ♀, 2 juv. ♂, 2 juv. ♀ (AMNH), 2 juv. ♂ (AMCC [LP 16764]), 3 juv. ♂, 7 juv. ♀ (AMCC [LP 16767]). *Dindigul Distr.*: MelPallam, ca. 9 km N on Palani Ghat/Perumal Malai/Palani–Kodaikanal Road, 10°21'N 77°34'E, 1133 m, 17.xi.2009, R. Datta, 1 subad. ♂ paratype (AMCC [LP 16760]); Uslampatti (route a Tandikoudi, Mts. Palni) [Palni Mts., route from Uslampatti to Thandikudi, 10°19'N 77°39'E], 14.viii.1945, Dr. K. Lindberg, 1 ♂, 1

subad. ♂ paratypes (MNHN RS 3266); Monts Palni, pente au dessus de la riviere Tanaikadu [Palni Mts., slope above Tanaikadu River, Pannaikadu, 10°17'N 77°38'E], 1200 m, viii.1943, Dr. K. Lindberg, 3 juv. ♂ paratypes (MNHN RS 3142).

Srilankametrus caesar (C.L. Koch, 1841): 1 ♀, 1 subad. ♀, 2 ♂, 1 juv. ♀ (ZMB 7244). “Amerique?”, 1 ♀ (MNHN RS 0120). “India?”, 1 ♂ (ZMB 2283), 1 ♀ (MNHN RS 0149 [Simon coll. 8212]). INDIA: 1 ♀ (BMNH). “India Orientalis” [East India], *Buthus caesar*: lectotype: 1 subad. ♀ (ZMB 63), Ind. Or. [India Orientalis], “Ostindien” [East India], 2 ♂ (ZMB 2282). Puducherry: *Puducherry Distr.*: Pondichéry [Pondicherry/Puduchcheri/Puducherry, 11°59'N 79°49'E], viii.1901, M. Maindron, *Heterometrus (Chersonesometrus) granulomanus*: holotype ♂, paratype ♀ (MNHN RS 0087), 1 ♂, 1 ♀ (MNHN RS 0079), 1 ♀, 1 subad. ♂, 2 subad. ♀, 1 juv. ♀ (MNHN RS 0087), 5 ♂, 10 ♀, 3 subad. ♂, 7 subad. ♀, 3 juv. ♀ (MNHN RS 6168), 1902 [probably viii.1901], M. Maindron, 1 ♀ (MNHN RS 3246).

Srilankametrus couzjini Prendini and Loria, 2020: INDIA: Tamil Nadu: *Dindigul Distr.*: Palni Hills [Palani/Pullney/Pullnay Hills, 10°12'N 77°28'E], Kadai canal [Kodaikanal, 10°14'N 77°29'E], 1889, T.H. Kolb, paratype ♂ (SMF 5332/1). *Thanjavur Distr.*: Tanjore [Thanjavur, 10°47'N 79°08'E], E.P. Popert, holotype ♂ (BMNH 1896.7.30.121 [BNHS 317]).

Srilankametrus gravimanus (Pocock, 1894): SRI LANKA: North Central Prov.: *Polonnaruwa Distr.*: Parakramabahu, S of Polonnaruwa, near Samudraya Dam, 07°56'N 81°00'E, 69 m, 27–30.i.2014, L. Prendini and P. Horsley, 5 ♀, 3 subad. ♂, 1 subad. ♀, 3 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 12279]). North Western Prov.: *Puttalam Distr.*: Wilpattu, Kokmotte on Modaragam Aru River, 08°32'N 80°01'E, 30 m, 24–25.i.2014, L. Prendini and P. Horsley, 1 ♂, 2 ♀, 1 subad. ♂, 2 subad. ♀, 3 juv. ♂, 1 juv. ♀ (AMNH), 1 juv. ♂ (AMCC [LP 12280]).

Srilankametrus indus (DeGeer, 1778): Madras, *Heterometrus (Srilankametrus) indus laevitens*: holotype ♂ (MNHN RS 0089). SRI LANKA: Netherlands Indies [label in bottle: “Ceylan”], *Scorpio crassimanus*: 1 subad. ♀ holotype (MRHNB [S.G. 38]); i.1965, F. Layard, 2 ♂ (MCZ); Ceylan, 1 ♂ (MNHN RS 3245 [Simon coll. 5770]); Ceylan?, Dr. H. Mutvei, 1 ♀ (MNHN RS 8117). Central Prov.: *Kandy Distr.*: Kandy [07°17'N 80°38'E], 1902, Dr. Bedeker, 1 ♂ (MNHN RS 3263); Kandy, outskirts of Udawattakele, 07°18'N 80°38'E, 217 m, 30.i.2014, L. Prendini and P. Horsley, 3 ♀, 2 juv. ♂, 3 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 12282]); Kandy, Udawattakele, 07°18'N 80°38'E, 560 m, 31.i–4.ii.2014, L. Prendini and P. Horsley, 6 ♂, 9 ♀, 4 subad. ♂, 4 subad. ♀, 5 juv. ♂, 10 juv. ♀ (AMNH), 1 juv. ♀ (AMCC [LP 12283]); Paradeniya [Peradeniya, 07°16'N 80°36'E], vii.1914, B.H. Buxton, 2 ♂, 1 subad. ♂, 1 subad. ♀, 1 juv. ♀ (MNHN RS 0081).

Srilankametrus pococki Prendini and Loria, 2020: SRI LANKA: North Central Prov.: *Anuradhapura Distr.*: Ritigala, Kodigala Mountain, summit and rocky upper slopes, 08°07'N 80°39'E, 762 m, 6.ii.2014, L. Prendini and P. Horsley, holotype ♀ (AMNH), paratypes: 3 subad. ♂, 2 subad. ♀, 3 juv. ♂, 3 juv. ♀ (AMNH), 1 juv. ♂, 1 juv. ♀ (IFS), 2 juv. ♀ (AMCC [LP 12281]).

Srilankametrus serratus (Pocock, 1900): SRI LANKA: Ceylon, Dr. Ondaatje, holotype ♂ (BMNH 1888.55), E.C. Fernando, 1 ♂, 1 subad. ♀, 1 juv. ♂ (FMNH 086 264), 21.iv.1969, E.C. Fernando, 3 ♂ (FMNH 086 916), 1899, Dr. H. Driesch, 1 ♂, 1 juv. ♂, 1 juv. ♀ (ZMH), x.1912, 1 ♀, 1 juv. ♂ (ZMH ex Mus. Calcutta). Southern Prov.: *Galle Distr.*: between Bentota and Haburugala, 06°24'N 80°01'E, x.2016, A. Ullrich, 1 juv. ♀ (AMCC [LP 14058]). Uva Prov.: *Moneragala Distr.*: Warahana [06°27'N 81°27'E], Uva, Ceylon, xi.1962, E.C. Fernando, 1 subad. ♂, 1 subad. ♀ (FMNH 086 251).

Appendix 2.

Distribution of 186 morphological characters used for phylogenetic analysis of the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879). Character states scored 0–4, inapplicable (–) or

unknown (?). Numbers following terminal taxon names correspond to samples from which DNA was extracted and sequenced (Appendix 1, Table 4). Character descriptions in Appendix 3.

Outgroup

Nebo hierichonticus 11138

0000020000	0000000100	0000000000	0000000000
0000000010	0010000011	0011000000	0000001000
0000000000	0000000000	0000000001	0100101100
1011000000	000–000100	0001000000	0000000000
0000000110	0000000001	000000	

Opisthophthalmus capensis 1046

0211240020	1101111111	1113010011	1010012000
1000111101	0100100012	0111100001	1000000001
1122011110	0100111001	1200013101	0111101111
0011101111	0011010120	1102112221	2100101101
1101111111	1001110001	002100	

Pandinus imperator 1601

1100000011	0000100320	0102000000	0001002210
1101100121	1120111133	1122000001	1000020102
1022111100	1–11020011	0122002010	1100011001
0111010000	0110011111	0101110101	0000001101
1100000111	0011100001	001100	

Scorpio fuscus 1556

0100210010	1000000300	0112000000	0000002110
1010100101	0110000033	1122000001	1001000001
1021011100	0100110001	0111111110	1110011101
0011100000	0010011120	0101101011	1000000001
1101111111	0100001110	000101	

Ingroup

Chersonesometrus bastawadei 16785 16787

0100210000	0000100220	0002000000	0001102220
1100101100	1010000023	0122020110	0110122202
1021111100	0101120001	0122002011	1100111001
1001010000	0110111111	0101110101	0000010000
1100000011	0011100000	001000	

Chersonesometrus beccaloniae 13133 13134

0100010001	0000100210	0002000000	0001102221
1100101000	1010000012	0112010111	0111112212
1022111100	0101120011	0122002011	1100111001
1001010000	0110111111	0111100001	0001011100
1100000101	0011100000	001000	

***Chersonesometrus fulvipes* 8268 13125 13981 13990 13997 14031**

0100210010	0000100210	0002000000	0001102220
1100101100	1010000023	0122020110	0110122202
1021111100	0101120101	0122002011	1100111001
1001010000	0110111111	0101110101	0000011101
1100000011	0011100000	001000	

***Chersonesometrus hendersoni* 16776 16782**

1100010001	0000100210	0002000000	0101102220
1100101000	1010000044	0022020110	0111112212
1022111100	0101120001	0122002011	1100111001
1001010000	0110111111	0111100001	0000010000
1100000111	0011100000	001000	

***Chersonesometrus madraspatensis* 1533 4324 13127**

0100210010	0000100220	0002000000	0001102221
1100101100	1010000023	1112020110	0111122202
1021111100	0101120101	0122002010	1100011001
1001010000	0110111111	0101110101	0000011101
1100000111	0011100000	001000	

Chersonesometrus nathanorum

?100010001	0000100210	0002000000	010110222?
?100101101	0100000011	0112010110	0112102202
1022?01100	0101120011	0122002011	1100111001
1101010000	01???11111	0100100101	0000011110
1100000101	0011100000	00100?	

***Chersonesometrus pelekomanus* 16773 16774 types**

0100010001	0000100220	0002000000	0101102220
1100101000	1010000044	0022020110	0111112212
1022111100	0101120011	0122002011	1100111001
1001010000	0110111111	0111100001	0000010000
1100000111	0011100000	001000	

Chersonesometrus shivashankari

0100210010	0000100220	0002000000	0101101220
1100101100	1010000023	1122020110	0111122202
1021111100	0101120101	0122002010	1100011001
1001010000	0110111111	0101110101	0000011101
1100000011	0011100000	001?00	

***Chersonesometrus tristis* 2276 13135 13136**

0100010001	0000100210	0002000000	0101102221
1100101000	1010000023	0012020111	0111112212
1022111100	0101120011	0122002011	1100111001
1001010000	0110111111	0110100101	0011011100
1100000111	0011100000	001000	

Chersonesometrus wroughtoni

0100210000	0000100210	0002000000	0101102220
1100101100	1010000023	0122020110	0111112202
1021111100	0101120011	0122002011	1100111001
1001010000	01???11111	0111100101	0000010100
1100000001	0011100000	00100?	

Deccanometrus bengalensis

0100210001	0000100120	0001000000	0001102221
1100101111	0110000011	1100100001	0010120102
1022100100	0101120011	0122002011	1100111001
0011010000	0110111111	0111110101	0000011101
1100000001	0011100000	001000	

***Deccanometrus bengalensis* 14238**

0100200001	0000100120	0001000000	0001102221
1100101111	0110000011	1100100001	0010120102
1022100100	0101120011	0122002011	1100111001
0011010000	0110111111	0111110101	0000011101
1100000001	0011100010	001000	

***Deccanometrus latimanus* 11568 11569 11601**

0100220010	0000100110	0011000000	0001101220
1100100121	0120000033	1122100001	1010020102
1122100100	0101120101	0122002010	0100001000
0011000000	0110110111	0110110001	0022011111
1110000001	0111100011	001110	

Deccanometrus liurus

0100130001	1000100110	0011020000	0001121221
1101101121	0120000012	1100100001	1000020102
1022100100	0101120101	0122002010	1100011001
0011010000	01???10111	0110110001	0033011110
1210000001	0111100001	001?0?	

***Deccanometrus obscurus* 6225 13974 16817**

0100200001	0000100110	0001000000	0101102221
1100101110	1010000012	1101000001	0010120102
1022111100	0101120011	0122002011	1100111001
1001010000	0110111111	0111110101	0000011101
1100000001	0011100000	001000	

***Deccanometrus obscurus* 16800**

0100200001	0000100110	0001000000	0101102221
1100101110	1010000012	1101000001	0010120102
1022111100	0101120011	0122002011	1100111001
1001010000	0110111111	0111110101	0000011101
1100000001	0011100010	001000	

***Deccanometrus phipsoni* types**

0100210001	0000100110	0001000000	0101102221
1100101110	1010000012	1100000001	0010120102
1022111100	0101120011	0122002011	1100111001
1001010000	0110111111	0110100101	0000011101
1100000001	0011100010	001000	

***Deccanometrus phipsoni* E Ghats**

0100210001	0000100110	0001000000	0101102221
1100101110	1010000012	0100000001	0010120102
1022111100	0101120011	0122002011	1100111001
1001010000	0110111111	0111100101	0000011101
1100000001	0011100000	001000	

Deccanometrus ubicki

0100110001	0000100220	0011000000	0101102221
1100101121	0120000012	1100100001	1000020102
1022100011	0101120101	0122002011	1100111001
1011010000	0110111011	0110110001	0000011100
1210000001	0111100001	001?0?	

***Deccanometrus xanthopus* types**

0100230010	0000100320	0001000000	0101101221
1101101111	0120000023	1111100000	0110120102
1121100100	0101120101	0122002010	0100001001
0011010000	0110110011	0100100001	0000011110
1100000001	0111100001	001110	

***Deccanometrus xanthopus* 13126 13971 16814**

0100231010	0000100220	0001000000	0101101221
1101101111	0120000022	1111100000	0110120102
1121100100	0101120101	0122002010	0100011001
0011010000	0110110111	0100100001	0000011110
1100000001	0111100001	001110	

***Gigantometrus swammerdami* 1607 13129**

1100220110	0000100220	0001000000	0001101220
1100101101	1120000044	1122010110	0111111102
1021111100	0101120011	0122002011	1100111001
1001010000	0110111111	0101110100	0000020000
1100000001	0001100000	101100	

***Gigantometrus swammerdami* 13128**

1100220110	0000100220	0001000000	0001101220
1100101101	1120000044	1122010110	0111111102
1021111100	0101120001	0122002011	1100111001
1001010000	0110111111	0101110100	0000020000
1100000001	0001100000	101100	

***Gigantometrus titanicus* 12287 12288**

1100210110	0000100220	0001000000	0001102220
1100101101	1120000044	1122010110	0111111102
1021111100	0101120011	0122002011	1100111001
1001010000	0110111111	0101110100	0000020000
1100000001	0001100000	101100	

***Heterometrus glaucus* 13922 13928**

1100000111	0010100220	0001120000	0001012221
1100101121	1121000001	1101100001	1000020102
1011000011	0101120011	0000002011	1000111001
1101010000	1110111111	0111110101	0000001101
1100000001	0011100001	021100	

***Heterometrus laevigatus* 1789 12335 12337 13045 13047 13048**

0100000111	0010100010	0001120000	0001012221
1100101121	1121000012	1101100001	1000020102
1111000011	0101020011	0000002011	1100111001
1001010000	1110111111	0110110101	0000001101
1100000001	0011100010	021100	

***Heterometrus laoticus* 11330 11331 11352 12336 13046 15584 16489**

0100000111	0010100000	0001121100	0101012220
1100101121	1121000021	1111100001	1000020102
1011011011	0101020011	0000002011	1000110001
1001010000	1110111111	0110110101	0000001101
1100001011	0001100010	021100	

***Heterometrus longimanus* 11948**

0100000111	0010100110	0001120000	0101012221
1100101121	1121000000	1100100001	1000020102
1011011011	0101120011	0000002011	1000111001
1001010000	1110111111	0111110101	0000001100
1100000001	0011100001	021100	

***Heterometrus longimanus* 12114 12115 12116 12139**

0100000111	0010100220	0001120000	0101012221
1100101121	1121000000	1100100001	1000020102
1011011011	0101120011	0000002011	1000110001
1001010000	1110111111	0111110101	0000001100
1100000001	0011100001	021100	

***Heterometrus petersii* 1604 8269**

0100000111	0010100110	0001120000	0001012220
1100101121	1121000022	1111100001	1000020102
1011011011	0101020011	0000002011	1000110001

1101010000 1110111111 0111110101 0000001101
1100000001 0011100000 021100

***Heterometrus petersii* 16458**

1100000111 0010100110 0001120000 0001012220
1100101121 1121000022 1111100001 1000020102
1011011011 0101020011 0000002011 1000110001
1101010000 1110111111 0111110101 0000001101
1100000001 0011100000 021100

***Heterometrus silenus* 1603 1888 11248 11252 11353 15590**

0100000111 0010100110 0001120000 0001012220
1100101121 1121000032 1121100001 1000020102
1011011011 0101020011 0000002011 1000110001
1001010000 1110111111 0111110101 0000001101
1100000001 0001100010 021100

***Heterometrus spinifer* 1906 10581 11920 11949 11950 11974**

1100000111 0010100110 0001120000 0001012220
1100101121 1121000022 1111100001 1000020102
1011011011 0101020011 0000002011 1000110001
1101010000 1110111111 0111110101 0000001101
1100000001 0011100000 021100

***Heterometrus thorellii* 2704 13049 13050 13051 13052 13153 13155 15604**

0100000111 0010100220 0001120000 0001012221
1100101121 1121000001 1101100001 1000020102
1011000011 0101120011 0000002011 1100111001
1001010000 1110111111 0110110101 0000001101
1100000001 0011100010 021100

***Javanimetrus cyaneus* 2309 8270 12112 12113 12117**

0100201011 0000100220 0001020000 00012-2220
1100101121 1120000023 1122100001 1000120112
1021100011 0101120011 0122002010 1100011001
1101010000 0110111111 0111110101 0000011101
1100000111 0011100000 011100

Sahyadrimetrus barberi

0100200001 0000100?20 0001020000 010?2-222?
110010?121 01100000?1 11?110000? 0?1?12?102
102?1?01?0 0101?20011 0122002011 1100111001
1101010000 0??????11? 01?0?1?101 000001?111
110?0?0?01 0011100000 011???

***Sahyadrimetrus kanarensis* 2289 2902 13960 13963 14029 16795 16797**

0100200101 0000100220 0001020000 01012-2220
1100101121 1120000012 1111100001 1000120102
1022100011 0101120011 0122002010 1100011001
0111010000 0110111111 0111110101 0000011101
1100000001 0011100000 011100

***Sahyadrimetrus mathewi* 13130 13131 13132 16752 16759**

0100200121 0000100330 0001020000 00012-2221
1100101121 1120000012 1101100001 1000120102
1022100101 0101120011 0122002010 1100011001
1101010000 0110111111 0101110101 0000011101
1100000001 0011100000 011100

***Sahyadrimetrus rugosus* 14007 14028 16768**

0100200001 0000100120 0001020000 01012-2221
1100101110 0010000011 1101100000 0110120102
1022100110 0101120011 0122002011 1100111001
1101010000 0110111111 0110110101 0000011101
1100000001 0011100000 011000

***Sahyadrimetrus scaber* 13979 14023 16771 16772**

0100200111 0000100220 0001020000 00012-2220
1100101121 1120000022 1111100001 1000120102
1022100101 0101120011 0122002010 1100011001
1101010000 0110111111 0101110101 0000011101
1100000001 0011100000 011100

***Sahyadrimetrus tikaderi* 16760 16764 16766 16767**

0100200001 0000100220 0001020000 01012-2220
1100101121 0110000011 1101100000 0110120102
1022100010 0101120011 0122002011 1100111001
1101010000 0110111111 0111110101 0000011101
1100000111 0011100000 011000

Srilankametrus caesar types

0100211010 0000100220 0101000000 0001102220
1100101101 0110000023 1122101001 1111121102
1021111100 0101120011 0122002011 1100111001
1001010000 0110111211 0111110101 0000010001
1100000111 0011100010 021110

Srilankametrus caesar

0100211010 0000100220 0101000000 0001102220
1100101101 0110000023 1122101001 1111121102

1021111100	0101120001	0122002011	1100111001
1001010000	0110111211	0111110101	0000010001
1100000111	0011100010	021110	

Srilankametrus couziji

0100211010	00001000?0	0101000?00	0?01102220
?1001011??	???000004?	112?101001	?1?1102102
??2111?0?	0101120011	0122002010	1100011001
1001010000	0110111?11	011?1101?1	0000010?01
1100?0?1?1	0011100010	021110	

***Srilankametrus gravimanus* 12279 12280**

0100211010	0000100000	0101000000	0101102220
1100101121	1120000033	1112101001	1112102102
1021111100	0101120011	0122002011	1100111001
1001010000	0110111211	0111110101	0000010001
1100000101	0011100010	021110	

***Srilankametrus serratus* 14058**

0100211010	0000100000	0101000000	0001102220
1100101121	1110000033	1122100001	1001111102
1021111100	0101120011	0122002011	1100111001
1001010000	0110111211	0111110101	0000010101
1100000001	0011100000	001110	

***Srilankametrus indus* 12282 12283**

0100201000	0000100000	0101021100	00012-2220
1100101121	0110000033	1122100001	1000120112
1022111100	0101120011	0122002010	1100011001
1001010000	0110111211	0120-10101	0022011111
1100000001	0011100010	001100	

***Srilankametrus pococki* 12281**

0100201000	0000100000	0101020000	00012-2220
1100101121	0110000033	0022100001	1000120112
1022111100	0101120011	0122002010	1100011001
1001010000	0110111211	0120-10101	0000011111
1100000001	0011100010	001100	

Srilankametrus yaleensis

0100211010	0000100000	0101000000	0101102220
1100101121	1120000022	1122102001	1112102102
1021111100	0101120011	0122002010	1100011001
1001010000	0110111211	0120-10101	0011010101
1100000001	0011100000	021110	

Appendix 3.

Morphological characters and character states used in phylogenetic analysis of the Asian forest scorpions (Scorpionidae:

Heterometrinae Simon, 1879). Terminology follows previous papers on Scorpionidae Latreille, 1802 by the first author (e.g., Prendini, 2000b, 2001a; Prendini et al., 2003; Tahir and Prendini, 2014; Prendini, 2016). Characters corresponding to the lists of Lamoral (1979: Table 8), Couzijn (1981: Tables 2 And 6) and Prendini (2000a: Appendix 3) and Prendini et al. (2003: Appendix 2) are denoted, respectively, by the abbreviations BHL, HWC, LP, and PEA, followed by the corresponding number. Character matrix in Appendix 2. Fifty-two uninformative characters are indicated by †.

Total Length

1. Adult length, measured from anterior margin of carapace to tip of aculeus: medium, minimum length of 50 mm to maximum length of 120 mm (0); extremely large, minimum length > 120 mm (1). [PEA 1]

Chelicerae

2. Chelicerae dorsal surfaces, infuscation: entirely pale or nearly so, fingers may be very slightly infuscate (cheliceral manus similar to, or paler than carapace interocular surface) (0); bicoloured, prodorsal half of cheliceral manus paler than retrodorsal half and fingers (manus similar to, or paler than carapace interocular surface) (1); entirely dark (cheliceral manus darker than carapace interocular surface) (2). [PEA 90]
- †3. Cheliceral coxae prodorsal surfaces, scaphotrix (stridulatory setae): absent (0); present (1). [LP 13, PEA 3]
- †4. Cheliceral coxae promedian surfaces, trichocopae (chemoreceptive lamelliform setae): absent (0); present (1). [LP 14, PEA 4]
5. Cheliceral movable finger, prodistal (DI) and retrodistal (DE) teeth: subequal, DE tooth only slightly smaller than DI tooth, and opposable, i.e., forming a bicuspid (0); unequal, DE tooth considerably smaller than DI tooth, and opposable, i.e., forming a bicuspid (1); unequal, DE tooth considerably smaller than DI tooth, aligned longitudinally and not opposable (2). [LP 11, PEA 2]

General Color

6. Carapace, pedipalps, tergites and metasoma, base colouration: black (0); very dark reddish- or brownish-black (1); dark reddish-brown (2); light yellowish-brown (3); light yellowish (4).

Carapace

7. Carapace shape, anterolateral margins, dorsal aspect: converging gradually anteriorly (0); converging slightly to subparallel anteriorly (1)
8. Carapace shape, mediolateral margins, dorsal aspect: diverging or converging slightly posteriorly (at posterolateral sulci) (0); converging markedly posteriorly (at posterolateral sulci) (1)
9. Carapace shape, lateral aspect: slightly to markedly dorsoventrally compressed, lateral surfaces sloping gently (0); vaulted, lateral surfaces sloping steeply (1); vaulted, lateral surfaces sloping very steeply (2)
10. Carapace, anterior margin/frontal lobes, shape: anterior margin straight, frontal lobes evenly rounded or truncate (0); anterior margin concave, frontal lobes uneven, sloping medially (towards notch/midline) (1)
11. Carapace anterior margin, median notch: deeply excavated (0); shallow (1). [LP 3, PEA 5]

- †12. Carapace anterior margin, median projection: absent (0); present (1). [PEA 6]
13. Carapace rostrrolateral margin: entire (0); distinct incision adjacent to posterior lateral ocelli (1). [PEA 13]
- †14. Carapace anterior median depression: absent (0); present (1). [PEA 7]
15. Carapace median longitudinal sulcus, infuscation: bordered by infuscation from anterior median depression to anterocular depression (0); immaculate (1).
- †16. Carapace interocular surface, base colouration: similar to, or darker than lateral and posterior surfaces (0); noticeably paler than lateral and posterior surfaces (1). [PEA 91]
- †17. Carapace interocular surface, infuscation: infusate (0); immaculate (1).
18. Carapace interocular surface, macrosculpture distribution (σ): entirely smooth (0); granular along median longitudinal and anterior furcated sulci only (1); frontal lobes and medial region granular, with smooth areas (2); entire surface uniformly granular (3); unknown (?). [Modified BHL 4 by adding two states to account for variation in the distribution of granulation, PEA 17]
19. Carapace interocular surface, macrosculpture distribution (φ): entirely smooth (0); granular along median longitudinal and anterior furcated sulci only (1); frontal lobes and medial region granular, with smooth areas (2); entire surface uniformly granular (3). [Modified BHL 5 by adding two states to account for variation in the distribution of granulation, PEA 18]
- †20. Carapace anterocular depression: absent, anterior portions of superciliary carinae parallel (0); present, anterior portions of superciliary carinae subparallel or diverging (1). [PEA 8]
- †21. Carapace circumocular depressions: converging anteriorly (0); diverging anteriorly (1). [PEA 9]
22. Carapace superciliary carinae, anterocular extensions: present (0); absent (1). [Incorporates HWC 7, PEA 11]
23. Carapace superciliary carinae: higher than median ocelli (0); lower than median ocelli (1). [merged BHL 9 and 10, PEA 12]
24. Carapace median ocular tubercle position, expressed as ratio of distance from carapace anterior margin : carapace length (CAM:CL): distinctly anterior, CAM:CL < 0.40 (0); anteromedial to medial, CAM:CL 0.40–0.50 (1); posteromedial, CAM:CL 0.51–0.62 (2); distinctly posterior, CAM:CL 0.63–0.78 (3). [Modified BHL 3, PEA 10]
25. Carapace median ocelli size relative to ocellus width: small, distance between ocelli equal to or greater than an ocellus width (0); large, distance between ocelli less than an ocellus width (1).
26. Carapace interocular suture, presence and breadth: slender (0); broad (1); absent (2). [PEA 14]
27. Carapace anterolateral and mediolateral surfaces, macrosculpture (σ): granular (0); smooth or nearly so (1). [part of BHL 5: separated into two characters for antero- and posterolateral surfaces]
28. Carapace anterolateral and mediolateral surfaces, macrosculpture (φ): granular (0); smooth or nearly so (1). [part of BHL 6: separated into two characters for antero- and posterolateral surfaces, PEA 19]
- †29. Carapace mediolateral surfaces, infuscation: infusate (0); immaculate (1).
- †30. Carapace posterolateral and, often, posteromedian surfaces, base colouration: as dark as or darker than tergites (0); as pale as tergites, if tergites immaculate, or paler than tergites, if tergites infusate (1). [PEA 92]
- †31. Carapace posterolateral surfaces, infuscation: infusate (0); immaculate (1).
32. Carapace posterolateral surfaces, macrosculpture (φ): granular (0); smooth or nearly so (1). [Part of BHL 6: separated into two characters for antero- and posterolateral surfaces]
- †33. Carapace posteromedian surface, infuscation: infusate (0); immaculate (1).

34. Carapace posteromedian surface, macrosculpture (σ): granular (0); smooth or nearly so (1); unknown (?). [BHL 7, PEA 20]
35. Carapace posterior sutures, presence and length relative to median ocular tubercle: present, extending to tubercle (0); present, extending past tubercle (1); absent (2). [PEA 15]
36. Carapace posterior sutures, connection: connected anteriorly to posterior furcations of interocular suture (0); connected by short cross-suture anterior to postocular depression, regardless of presence or absence of interocular suture (1); disconnected (2); absent (-). [PEA 16]

Pedipalps

37. Pedipalps infuscation: immaculate (0); femur and patella infusate only (1); femur, patella and chela infusate (2). [PEA 93]
38. Pedipalps and first walking legs, opposing surfaces of coxae, stridulatory surface, presence and development: absent (0); weakly developed (1); well developed (2). [LP 74, PEA 48]
39. Pedipalps and first walking legs, opposing surfaces of coxae, stridulatory surface, granular tubercles of “rasp” and stridulatory setae (scaphotrix) of “scraper” relative positions: absent (0); “rasp” and “scraper” situated on coxae of pedipalp and first leg, respectively (1); “rasp” and “scraper” situated on coxae of first leg and pedipalp, respectively (2). [LP 75, PEA 49]
40. Pedipalps length (σ), expressed as ratios of pedipalp femur length : carapace posterior width (FL:PCW) and pedipalp femur length : carapace length (FL:CL): short, FL:PCW < 0.77, FL:CL < 0.74 (0); long, FL:PCW $\geq$ 0.80, FL:CL $\geq$ 0.76 (1); unknown (?). [Replaces BHL 16, PEA 21]
- †41. Pedipalps length (φ), expressed as ratios of femur length : posterior carapace width (FL:PCW) and femur length : carapace length (FL:CL): long, FL:PCW $\geq$ 0.80, FL:CL $\geq$ 0.76 (0); short, FL:PCW < 0.77, FL:CL < 0.74 (1). [Replaces BHL 16, PEA 21]
42. Pedipalp femur retrodorsal carina, development relative to prodorsal carina: less developed (0); similar or more strongly developed (1). [Modified HWC 28: corrected entry for *Opisththalmus*, PEA 44]
- †43. Pedipalp femur trichobothrium *i*, position: prolateral surface (0); dorsal surface (1). [LP 40, PEA 56]
44. Pedipalp femur prodorsal carina, vertical development: present and distinct (0); absent or obsolete (1). [PEA 45]
- †45. Pedipalp femur prodorsal and promedian carinae, orientation relative to dorsal secondary and proventral carinae: parallel to dorsal secondary and proventral carinae (0); diagonally from proventral carina to dorsal secondary carina (1). [PEA 46]
- †46. Pedipalp femur secondary accessory carina, macrosculpture: absent (0); present, comprising weakly developed granular row (1).
47. Pedipalp femur retroventral carina, longitudinal development (σ): absent or obsolete (0); present, extending partly or entirely to distal edge of segment (1); unknown (?). [PEA 47]
- †48. Pedipalp patella dorsal surface, shape: flat (or nearly so), axes of dorsomedian and retrodorsal carinae in same plane (0); convex, axis of dorsomedian carina dorsal to axis of retrodorsal carina (1). [LP 15, PEA 38]
49. Pedipalp patella dorsomedian carina, macrosculpture (φ): entirely to predominantly costate (0); entirely to predominantly granular (1); absent or obsolete (2). [PEA 39]
50. Pedipalp patella dorsal intercarinal surfaces, macrosculpture (φ): granular (0); smooth or nearly so (1). [HWC 37, PEA 40]
51. Pedipalp patella retrodorsal carina, development relative to retromedian carinae (φ): absent or obsolete (0); as strongly developed as, or more strongly developed than retromedian carinae (1). [PEA 42]

52. Pedipalp patella retrodorsal and retroventral intercarinal surfaces, macrosculpture (♀): granular (0); smooth or nearly so (1). [Merged BHL 32 and 33; incorporates HWC 37, PEA 41]
53. Pedipalp patella retromedian carinae, macrosculpture (♀): costate (0); granular (1); absent or obsolete (2). [PEA 43]
54. Pedipalp patella proventral carina, prominent spiniform granule: absent (0); present (1).
55. Pedipalp patella *est* trichobothria, count: 1 (0); 2 (one accessory) (1). [Split PEA 54 into three characters]
- [†]56. Pedipalp patella *em* trichobothria, count: 2 (0); 3 (one accessory) (1). [Split PEA 54 into three characters]
- [†]57. Pedipalp patella *esh* trichobothria, count: 2 (0); 3 (one accessory) (1). [Split PEA 54 into three characters]
- [†]58. Pedipalp patella *v* trichobothria, count: single row of 3 (0); two or more rows of > 25 (1). [Modified BHL 28 and LP 43 with extra states, PEA 55]
59. Pedipalp chela manus dorsal surface, proximal margin (lobe), curvature (♂): dorsal surface, proximal margin distal to retrolateral surface, proximal margin (0); dorsal surface, proximal margin aligned with retrolateral surface, proximal margin (1); dorsal surface, proximal margin proximal to retrolateral surface, proximal margin but distal to manus condyle (articulation with patella), proximal margin (2); dorsal surface, proximal margin aligned with manus condyle, proximal margin (3); dorsal surface, proximal margin proximal to manus condyle, proximal margin (4).
60. Pedipalp chela manus dorsal surface, proximal margin (lobe), curvature (♀): dorsal surface, proximal margin distal to retrolateral surface, proximal margin (0); dorsal surface, proximal margin aligned with retrolateral surface, proximal margin (1); dorsal surface, proximal margin proximal to retrolateral surface, proximal margin but distal to manus condyle (articulation with patella), proximal margin (2); dorsal surface, proximal margin aligned with manus condyle, proximal margin (3); dorsal surface, proximal margin proximal to manus condyle, proximal margin (4).
61. Pedipalp chela manus dorsal surface (between dorsomedian and digital carinae), curvature (♂): flat (0); curved, slightly to markedly convex (1).
62. Pedipalp chela manus dorsal surface (between dorsomedian and digital carinae), curvature (♀): flat (0); curved, slightly to markedly convex (1).
63. Pedipalp chela manus dorsal intercarinal surface, maximum distance between dorsomedian and dorsal secondary carinae (DMC–DSC) relative to maximum distance between dorsal secondary and digital carinae (DSC–DC) or rows of spiniform macrosetae demarcating carinae (♂): DMC–DSC less than DSC–DC (0); DMC–DSC similar to DSC–DC (1); DMC–DSC greater than DSC–DC (2).
64. Pedipalp chela manus dorsal intercarinal surface, maximum distance between dorsomedian and dorsal secondary carinae (DMC–DSC) relative to maximum distance between dorsal secondary and digital carinae (DSC–DC) or rows of setae demarcating carinae (♀): DMC–DSC less than DSC–DC (0); DMC–DSC similar to DSC–DC (1); DMC–DSC greater than DSC–DC (2).
65. Pedipalp chela manus dorsomedian carina, longitudinal development: continuous to proximal edge of manus (0); becoming obsolete proximally (1). [PEA 27]
66. Pedipalp chela manus dorsomedian carina, macrosculpture: obsolete proximally with irregular spiniform granules distally, or irregular spiniform granules along entire length (0); costate to costate-granular proximally, becoming irregular row of spiniform granules distally (1); costate to costate-granular proximally, becoming regular row of spiniform granules along entire length (2).
67. Pedipalp chela manus, distal half, and fixed finger, dorsomedian carina, vertical development: obsolete (0); pronounced, costate on fixed finger and distally on manus (1); pronounced, costate on fixed finger and distal three-quarters of manus (2).
68. Pedipalp chela manus dorsal secondary, subdigital and digital carinae, relative orientation of ridges or setal rows: dorsal secondary, subdigital and digital carinae well separated along entire length (0); proximal half of dorsal secondary carina, distal half of subdigital carina and distal third of digital carina closely adjacent (1).
69. Pedipalp chela manus dorsal secondary carina, macrosculpture: single row (0); double row (1).
70. Pedipalp chela manus dorsal secondary carina, sulcus accommodating macrosetae (regardless of presence or absence of carina) (♂): present (0); absent or obsolete (1).
71. Pedipalp chela manus dorsal secondary carina, sulcus accommodating macrosetae (regardless of presence or absence of carina) (♀): present (0); absent or obsolete (1).
72. Pedipalp chela manus, sulcus between dorsal secondary and subdigital carinae or rows of setae demarcating carinae (♂): absent or obsolete (0); present (1).
73. Pedipalp chela manus, sulcus between dorsal secondary and subdigital carinae or rows of setae demarcating carinae (♀): absent or obsolete (0); present (1).
74. Pedipalp chela manus dorsal secondary and subdigital carinae, macrosculpture (♂): absent or obsolete (0); entirely to predominantly granular (1); entirely to predominantly costate (2); unknown (?). [Incorporates HWC 61, PEA 31]
75. Pedipalp chela manus, sulcus between subdigital and digital carinae or rows of setae demarcating carinae: absent or obsolete (0); partial (1); complete (2).
76. Pedipalp chela manus digital carina, macrosculpture: entirely to predominantly costate (0); entirely to predominantly granular (1); absent or obsolete (2). [Merged BHL 19 and 20, PEA 32]
77. Pedipalp chela manus retromedian carina, macrosculpture (♂): absent or obsolete (0); entirely to predominantly granular (1); entirely to predominantly costate (2); unknown (?).
78. Pedipalp chela manus retromedian carina, development relative to digital carina (♂, ♀): less developed (0); similar (1); more developed (2).
79. Pedipalp chela manus ventral surface, shape: flat, axes of retroventral and ventromedian carinae in approximately same plane (0); angular, axis of retroventral carina ventral to axis of ventromedian carina (1).
80. Pedipalp chela manus ventromedian carina, macrosculpture: absent or reduced to few proximal granules (0); present and strongly sclerotized (1); present and weakly sclerotized (2). [Modified LP 28 with new state to distinguish between weak and strong sclerotization, PEA 37]
81. Pedipalp chela manus proventral carina, development relative to promedian carina (♀): more strongly developed than promedian carina, which may be obsolete (0); similarly developed (1). [LP 29, PEA 36]
82. Pedipalp chela manus promedian carina, macrosculpture (♀): present, granular (0); absent or obsolete (1). [PEA 35]
83. Pedipalp chela manus prolateral surface, distance between dorsomedian and promedian carinae (DMC–PMC) relative to distance between promedian and proventral carinae (PMC–PVC) or rows of setae demarcating carinae: DMC–PMC less than PMC–PVC (0); DMC–PMC similar to PMC–PVC (1); DMC–PMC slightly to markedly greater than PMC–PVC (2).
84. Pedipalp chela setation (♂): virtually asetose (0); sparsely setose (1); moderately to densely setose (2). [PEA 22]
85. Pedipalp chela manus dorsal surface, depression proximal to fixed finger (♂, sometimes evident in ♀): present and distinct (0); absent or obsolete (1).
86. Pedipalp chela manus dorsal intercarinal surface, macrosculpture (♂): shallowly reticulate (0); reticulation absent (1); unknown (?). [State 0 of BHL 17; incorporates HWC 61, PEA 29]

87. Pedipalp chela manus dorsal surface, macrosculpture (♀): shallowly reticulate (0); reticulation absent (1). [State 0 of BHL 18; incorporates HWC 61, PEA 30]
88. Pedipalp chela manus dorsal intercarinal surface, macrosculpture (♀): smooth (0); finely to coarsely granular (1). [State 1 of BHL 17 and 18; incorporates HWC 60 and 61, PEA 28]
89. Pedipalp chela manus retrolateral intercarinal surfaces, macrosculpture (♂): granular (0); smooth or nearly so (1); unknown (?). [PEA 33]
90. Pedipalp chela manus retrolateral intercarinal surfaces, macrosculpture (♀): granular (0); smooth or nearly so (1). [PEA 34]
- †91. Pedipalp chela manus *V* trichobothria, count: 4 (0); 8–10 (1). [Modified BHL 26 and LP 49 with extra states, PEA 51]
- †92. Pedipalp chela manus *V* trichobothria, longitudinal position of trichobothrium *V*₃ if only four *V* trichobothria: distal third (0); proximal third (1); inapplicable if more than four *V* trichobothria (-). [BHL 25 with extra state for *Nebo*, PEA 52]
- †93. Pedipalp chela *i* trichobothria, count: two (0); three (1). [LP 46, PEA 53]
94. Pedipalp chela fingers retrolateral intercarinal surface, macrosculpture: smooth (0); granular (1). [PEA 26]
95. Pedipalp chela movable finger second lobe, development relative to other lobes on finger (♂): more strongly developed than other lobes, with correspondingly well-developed notch in fixed finger (0); not noticeably larger or only slightly larger than other lobes (1); unknown (?). [Modified LP 35: also recognized in *Nebo*, PEA 25]
96. Pedipalp chela fingers terminal teeth, interlocking when closed: evenly, fixed and movable fingers aligned (0); unevenly, movable finger displaced retrolaterally, with weakly developed notch near tip of fixed finger (1); unevenly, movable finger displaced retrolaterally, with strongly developed notch near tip of fixed finger (2). [PEA 23]
- †97. Pedipalp chela fixed finger trichobothria *est*–*esb*, distance relative to trichobothria *esb*–*eb*, distance: *est*–*esb* greater than half *esb*–*eb* (0); *est*–*esb* ca. half *esb*–*eb* (1) [Merged BHL 23 and 24 because these have the same distribution, PEA 50]

Coxosternum

98. Coxosternal region, infuscation: maxillary lobes darker than rest (0); uniformly pale, immaculate (1).

Legs

99. Legs, colour: pale (though perhaps very lightly infuscate) (0); very dark or heavily infuscate (1). [PEA 99]
- †100. Legs I–IV femora, ventromedian surfaces, macrosculpture: acarinate, little or no discernable carina demarcating pro- and retrolateral surfaces (0); uncarinate, distinct pro- and retrolateral surfaces demarcated by macrosetae and, usually, granules (1). [PEA 65]
- †101. Legs I and II patellae, pro- and retrolateral surfaces, setation: scattered macrosetae, not arranged in definite row (0); comb-like row of more than three long to very long macrosetae (1).
102. Legs I and II tibiae, pro- and retrolateral surfaces, setation: scattered macrosetae, not arranged in definite row (0); row of spiniform macrosetae only (1); comb-like row of more than three long to very long macrosetae (2). [PEA 66]
103. Leg I tibia, retrolateral surface, spiniform macrosetae: absent (0); one (1); two or three (2).

104. Leg II tibia, retrolateral surface, spiniform macrosetae: absent (0); one (1); two or three (2).

†105. Legs I–IV basitarsi, shape in cross-section: I–IV dorsoventrally compressed (0); I and II dorsoventrally compressed, III and IV terete (1).

106. Legs I–IV basitarsi, retrolateral margins: unmodified, rounded, similar on I–IV (0); produced into blade-like edge on I and II, unmodified, rounded on III and IV (1).

107. Legs I and II basitarsi, pro- and retrolateral surfaces, setation: scattered macrosetae, not arranged in definite row (0); row of three or more spiniform macrosetae (1); only two spiniform macrosetae (2); comb-like row of more than three setiform macrosetae with or without elongated spiniform macrosetae (3) [Modified BHL 76 and merged LP 72 and 73, PEA 67]

108. Leg I basitarsus, retrolateral surface, macroseta *st*: setiform (0); spiniform (1).

109. Leg I basitarsus, retrolateral surface, macroseta *sb*: setiform (0); spiniform (1).

110. Leg I basitarsus, retroventral surface, macroseta *st*: setiform (0); spiniform (1).

111. Leg I basitarsus, retroventral surface, macroseta *sb*: setiform (0); spiniform (1).

112. Leg I basitarsus, proventral surface, macroseta *st*: setiform (0); spiniform (1).

113. Leg II basitarsus, retrolateral surface, macroseta *st*: setiform (0); spiniform (1).

†114. Leg II basitarsus, retrolateral surface, macroseta *m*: setiform (0); spiniform (1).

115. Leg II basitarsus, retroventral surface, macroseta *st*: setiform (0); spiniform (1).

116. Leg II basitarsus, retroventral surface, macroseta *sb*: setiform (0); spiniform (1).

117. Leg II basitarsus, proventral surface, macroseta *st*: setiform (0); spiniform (1).

118. Leg II basitarsus, prolateral surface, macroseta *t*: setiform (0); spiniform (1).

†119. Leg III basitarsus, retrolateral surface, macroseta *st*: setiform (0); spiniform (1).

120. Leg III basitarsus, retrolateral surface, macroseta *sb*: setiform (0); spiniform (1).

121. Leg III basitarsus, retroventral surface, macroseta *st*: setiform (0); spiniform (1).

122. Leg III basitarsus, retroventral surface, macroseta *sb*: setiform (0); spiniform (1).

123. Leg III basitarsus, proventral surface, macroseta *st*: setiform (0); spiniform (1).

†124. Leg IV basitarsus, retrolateral surface, macroseta *t*: setiform (0); spiniform (1).

125. Legs I and II telotarsi, pro- and retrolateral surfaces, setation: long and short, densely scattered macrosetae, not arranged in definite row (0); two or more, dense brush-like rows of long to very long setiform macrosetae (1).

126. Legs I–IV telotarsi, laterodistal lobes, setation: three or more spiniform macrosetae on each laterodistal lobe (0); two spiniform macrosetae on each laterodistal lobe (1). [PEA 69]

†127. Legs I–IV telotarsi, number of spiniform macrosetae in proventral rows: constant or increasing from I–IV (0); constant or increasing from I–III but decreasing on IV (1). [PEA 70]

†128. Legs I–IV telotarsi, proventral row of spiniform macrosetae (excluding macrosetae on laterodistal lobes): present on I–IV (0); present on I–III, absent on IV (1). [Incorporates BHL 77 and 79, LP 71, PEA 71]

†129. Legs I–IV telotarsi, ungues, shape: short and distinctly curved (0); elongated and curved (proximally or distally) to sub-linear (1). [Modified BHL 82 with extra states, PEA 72]

- †130. Legs I–IV telotarsi, ungues, relative length within each pair: equal to subequal on I–IV (0); unequal on I and II, equal to subequal on III and IV (1). [Merged BHL 83 and 84, PEA 73]
 131. Legs I–IV telotarsi, pseudonygium: reduced, rounded (0); prominent, acuminate (1).

Genital Operculum

132. Genital opercula (♂): not overlapping (0); overlapping (1); unknown (?). [PEA 57]

Hemispermaphore

- †133. Distal lamina, distal crest: tapering (0); truncate (1); unknown (?). [PEA 58]
 †134. Hook notch, position on distal lamina: basal (0); medial (1); unknown (?); inapplicable (no hook notch in *Nebo*) (-).
 135. Accessory distal lobe protruding between articular suture and distal lobe (hook): absent (0); present (1); unknown (?). [PEA 59]

Pectines

- †136. Pecten length, expressed relative to length of leg IV coxa and trochanter (♂): distal edge extending past distal edge of coxa but not extending to distal edge of trochanter (0); distal edge extending to distal edge of coxa (1); unknown (?). [PEA 60]
 137. First proximal median lamella (scape), angle (♂): approximately 90° (0); obtuse, > 90° but < 180° (1); unknown (?). [PEA 61]
 138. First proximal median lamella (scape), angle (♀): distinctly angular, approximately 90° (0); distinctly angular, > 90° but < 180° (1); straight or shallowly curved (2). [PEA 62]
 139. Internal fulcral plates, setation: smooth to sparsely setose (3–4 microsetae distally) (0); smooth proximally but densely setose (microsetae only) distally (1); sparsely setose (scattered macrosetae) across entire surface (2). [Modified HWC 86: redefined states, PEA 64]
 140. Pectinal teeth, shape (♂): short, curved (0); long, straight (1); unknown (?). [PEA 63]

Tergites

- †141. Tergites I–VII, base colouration: uniform (0); posterior third noticeably paler (1). [PEA 94]
 †142. Tergites I–VII, infuscation: entirely or mostly (more than two-thirds) immaculate (0); entirely or mostly (more than two-thirds) infuscate (1). [PEA 95]
 143. Tergites I–VI, macrosculpture distribution (♂): mesial and lateral surfaces granular (0); mesial surfaces smooth, lateral surfaces granular (rarely smooth) (1); mesial and lateral surfaces smooth (2); unknown (?). [Modified BHL 34 states 0, 1, and 2; incorporates HWC 91, PEA 74]
 144. Tergites I–VI, macrosculpture distribution (♀): mesial and lateral surfaces smooth (0); mesial surfaces smooth, lateral surfaces granular (1); mesial and lateral surfaces granular (2). [HWC 91, PEA 75]
 †145. Tergites I–VI, macrosculpture (♂): uniformly finely granular (0); unevenly finely and coarsely granular to predominantly

coarsely granular (1); unknown (?); inapplicable (-). [BHL 34 states 1 and 2, with state 0 transferred to another character, PEA 76]

Sternites

146. Sternites III–VII, infuscation: III–VI immaculate, VII occasionally infuscate (0); III–VII infuscate (1). [PEA 98]
 †147. Sternites III–VII surfaces, macrosculpture (♂): smooth (0); III–VI, finely rugose, VII, scattered granules (1); III–VII, tuberculate (2); unknown (?). [PEA 101]
 148. Sternite VII surface, ventrosubmedian and ventrolateral carinae: two pairs of moderately to strongly developed ventrosubmedian and ventrolateral carinae (0); pair of weakly developed ventrolateral carinae only (1); acarinate (2). [PEA 100]
 †149. Sternite VII surface, and metasomal segment I ventral intercarinal surface, macrosculpture (♀): smooth (0); scattered granules (1); tuberculate (2). [Modified BHL 36, PEA 101]

Metasoma

150. Metasomal segments I and II, length relative to width: markedly greater than (0); approximately equal to or less than (1).
 †151. Metasomal segments I–V, colour: segments similar, dark (0); posterior segments darker than anterior segments (1); segments similar, pale (2).
 †152. Metasomal segments I–V, dorsal intercarinal surfaces, colour (♂, ♀): as dark as tergites, if pigmented (0); at least segments I and II or I–III as pale as tergites, if tergites immaculate, or paler than, if tergites infuscate (1). [PEA 96]
 153. Metasomal segments I–IV dorsosubmedian carinae, macrosculpture (♂): distinct, granular or costate-granular on segments I–IV (0); distinct, granular or costate-granular on segments III–IV or IV, costate on I and II or I–III (1); distinct, costate on segments I–IV (2); obsolete (3).
 154. Metasomal segments I–IV dorsosubmedian carinae, macrosculpture (♀): distinct, granular or costate-granular on segments I–IV (0); distinct, granular or costate-granular on segments III–IV or IV, costate on I and II or I–III (1); distinct, costate on segments I–IV (2); obsolete (3).
 †155. Metasomal segments II–IV dorsosubmedian carinae, posterior spiniform granules, development: weak (not noticeably larger than preceding granules) (0); moderate to strong (distinctly larger than preceding granules) (1). [Modified BHL 58: merged states 1 and 2, PEA 78]
 156. Metasomal segment V dorsosubmedian carinae, longitudinal development: absent (0); vestigial (1); partial (2).
 157. Metasomal segment V dorsal intercarinal surface, macrosculpture (♂): granular (0); smooth (1); unknown (?). [BHL 64, PEA 79]
 158. Metasomal segment V dorsal intercarinal surface, macrosculpture (♀): granular (0); smooth (1). [BHL 65, PEA 80]
 159. Metasomal segment V dorsolateral carinae, development: strong, continuous (0); weak, discontinuous to absent (1). [Merged BHL 66 and 67, PEA 77]
 160. Metasomal segments I–IV lateral surfaces, curvature, dorsal and ventral aspect: straight, subparallel (0); convex (1)
 †161. Metasomal segments I and II lateral carinae, orientation relative to dorsolateral and ventrolateral carinae (inferred from carina or setal row if carina absent): parallel to carinae (0); diagonal between carinae (1).

162. Metasomal segment I lateral carinae, longitudinal development: complete (0); partial (1); vestigial (2).
163. Metasomal segment V lateral carinae, longitudinal development: partial (0); absent or obsolete (1).
164. Metasomal segment II ventral surface, macrosculpture (σ): smooth (0); scattered granules (1); unknown (?). [Components of BHL 35, 42, 48, 55, and 61, PEA 102]
165. Metasomal segment II ventral surface, macrosculpture (φ): smooth (0); scattered granules (1). [Modified BHL 36, PEA 102]
166. Metasomal segment III ventral surface, macrosculpture (σ): smooth (0); scattered granules (1); unknown (?). [Components of BHL 35, 42, 48, 55, and 61, PEA 103]
167. Metasomal segment III ventral surface, macrosculpture (φ): smooth (0); scattered granules (1). [Modified BHL 36, PEA 103]
168. Metasomal segment IV ventral surface, macrosculpture (σ): smooth (0); scattered granules (1); unknown (?). [Components of BHL 35, 42, 48, 55, and 61]
169. Metasomal segment IV ventral surface, macrosculpture (φ): smooth (0); scattered granules (1). [Components of BHL 35, 42, 48, 55, and 61]
- [†]170. Metasomal segments I–V ventrosubmedian and/or ventrolateral carinae, infuscation (may be present even if carinae obsolete or absent) (σ): immaculate (0); infusate (1). [PEA 97]
- [†]171. Metasoma segments I–IV ventrosubmedian carinae, development: present and distinct on segments I–IV (0); absent or obsolete on segments I, I and II, or I–IV (1). [Merged BHL 40, 45, 50, 51, and 59 and modified, by merging states 1, 2 and 3, PEA 81]
172. Metasomal segments I–IV ventrosubmedian and ventrolateral carinae, development on segments I and II relative to III and IV: more strongly developed on segments III and IV than on I and II (0); more strongly developed on segments I and II than on III and IV (1); inapplicable (-). [LP 96, PEA 82]
173. Metasomal segments I–IV, ventrosubmedian and ventrolateral carinae: granular on segments I–IV, II–IV (costate on I), or III and IV (costate on I and II) (0); granular or costate-granular on segment IV only, costate on segments I–III (1). [Merged states 1–3 from BHL 40, 41, 45, 46, 50–53, 59 and 60]
174. Metasomal segment V ventrolateral carinae, macrosculpture: rounded granules (0); spiniform or denticulate granules (1). [BHL 69, PEA 84]

175. Metasomal segment V ventrolateral carinae, posterior spiniform granules, development: obsolete (not noticeably larger than preceding granules) (0); weak (slightly larger than preceding granules) (1). [Incorporates HWC 113, PEA 85]

[†]176. Metasomal segment V ventrolateral carinae, posterior orientation: divergent (0); subparallel to convergent (1). [BHL 70, PEA 83]

[†]177. Metasomal segment V ventromedian carina, posterior macrosculpture: unmodified moniliform row (0); breaking into numerous granules (1); inapplicable (-). [Part of LP 99, PEA 86]

[†]178. Metasomal segment V anal arch, anterior carina, macrosculpture: costate (0); serrate (1). [PEA 87]

Telson

179. Telson colour relative to metasomal segment V: paler than segment V (0); as dark as segment V (1).
180. Telson vesicle, width relative to metasomal segment V, width (σ): approximately equal to or less than (0); greater than (1).
181. Telson vesicle, width relative to metasomal segment V, width (φ): approximately equal to or less than (0); greater than (1).
182. Telson vesicle shape: globose (0); intermediate (1); elongate (2). [PEA 88]
183. Telson vesicle surfaces, macrosculpture: longitudinal rows of granules on lateral and ventral surfaces (0); longitudinal rows of granules on all or part of ventral surface only (1); entirely smooth (2). [Merged BHL 73 and 74, PEA 89]

Ecology and Behavior

184. Burrows constructed in the open: absent (0); present (1); unknown (?). [PEA 104]
185. Burrows constructed under stones: present (0); absent (1); unknown (?). [PEA 105]
- [†]186. Mesosomal percussion (σ): absent (0); present (1); unknown (?). [LP 115, PEA 106]

Appendix 4

Remote-sensing data for mean monthly precipitation (mm) from WorldClim ver. 2 and mean normalized difference vegetation index (NDVI) from NASA Earth Observatory associated with georeferences for collection sites of terminal taxa used to reconstruct ancestral ecomorphotypes for the Asian forest scorpions (Scorpionidae: Heterometrinae Simon, 1879). Tissue samples deposited in the Ambrose Monell Collection for Molecular and Microbial Research (AMCC) at the American Museum of Natural History, New York.

Species	AMCC	Country	Lat.	Long.	mm	NDVI
<i>Nebo hierichonticus</i>	LP 11138	Israel	32°43'N	35°01'E	56.3	0.39
<i>Opisthophthalmus capensis</i>	LP 1046	South Africa	33°56'S	18°25'E	47.1	0.37
<i>Pandinus imperator</i>	LP 1601	Ghana	05°20'N	01°23'W	120.8	0.76
<i>Scorpio fuscus</i>	LP 1556	Israel	32°33'N	34°57'E	46.3	0.39
<i>Chersonesometrus bastawadei</i>	LP 16787	India	15°54'N	75°44'E	57.3	0.38
<i>Chersonesometrus beccaloniae</i>	LP 13134	India	12°35'N	78°39'E	69.9	0.61
<i>Chersonesometrus fulvipes</i>	LP 13125	India	18°01'N	74°01'E	63.5	0.35
<i>Chersonesometrus hendersoni</i>	LP 16676	India	15°22'N	76°26'E	54.3	0.38
<i>Chersonesometrus madraspatensis</i>	LP 4324	India	12°43'N	80°13'E	105.4	-
<i>Chersonesometrus pelekomanus</i>	LP 16773	India	14°11'N	76°24'E	48.5	0.46
<i>Chersonesometrus tristis</i>	LP 13135	India	13°35'N	79°15'E	70.4	0.61
<i>Deccanometrus bengalensis</i>	LP 14238	India	21°55'N	86°45'E	134.1	0.50
<i>Deccanometrus latimanus</i>	LP 15569	Pakistan	34°31'N	72°28'E	79.6	0.45
<i>Deccanometrus obscurus</i>	LP 6225	India	19°12'N	72°54'E	224.8	0.59
<i>Deccanometrus xanthopus</i>	LP 13126	India	18°01'N	74°01'E	63.5	0.35
<i>Gigantometrus swammerdami</i>	LP 13128	India	13°23'N	79°06'E	69.9	0.48
<i>Gigantometrus titanicus</i>	LP 12287	Sri Lanka	07°56'N	81°00'E	140.6	0.67
<i>Heterometrus glaucus</i>	LP 13922	Indonesia	01°12'N	97°25'E	272.8	0.84
<i>Heterometrus laevigatus</i>	LP 13047	Thailand	07°41'N	99°52'E	165.3	0.84
<i>Heterometrus laoticus</i>	LP 11131	Laos	17°24'N	104°50'E	190.7	0.69
<i>Heterometrus longimanus</i>	LP 12116	Malaysia	01°36'N	110°12'E	306.8	0.79
<i>Heterometrus petersii</i>	LP 1604	Singapore	01°22'N	103°48'E	190.0	0.69
<i>Heterometrus silenus</i>	LP 11252	Vietnam	11°16'N	107°25'E	194.8	0.70
<i>Heterometrus spinifer</i>	LP 11949	Malaysia	03°18'N	101°42'E	213.2	0.76
<i>Heterometrus thorellii</i>	LP 13050	Myanmar	17°12'N	96°10'E	228.6	0.65
<i>Javanimetrus cyaneus</i>	LP 12112	Indonesia	06°42'S	107°02'E	256.7	0.72
<i>Sahyadrimetrus kanarensis</i>	LP 14029	India	14°59'N	74°08'E	276.5	0.72
<i>Sahyadrimetrus mathewi</i>	LP 13132	India	10°37'N	76°14'E	264.8	0.71
<i>Sahyadrimetrus rugosus</i>	LP 14007	India	11°32'N	76°14'E	284.2	0.77
<i>Sahyadrimetrus scaber</i>	LP 16771	India	12°38'N	75°18'E	383.6	0.72
<i>Sahyadrimetrus tikaderi</i>	LP 16764	India	10°26'N	76°59'E	151.2	0.72
<i>Srilankametrus gravimanus</i>	LP 12279	Sri Lanka	07°56'N	81°00'E	140.8	0.70
<i>Srilankametrus indus</i>	LP 12282	Sri Lanka	07°18'N	80°38'E	190.1	0.77
<i>Srilankametrus pococki</i>	LP 12281	Sri Lanka	08°07'N	80°40'E	154.5	0.68
<i>Srilankametrus serratus</i>	LP 14058	Sri Lanka	06°24'N	80°01'E	259.5	0.75